

Spectral Volume Models: Universal High-Frequency Periodicities in Intraday Trading Activities*

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Abstract

We develop spectral volume models to systematically estimate, explain, and exploit the high-frequency periodicity in intraday trading activities using Fourier analysis. The framework consistently recovers periodicities at specific frequencies in three steps, despite their low signal-to-noise ratios. This reveals persistent and universal high-frequency periodicities in the United States (US) and Chinese stock markets in recent years, and the dominant frequencies explain a significant fraction of the total variance of intraday volumes. We provide evidence that this phenomenon likely reflects the behaviors of trading algorithms with repeated and regular trading instructions. Finally, we demonstrate that uncovering such high-frequency periodicities improves intraday volume predictions and VWAP execution qualities, yields insights for price informativeness of algorithmic trading, and generates excess returns.

Keywords: Trading volume; Periodicity; Algorithmic trading; VWAP execution; Price informativeness; Excess return.

JEL Classification: C32, C55, G12, G14

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1 Introduction

Understanding trading volume has long been a fundamental question for both academics and practitioners.¹ Specific patterns of trading volume provide valuable information about a security, including the magnitudes of future price changes (Karpoff, 1987; Wang, 1994), price momentum (Jain and Joh, 1988; Lee and Swaminathan, 2000), trading cost (Goyenko et al., 2024), and serial and cross correlations in stock returns (Campbell, Grossman, and Wang, 1993; Andersen, 1996; Chordia and Swaminathan, 2000).

Trading volumes have been documented to contain periodic patterns that yield important applications for predicting future prices and volumes. Hong and Yu (2009) find that a seasonal effect of lower trading volumes in the summer is associated with lower stock returns. Heston, Korajczyk, and Sadka (2010) find that stock returns and trading volumes in half-hour intervals show periodicities at daily frequencies, a phenomenon rationalized by infrequent rebalancing (Bogousslavsky, 2016).² Chincio and Ye (2017) find that stocks dominated by short-run fluctuations in trading volume have abnormal excess returns. Despite their importance, most known periodicities are at the daily, weekly, or monthly frequency.

In the analysis of high-frequency transaction-level data, Hasbrouck and Saar (2013) observe that messages arrive at the market with large peaks in several milliseconds shortly after the one-second boundary. Broussard and Nikiforov (2014) find a similar pattern for 36 stocks in which the trading volume spikes at regular intervals throughout the trading day.

These empirical findings suggest that there may be important, but less well-known, periodic patterns in high-frequency trading volumes. In addition, they raise questions about whether these patterns persist across stocks and markets, how to uncover them reliably, why they occur, and whether they matter for trading and asset returns.

In this article, we develop a framework to systematically answer these questions. We model, estimate, explain, and exploit the periodicity in high-frequency intraday trading activities using a spectral representation of trading volumes and Fourier analysis. We show that universal high-frequency periodicities exist at second and minute levels across a large sample of stocks in both

¹Theories of trading volume have been explored from the informational (Karpoff, 1986; Admati and Pfleiderer, 1988; Wang, 1994), behavioral (Statman, Thorley, and Vorkink, 2006; Hasbrouck, 2018), and portfolio (Lo and Wang, 2000) perspectives.

²See also the well-known intraday U-shaped pattern (Admati and Pfleiderer, 1988; Jain and Joh, 1988).

the US and the Chinese stock markets. We provide evidence suggesting that these patterns reflect the behaviors of trading algorithms with repeated and regular trading instructions.³ We show that accounting for these periodic patterns improves volume prediction and generates excess returns.

Our dataset contains tick-level transaction data for constituents of the S&P 500 index with available data (around 500 stocks) from 1993 to 2023 in the US and all stocks traded on the Shenzhen Stock Exchange (more than 2000 stocks) in China from 2014 to 2023. Our analysis of the Chinese stock market provides not only validation of the US results in a different market, but also useful insights in its own right because of its growing size and increasing centrality in the global financial system (Billio et al., 2022).⁴ We compute three different versions of trading volumes every three seconds for all stocks over all trading days, including the number of trades, the number of traded shares, and the volume in monetary terms. This leads to approximately 54 billion samples for each version of the trading volume.

We use a spectral representation to model the intraday time series of trading volumes of individual stocks, which contains a trend term, an idiosyncratic noise term, and multiple spectral terms at different frequencies. We demonstrate that periodicities at specific frequencies—despite their low signal-to-noise ratios—can be consistently recovered in a three-step process. This includes removing the intraday trend, estimating the autocovariance function of the de-trended series, and estimating the *intensity coefficient* of each periodic term using inverse discrete Fourier transform (DFT). We validate the effectiveness of our framework with synthetic data and show its robustness against misspecifications of the periodic functional forms.

Using this framework, we uncover important universal periodicities in trading activities in both the US and Chinese markets. We compute the *frequency variance ratio* (fVR) for a periodic term as the fraction of variance in trading volumes explained by that term, which is a function of the intensity coefficients. In the US market, we find the strongest periodicities at the 10-second, 15-second, 20-second, 30-second, 1-minute, and 5-minute frequencies, while the Chinese market shows the strongest periodicities at lower frequencies including 30-second, 1-minute, 2.5-minute, 5-minute, and 10-minute ones. These frequencies each explain 2 to 10 percent of the variance of de-trended intraday volumes on average, compared to the benchmark of 0.2 percent if no periodicity exists.

³This includes all investors who rely on algorithms to automate their trade, including, for example, both sophisticated quantitative hedge funds and traditional asset managers who rely on brokers’ algorithms for execution.

⁴China’s stock market is the second-largest in the world after the US in terms of market capitalization.

These periodicities have persisted in the US market since at least 2008 and in the Chinese market since at least 2014, with an increasing strength of periodicity over time driven by trading at higher frequencies such as 1 minute. We also demonstrate that our model can be used to uncover periodic patterns in other financial time series such as the bid-ask spread and volatility.

To understand these periodicities, we conduct a range of analyses which together suggest that they likely reflect the behaviors of trading algorithms with repeated and regular trading instructions, including their methods to obtain and process information. Examples include the VWAP execution strategy (Konishi, 2002; Frei and Westray, 2015) and the intraday contrarian strategy (Grant, Wolf, and Yu, 2005; Khandani and Lo, 2011). This is consistent with Bogousslavsky’s (2016) model of infrequent rebalancing as well as equilibrium models of regular or clustered trading volumes to reduce trading costs (e.g. sunshine trading) (Pagano, 1989; Admati and Pfleiderer, 1991). It can also be regarded as the high-frequency version of the well-known fund flow patterns at the low frequency (Guercio and Tkac, 2002; Frazzini and Lamont, 2008; Campbell, Ramadorai, and Schwartz, 2009; Lou, 2012; Kamstra et al., 2017).

In particular, we first demonstrate that the strength of periodicity is positively associated with the level of algorithmic trading by exploring variations in the measures, cross-section, and time-series of trading volume. We show that high-frequency periodicities are strongest for volumes measured by the number of trades, compared to traded shares or monetary volume,⁵ which is consistent with the fact that algorithmic trading tends to split large orders into smaller quantities (O’Hara, Yao, and Ye, 2014). Using a regression analysis of the strength of periodicities on several stock-day level characteristics, we find that the level of algorithmic trading is positively associated with the strength of periodicities. Our analysis based on the full sample (1993–2023 in the US and 2014–2023 in China) shows that the time-series patterns of the strength of periodicity in both markets are highly consistent with the evolution of the popularity of algorithmic trading. Notably, no significant periodicities are observed in the pre-decimalization period before 2001, after which periodicities start to emerge and stay robust since at least 2008 in the US.

Second, we directly link the strength of periodicity to the level of common algorithmic trading strategies in terms of profitability and price impact from trades. The strength of periodicity

⁵Although periodicities are robust when using all three measures of trading volume, no periodicity exists in the average shares per trade or the average monetary value per trade.

is positively associated with the profitability of the VWAP execution strategy and the intraday contrarian strategy, after controlling for other proxies of algorithmic trading. We also show that trade clusters at round times have higher average price impact and, therefore, information content, compared to those at non-round times, providing additional support for behaviors from algorithmic trading because they tend to be informed traders.

Third, we disprove two alternative channels as drivers of observed periodic trading volume. The periodicities remain strong and robust after controlling for several retail trading proxies, including the removal of retail trades identified by the BJZZ algorithm, the removal of odd-lot trades, comparing Robinhood outage days with normal days, and the time-series pattern during the Covid outbreak in 2020. Furthermore, we find no periodicities in the published news articles using the Ravenpack dataset, which suggests that they are unlikely to be the direct cause of periodicity.

Finally, we demonstrate how to exploit the high-frequency periodicities in three applications. The first is to improve intraday volume predictions. We show that models that incorporate the periodic terms uncovered by our spectral model significantly outperform a baseline model that uses the U-shaped intraday trend. The dollar cost savings in volume-weighted average price (VWAP) execution are positive for most stocks, and the magnitude is 2–5% in the US and 3–7% in China compared to the benchmark for the median stock, yielding sizable economic gains.

In our second application, motivated by the recent literature that documents the tension between acquiring information (price informativeness) and incorporating it into asset prices (price efficiency), we show that stocks with stronger periodicity in volume are associated with lower price informativeness despite higher price efficiency. We provide consistent evidence using both the intraperiod timeliness measure and the price nonsynchronicity measure following Weller (2018), which confirms Weller’s (2018) and Gider, Schmickler, and Westheide’s (2019) results from a different angle—intraday periodicities as a proxy of algorithmic trading.

Our last application exploits excess returns from stocks with strong periodicities in trading volume. We show that long-short portfolios constructed based on the cross-sectional rank of the strength of periodicity in trades yield statistically significant alphas after controlling for Fama-French, momentum, and liquidity factors. In addition, stocks with the strongest periodicities change frequently from month to month, which also suggests that such excess returns reflect risks associated with short-term trading rather than traditional risk factors based on firm characteristics.

Related Literature. We contribute to the literature in several aspects. Our work is inspired by a growing literature on using spectral models in the analysis of economic and financial time series including, for example, asset pricing and risk factors (Carr and Madan, 1999; Bogousslavsky, 2015; Dew-Becker and Giglio, 2016; Bandi et al., 2021; Faria and Verona, 2021), intraday return volatility (Andersen and Bollerslev, 1997; Andersen, Thyrgaard, and Todorov, 2019; Andersen et al., 2024), decomposing trading profits (Hasbrouck and Sofianos, 1993; Chaudhuri and Lo, 2019), trading volumes (Chinco and Ye, 2017), business cycle (A’Hearn and Woitek, 2001), consumption correlation (Pakko, 2004), dynamic correlation (Croux, Forni, and Reichlin, 2001; Rua, 2010), and testing causality (Lemmens, Croux, and Dekimpe, 2008; Breitung and Candelon, 2006).

We contribute to this literature by providing a framework of spectral analysis for trading volume that is designed specifically for high-frequency transaction data. Our work provides a methodology for extracting periodic signals in nonstationary financial time series that is different from previous studies such as the Hilbert–Huang Transform (Huang et al., 2003), Wold representation (Bandi et al., 2019; Ortu et al., 2020), and wavelet analysis (Crowley, 2007; Chinco and Ye, 2017; Hasbrouck, 2018; Faria and Verona, 2021). Our framework can also be applied beyond trading volumes to study other high-frequency quantities such as the bid-ask spread, volatility, and lifetime of orders.

Previous studies of periodic patterns in trading volumes either focus on the daily, weekly, and monthly frequencies (Wood, McInish, and Ord, 1985; Harris, 1986; Baxter and King, 1999; Hong and Yu, 2009; Heston, Korajczyk, and Sadka, 2010; Brownlees, Cipollini, and Gallo, 2011; Bogousslavsky, 2016), or are based on high-frequency data for a limited number of sample stocks (Broussard and Nikiforov, 2014; Muravyev and Picard, 2022) or in one single market (Hasbrouck and Saar, 2013; Chinco and Ye, 2017).

Although high-frequency periodic trading patterns have appeared in the literature (Broussard and Nikiforov, 2014; Muravyev and Picard, 2022), our study offers fundamentally new insights in several aspects. First, we provide a rigorous and general framework based on Fourier analysis for trading volume, which allows us to systematically discover a range of frequencies with clustered trading volume, while Broussard and Nikiforov (2014) and Muravyev and Picard (2022) use dummy variable regressions to each identify a selected set, but not all, periodicities.⁶ Second, we conduct

⁶Broussard and Nikiforov (2014) find that volumes spike at regular intervals of 5 or 10 minutes. Muravyev and Picard (2022) focus on trade clusters in the first 100ms of each second.

empirical analysis based on 31 years of data in the US market and 10 years of data in the Chinese market across over 2,500 stocks, thereby establishing the universality of observed periodicities across stocks and markets, and useful time-series patterns.⁷ Third, we extensively analyze the underlying driver of observed high-frequency periodicities in trading volumes and show that they are linked with the level of algorithmic trading, which is not available in the literature. For example, Broussard and Nikiforov (2014, page 197) made a similar conjecture and called for “additional efforts that may help explain this pervasive pattern”. Finally, we provide three novel applications in the context of volume prediction and execution algorithms, price efficiency and price informativeness, and excess returns, which substantially expand the broader economic significance of our findings.

The high-frequency periodic effect in trading volume is related to several theoretical models of trading behaviors including infrequent rebalancing (Bogousslavsky, 2016), sunshine trading (Admati and Pfleiderer, 1991), and optimal execution that minimizes trading costs (Pagano, 1989; Bertsimas and Lo, 1998; Almgren and Chriss, 2001). It is also analogous to the price clustering effect whereby prices tend to fall more frequently on certain values than others, including the whole-dollar effect (Niederhoffer, 1965; Harris, 1991) and the stock-pinning effect on option strikes (Ni, Pearson, and Poteshman, 2005; Avellaneda, Kasyan, and Lipkin, 2012). Our results reveal a clustering effect for executed trades along the time-series dimension rather than the price dimension.

Considerable additional literature exists on the impact of algorithmic trading, high-frequency trading, and AI-powered trading on market liquidity, return volatilities, price efficiency, and price informativeness.⁸ Our results add to this literature by revealing a novel effect of algorithmic trading—periodicities in trading volumes. More generally, high-frequency price and volume data have been used in asset pricing and microstructure.⁹ Our spectral volume model provides a general framework for analyzing any high-frequency quantity with potential periodic patterns.

Our article is organized as follows. Section 2 describes our data and provides a motivating

⁷Broussard and Nikiforov’s (2014) analysis is limited to less than 40 US stocks based on 3.5 years of data and Muravyev and Picard’s (2022) analysis uses 120 US stocks based on 44 trading days.

⁸See, for example, Hendershott, Jones, and Menkveld (2011), Moallemi and Sağlam (2013), Foucault, Hombert, and Roşu (2016), Clark-Joseph, Ye, and Zi (2017), Weller (2018), Gider, Schmickler, and Westheide (2019), Roşu (2019, 2020), Baldauf and Mollner (2020), Bongaerts and Van Achter (2021), Li, Wang, and Ye (2021), Ernst, Spatt, and Sun (2022), Ait-Sahalia and Sağlam (2024), and Dou, Goldstein, and Ji (2024).

⁹Examples include intraday risk factors (Ait-Sahalia and Xiu, 2017; Pelger, 2020), price discovery and liquidity (Bogousslavsky, Collin-Dufresne, and Sağlam, 2021; Hasbrouck, 2021; Bogousslavsky and Collin-Dufresne, 2023), and the impact of market structure on trading (Biais, Hillion, and Spatt, 1995; O’Hara and Ye, 2011; O’Hara, Yao, and Ye, 2014; Bryzgalova, Pavlova, and Sikorskaya, 2023).

example. Section 3 develops the spectral volume model and validates the framework on synthetic data. Section 4 uncovers universal periodicities in both markets empirically. Section 5 provides additional analysis to understand the periodic trading behavior. Sections 6 demonstrate three applications in the context of intraday volume prediction, price informativeness, and excess returns, respectively. Section 7 concludes.

2 Data and Motivating Example

Our raw dataset contains tick-level trade and quote records for stocks in both the US and Chinese markets. We collect data from 1993 to 2023 in the US market and from 2014 to 2023 in the Chinese market. For the majority of our analysis, we focus on the period from 2019 to 2021, but we conduct analysis on robustness and time-series patterns using the full dataset whenever appropriate. We obtain data for the US market from the *NYSE Trade and Quote (TAQ) database*, from which we collected information for stocks that were constituents of the S&P 500 index,¹⁰ over 757 trading days in 2019–2021. We obtain data for the Chinese market from the *Shenzhen Stock Exchange (SZSE) Historical Tick Data*,¹¹ which contains all 2,081 stocks traded in SZSE over 730 trading days in 2019–2021. Our main dataset therefore contains intraday transaction data (trades and quotes) for a total of 2573 (492 + 2081) stocks.

From the raw data, we compute intraday trading volumes every three seconds for all stocks over all trading days. In the US market, open market trading occurs from 9:30 am to 4:00 pm. Therefore, the time series for each stock on each trading day includes 7,800 data points (390 minutes / 3 seconds). In the Chinese market, open market trading occurs from 9:30 to 11:30 am and 1:00 to 2:57 pm.¹² Therefore, the time series for each stock on each trading day includes 4,740 data points (237 minutes / 3 seconds). In total, our processed 3-second time series data contains approximately 10^{10} samples ($492 \text{ (stocks)} \times 757 \text{ (days)} \times 7800 + 2081 \text{ (stocks)} \times 730 \text{ (days)} \times 4740$).

In our analysis, we compute three versions of trading volumes, including the number of trades, the number of traded shares, and the volume in monetary terms (dollar in the US and yuan in

¹⁰More information about the data can be found at <http://wrds-www.wharton.upenn.edu/pages/about/data-vendors/nyse-trade-and-quote-taq>. Out of these stocks, 492 are available in the TAQ database.

¹¹More information about the data can be found at <http://www.szse.cn/English/services/dataServices/index.html>.

¹²There is a lunch break from 11:30 am to 1:00 pm and a closing call auction from 2:57 to 3:00 pm.

China). In this article, we broadly refer to all of them as trading volumes. We discuss their differences and why we use the number of trades as our main object of interest in Section 5.1.1.

Table A.1 in Appendix B provides summary statistics for our three-second intraday trading volume time series. On average, 4.38 trades were executed every three seconds for stocks in the US market, which is approximately 50% higher than the number in the Chinese market. In addition, the number of trades in three-second windows varies greatly across stocks and over time.

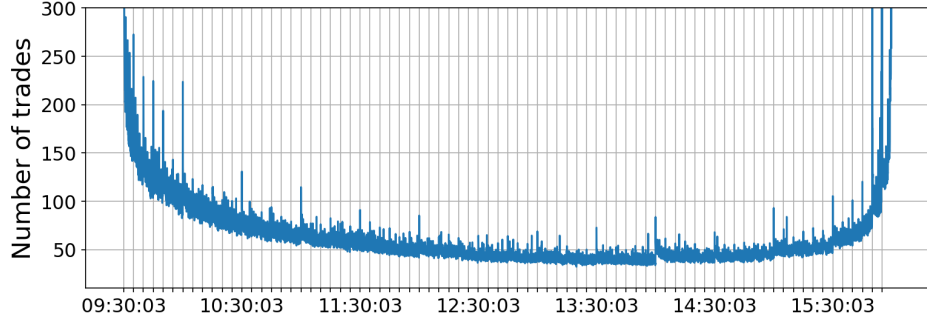
We additionally compute several characteristics to measure the price efficiency, volatility, and level of trading activity for each stock on each day, based on the intraday price and volume information obtained from the raw tick-level data. These characteristics are used in Section 5.1.2. We also use the raw tick-level quotes and trades in our analysis of the price impact regression in Section 5.2.2.

A Motivating Example. It is well documented that the intraday trading volume in the US stock market shows a U-shaped pattern. In the Chinese stock market, the intraday trading volume shows two U shapes due to the lunch break from 11:30 am to 1:00 pm. We demonstrate this pattern using two representative stocks, Apple Inc (AAPL) and Ping An Bank (000001.SZSE), which are two large-cap public companies in the two financial markets.

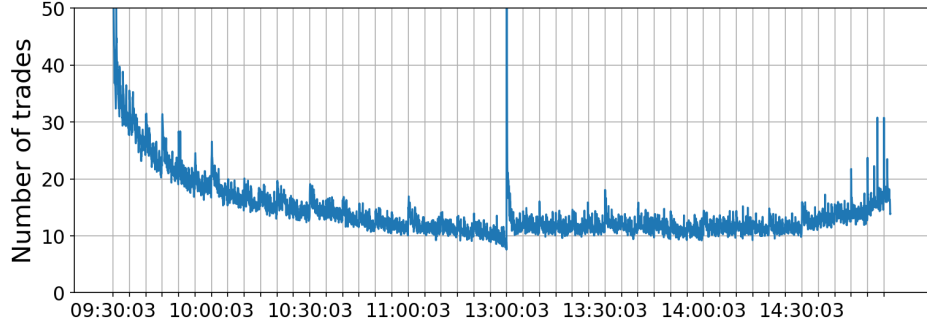
Figures 1a–1b show the intraday trading volumes for these two stocks as measured by the number of trades in each three-second window, averaged over all trading days in 2019–2021. Beyond the clear U-shaped (or double U-shaped) intraday patterns, the time series for both stocks seem noisy at least visually.

However, these patterns look very different when we focus on the cross-sectional average trading volumes. Figures 1c–1d show the intraday trading volumes as measured by the number of trades in each three-second window, averaged across all stocks and over all trading days. The cross-sectional average serves as a noise reduction mechanism. In the Chinese market (Figure 1d), significant spikes occur every five or ten minutes, which exactly match the boundaries of five-minute time intervals represented by the vertical grid. Though not as obvious as in the Chinese market, the same patterns also exist in the US market (Figure 1c), where spikes tend to happen at the boundaries of five-minute time intervals.

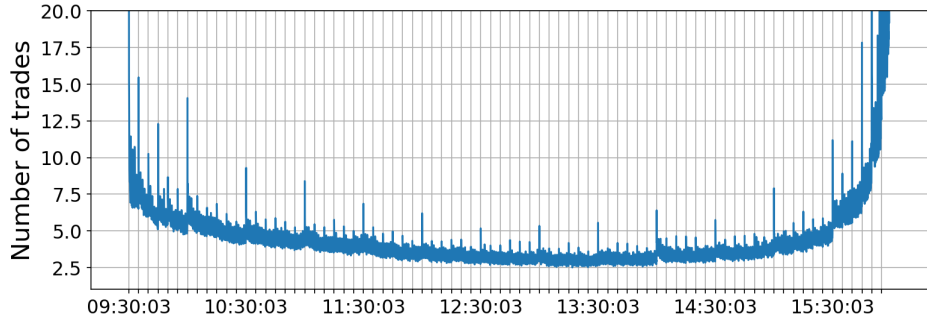
This simple example clearly demonstrates that important periodicities are present in the trading



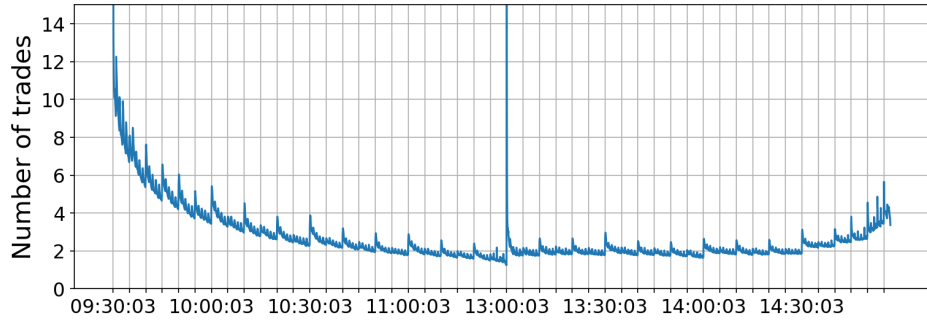
(a) Apple Inc.



(b) Ping An Bank.



(c) The US market.



(d) The Chinese market.

Figure 1: Average intraday trading volumes for two individual stocks (a–b) and the average in two stock markets (c–d) in 2019–2021. Each data point represents the average number of trades in the preceding 3-second windows over all trading days. For example, the value of Ping An Bank at 13:30:03 am is 18.08, which means that, on average, there are 18.08 trades each day from 13:30:00 to 13:30:03 am, over 730 trading days in 2019–2021. Vertical grey lines represent a 5-minute grid.

activities in both markets. However, uncovering these periodicities at the individual stock level may be difficult due to the low signal-to-noise ratio. In the next section, we develop a framework to systematically model and estimate the periodicity in intraday trading activities.

3 A Spectral Model for Trading Volume

Denote by V_t , $t = 1, 2, \dots, T$, the intraday time series of trading volume in terms of either the monetary volume, the number of traded shares, or the number of trades. We assume that V_t has the following spectral representation:

$$V_t := m_t + X_t = m_t + \sum_{j=1}^n a_j \cos(\lambda_j t) + \varepsilon_t, \quad 0 < \lambda_1 < \dots < \lambda_j < \dots < \lambda_n < \pi, \quad (1)$$

where m_t is the non-stochastic trend term and X_t is the de-trended process. X_t is further decomposed into n periodic terms where each term, $\cos(\lambda_j t)$, corresponds to a particular frequency, and a_j is the coefficient for the j -th periodic term, for $j = 1, \dots, n$. Finally, ε_t is a white noise process. We make the following technical assumptions:

Assumption 1. *The trend term, m_t , is uniformly bounded, i.e. there exists a constant M such that $|m_t| < M$, $\forall t$.*

Assumption 2. *The trend term, m_t , does not contain any periodic component that is already captured by the periodic terms, $\cos(\lambda_j t)$'s, i.e. $\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T m_t \cos(\lambda_j t) = 0$, $j = 1, \dots, n$.*

Assumption 3. *The white noise, ε_t , is independent and identically distributed with mean 0 and variance σ^2 over time.*

Assumption 2 requires that the terms $\cos(\lambda_j t)$ in (1) fully capture all periodic patterns at these frequencies. As a result, the trend term, m_t , is orthogonal to $\cos(\lambda_j t)$.

Equation (1) represents the volume time series as a linear combination of three parts. The first trend term captures the well-known intraday U shape (or W shape) in trading volumes. The second part contains n periodic terms with cosine basis functions, which represent the periodic fluctuations we observe in Section 2, driven by different frequencies. Their coefficients, a_j 's, represent the *intensity* of the corresponding frequencies. A higher absolute value of a_j represents a stronger periodicity for the j -th component. The last part is an idiosyncratic noise term.

3.1 Uncovering Periodicity

To uncover periodicities from our trading volume time series based on (1), we first define the sample *autocovariance function* for any of the three processes, $p_t \in \{V_t, m_t, X_t\}$:

$$\gamma_T^p(h) := \frac{1}{T-h} \sum_{t=1}^{T-h} (p_t - \bar{p}) (p_{t+h} - \bar{p}), \quad (2)$$

where $\bar{p} := \frac{1}{T} \sum_{t=1}^T p_t$ is the average over time and h represents a particular lag. Our specification (1) implies that the randomness in the volume time series comes from the idiosyncratic noise, ε_t . However, as a time series, the variation in V_t (and X_t) also comes from the periodic terms, $\cos(\lambda_j t)$.

The following theorem connects the autocovariance function (2) with the periodic terms in (1).

Theorem 1. *Under Assumptions 1–3, as $T \rightarrow \infty$, the autocovariance function, $\gamma_T^X(h)$, for the de-trended volume time series, X_t , converges almost surely as follows:*

$$\gamma_T^X(h) \xrightarrow{\text{a.s.}} \frac{1}{2} \sum_{j=1}^n a_j^2 \cos(\lambda_j h) + \mathbf{1}_{h=0} \sigma^2. \quad (3)$$

We provide proofs in Appendix A. Equation (3) connects the autocovariance function with the intensity of each periodic term, a_j^2 . This result provides a method to estimate a_j 's once we collect enough samples to estimate the autocovariance function, $\gamma_T^X(h)$.

In addition, the following result provides a useful decomposition for the variance of the intraday volumes, V_t , by taking $h = 0$ in (3).

Corollary 1. *Under Assumptions 1–3, as $T \rightarrow \infty$, the variance of the intraday volume process, V_t , can be decomposed by:*

$$\gamma_T^V(0) - \gamma_T^m(0) \xrightarrow{\text{a.s.}} \sigma^2 + \frac{1}{2} \sum_{j=1}^n a_j^2. \quad (4)$$

Corollary 1 shows that the total variance of the intraday trading volume can be decomposed into the variance of the trend term, the idiosyncratic variance, and n additional terms corresponding to different periodic frequencies. It also highlights an important interpretation of a_j^2 because it

represents the fraction of the total variance—beyond the trend and the idiosyncratic term—that can be explained by the j -th periodic term. We refer to a_j 's as the intensity coefficients henceforth.

3.2 Estimation Details

In order to estimate the intensity coefficients, we first estimate the autocovariance function in (2) and then apply the inverse discrete Fourier transform (DFT). We choose $\lambda_j = \frac{j\pi}{n}$, $j = 1, \dots, n$, which corresponds to a period of $\frac{2n}{j} \times 3$ seconds for the j -th term. Based on empirical observations in Figure 1, we choose $n = 500$ so that our periodic basis can capture frequencies that correspond to periods such as five minutes and ten minutes.

Given a dataset of $S \times D$ intraday volume time series for S stocks over D different days, $\{(V_{t,s,d})_{t=1}^T\}_{d=1}^D\}_{s=1}^S$, we explicitly re-write the spectral representation (1) for each stock s on each day d , by adding two additional subscripts:

$$V_{t,s,d} := m_{t,s,d} + X_{t,s,d} = m_{t,s,d} + \sum_{j=1}^n a_{j,s,d} \cos(\lambda_j t) + \varepsilon_{t,s,d}. \quad (5)$$

We then compute the average intraday volume time series over all D days, $\bar{V}_{t,s,\cdot} = \frac{1}{D} \sum_{d=1}^D V_{t,s,d}$, which has the same representation:

$$\bar{V}_{t,s,\cdot} = \bar{m}_{t,s,\cdot} + \bar{X}_{t,s,\cdot} = \bar{m}_{t,s,\cdot} + \sum_{j=1}^n \bar{a}_{j,s,\cdot} \cos(\lambda_j t) + \bar{\varepsilon}_{t,s,\cdot},$$

where $\bar{m}_{t,s,\cdot}$, $\bar{X}_{t,s,\cdot}$, $\bar{a}_{j,s,\cdot}$, and $\bar{\varepsilon}_{t,s,\cdot}$ are the corresponding average time series over all D days:

$$\bar{m}_{t,s,\cdot} = \frac{1}{D} \sum_{d=1}^D m_{t,s,d}, \quad \bar{X}_{t,s,\cdot} = \frac{1}{D} \sum_{d=1}^D X_{t,s,d}, \quad \bar{a}_{j,s,\cdot} = \frac{1}{D} \sum_{d=1}^D a_{j,s,d}, \quad \bar{\varepsilon}_{t,s,\cdot} = \frac{1}{D} \sum_{d=1}^D \varepsilon_{t,s,d}.$$

For each stock, we estimate the average intensity coefficient, $\bar{a}_{j,s,\cdot}$, over all D trading days in the following steps.

Step 1. Eliminating the trend. The first step in spectral estimation is to eliminate the trend term, which corresponds to the general U-shape (or double U-shape) in typical intraday trading volume in the US (or Chinese) market. Let q be a non-negative integer, which we set to 100 in our

empirical analysis. We estimate $\bar{m}_{t,s,\cdot}$ by a rolling-window moving average:

$$\hat{m}_{t,s,\cdot} = \frac{1}{2q+1} \sum_{j=-q}^q \bar{V}_{t+j,s,\cdot}, \quad q+1 \leq t \leq T-q.$$

The variance of the trend term can therefore be estimated by the sample variance of $\hat{m}_{t,s,\cdot}$:

$$\hat{\gamma}_T^{\bar{m}_{s,\cdot}}(0) = \frac{1}{T-2q} \sum_{t=q+1}^{T-q} \left(\hat{m}_{t,s,\cdot} - \frac{1}{T-2q} \sum_{t=q+1}^{T-q} \hat{m}_{t,s,\cdot} \right)^2.$$

Step 2. Estimating autocovariances. We take the de-trended time series, $\bar{V}_{t,s,\cdot} - \hat{m}_{t,s,\cdot}$, as the observed $\bar{X}_{t,s,\cdot}$ and estimate its autocovariance function, $\gamma_T^{\bar{X}_{s,\cdot}}(h)$, for $-n \leq h \leq n$ by:

$$\hat{\gamma}_T^{\bar{X}_{s,\cdot}}(h) = \frac{1}{T-2q-h} \sum_{t=q+1}^{T-q-h} \left(\bar{X}_{t,s,\cdot} - \frac{1}{T-2q} \sum_{t=q+1}^{T-q} \bar{X}_{t,s,\cdot} \right) \left(\bar{X}_{t+h,s,\cdot} - \frac{1}{T-2q} \sum_{t=q+1}^{T-q} \bar{X}_{t,s,\cdot} \right).$$

Step 3. Estimating intensity coefficients. Based on Theorem 1, when the number of sample is large, the limit in (3) is:

$$\begin{aligned} \gamma_T^{\bar{X}_{s,\cdot}}(-h) &= \gamma_T^{\bar{X}_{s,\cdot}}(h) \xrightarrow{\text{a.s.}} \frac{1}{2} \sum_{j=1}^n \bar{a}_{j,s,\cdot}^2 \cos(\lambda_j h) + \frac{\mathbf{1}_{h=0}}{D} \sigma_s^2 \\ &= \frac{1}{4} \sum_{j=1}^n \bar{a}_{j,s,\cdot}^2 \left(e^{i\lambda_j h} + e^{-i\lambda_j h} \right) + \frac{\mathbf{1}_{h=0}}{D} \sigma_s^2 \\ &= \sum_{j=-n}^{-1} \frac{1}{4} \bar{a}_{-j,s,\cdot}^2 e^{i\frac{\pi}{n} j h} + \frac{\mathbf{1}_{h=0}}{D} \sigma_s^2 + \sum_{j=1}^n \frac{1}{4} \bar{a}_{j,s,\cdot}^2 e^{i\frac{\pi}{n} j h}. \end{aligned}$$

Hence we can approximately take

$$\left(\hat{\gamma}_T^{\bar{X}_{s,\cdot}}(-n), \hat{\gamma}_T^{\bar{X}_{s,\cdot}}(-n+1), \dots, \hat{\gamma}_T^{\bar{X}_{s,\cdot}}(0), \dots, \hat{\gamma}_T^{\bar{X}_{s,\cdot}}(n-1), \hat{\gamma}_T^{\bar{X}_{s,\cdot}}(n) \right)$$

as the DFT of $(\bar{a}_{n,s,\cdot}^2, \bar{a}_{n-1,s,\cdot}^2, \dots, \sigma_s^2, \dots, \bar{a}_{n-1,s,\cdot}^2, \bar{a}_{n,s,\cdot}^2)$ when T is large. Therefore, we estimate intensity coefficients by applying the inverse DFT to the estimated autocovariance, $\gamma_T^{\bar{X}_{s,\cdot}}(h)$:

$$\hat{\bar{a}}_{j,s,\cdot}^2 = \frac{2}{n} \left(\sum_{h=1}^n \hat{\gamma}_T^{\bar{X}_{s,\cdot}}(h) \left(e^{-i\frac{\pi}{n} h j} + e^{i\frac{\pi}{n} h j} \right) + \hat{\gamma}_T^{\bar{X}_{s,\cdot}}(0) \right) = \frac{2}{n} \left(2 \sum_{h=1}^n \hat{\gamma}_T^{\bar{X}_{s,\cdot}}(h) \cos\left(\frac{j\pi}{n} h\right) + \hat{\gamma}_T^{\bar{X}_{s,\cdot}}(0) \right).$$

In Appendix C, we validate the effectiveness of our model with synthetic data.

4 Estimating Periodicity Empirically

In this section, we apply the spectral analysis in Section 3 to real data and focus on trading volumes measured by the number of trades. We first conduct an extensive analysis for single stocks in Section 4.1, followed by a market-wide analysis in Section 4.2. We discuss results for other measures of trading volumes and the reason behind our choice of the number of trades as the main target in Section 5.1.1.

4.1 Single Stocks

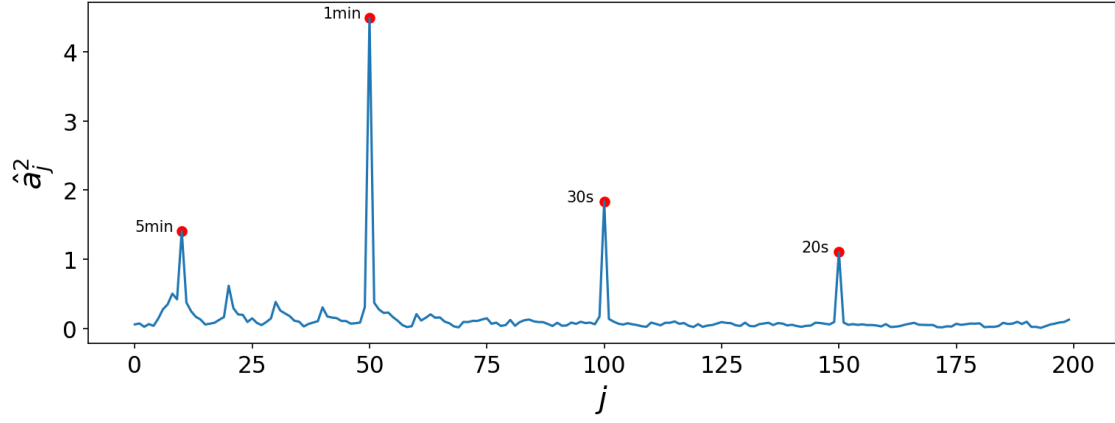
We first focus on two representative stocks, Apple Inc in the US market and Ping An Bank in the Chinese market. We use the de-trended average intraday trading volume for both stocks as our observed data X_t . Due to the lunch break in the Chinese market, we split the intraday time series data for Ping An Bank into the morning session and the afternoon session and estimate their intensity coefficients separately.

Uncovering Periodicity. By following the methodology in Section 3 and setting the number of basis functions, $n = 500$,¹³ we estimate the intensity coefficients which allow us to quantify the relative importance of these periodicities at different frequencies, shown in Figure 2.¹⁴ The estimated squared intensity coefficients, $\hat{a}_j^2$, are shown as a function of their indices, j , and we observe several peaks similar to the results for synthetic data (Figure A.2 in the Appendix). The red dots mark the corresponding frequencies, which happen to be round times such as 30 seconds, 1 minute, and 5 minutes.

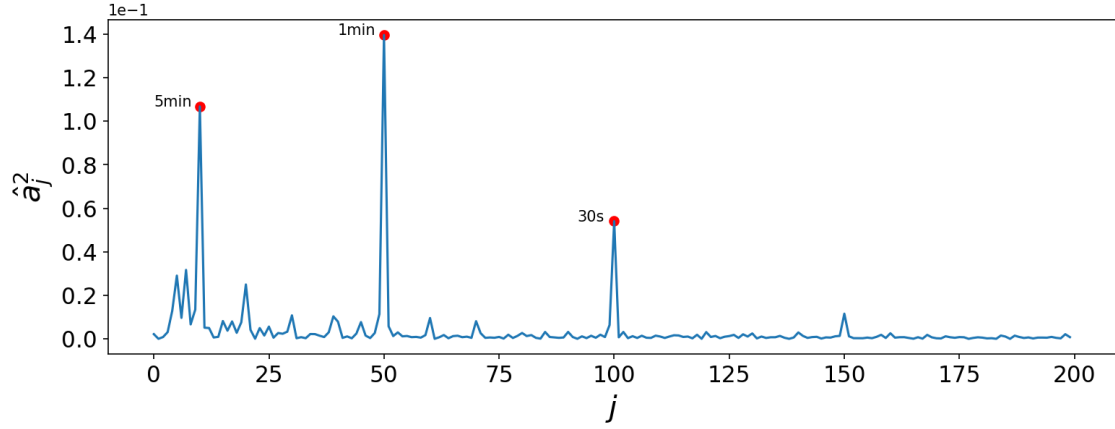
In particular, Apple Inc shows high intensity coefficients at the 5-minute, 1-minute, 30-second, and 20-second frequencies. Similarly, both the morning and the afternoon sessions of Ping An Bank show high intensity coefficients at the 5-minute, 1-minute, and 30-second frequencies. Though these two stocks are traded in different markets with very different compositions of market participants, they demonstrate strikingly similar periodic patterns.

¹³Under this setting, we can detect the frequencies with period in terms of 50 minutes/ j , $j = 1, 2, \dots, 500$.

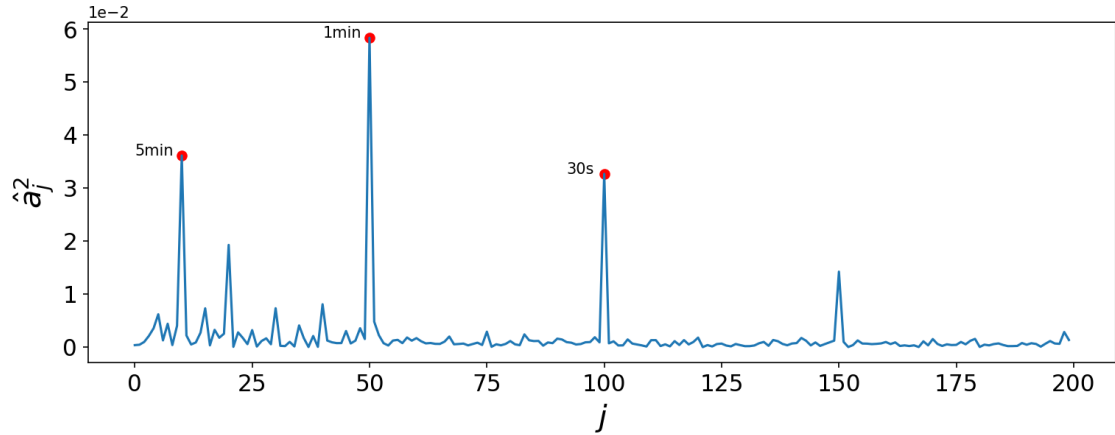
¹⁴Figure A.3 in the Appendix shows the estimated autocovariance function $\hat{\gamma}(h)$ for the de-trended time series, X_t , as an intermediate result.



(a) Apple Inc.



(b) Ping An Bank (morning session).



(c) Ping An Bank (afternoon session).

Figure 2: Estimated squared intensity coefficients, $\hat{a}_j^2$, for two individual stocks in 2019–2021. The red dots mark the period of the corresponding frequencies for the index j .

Spectral Decomposition. We have demonstrated so far the existence of significant periodicities in trading volumes. But how important are they? How much variation in trading activities can they explain? To answer these questions, we decompose the variance of intraday trading volumes into different spectral frequencies based on Equation (4) in Section 3, which allows us to quantify the importance of each periodic component.

We define the *frequency variance ratio* (fVR) for the j -th periodic component as:

$$\text{fVR}_j := \frac{a_j^2}{2\sigma^2 + \sum_{i=1}^n a_i^2}, \quad (6)$$

which measures the explained variance at each frequency as a fraction of the total variance of the de-trended volume time series. It is worth noting that we did not include the variance of the U-shaped (or double U-shaped) trend term, $\gamma_T^m(0)$, in the denominator because the U-shaped (or double U-shaped) pattern is well-known in the literature and therefore not our main focus. In addition, this allows us to focus on the relative importance of each periodic term in explaining the residuals from the well-known U-shaped (or double U-shaped) pattern, which leads to important applications in Section 6.1.

For stocks in the Chinese market, including Ping An Bank, because we estimate spectral models for the morning and the afternoon sessions separately, in practice we compute the fVR in (6) by combining the estimated intensity coefficients from the two sessions:

$$\text{fVR}'_j := \frac{a_{j,\text{morning}}^2 + a_{j,\text{afternoon}}^2}{2\sigma_{\text{morning}}^2 + \sum_{i=1}^n a_{i,\text{morning}}^2 + 2\sigma_{\text{afternoon}}^2 + \sum_{i=1}^n a_{i,\text{afternoon}}^2}. \quad (7)$$

Table 1 summarizes fVRs for the frequencies highlighted in Figure 2, for both Apple Inc and Ping An Bank. Because we included $n = 500$ periodic components in model (1), the benchmark fVR is $\frac{1}{500} = 0.2\%$ if no periodicity exists (see also Figure A.2a in Appendix C). However, the estimated fVRs of the 5-minute, 1-minute, 30-second, and 20-second frequencies for Apple Inc, and those of the 10-minute, 5-minute, 1-minute, and 30-second frequencies for Ping An Bank, are all significantly higher than 0.2 percent. These results confirm the observations in Figure 2 that important periodicities exist in intraday trading volumes, and highlight different relative importance at different frequencies between the two sample stocks from the US and the Chinese markets.

Table 1: Frequency variance ratio (fVR)—the fraction of the variance explained over the total variance of the de-trended time series—for two individual stocks. The benchmark fVR is $\frac{1}{500} = 0.2\%$ if no periodicity exists because we included $n = 500$ periodic components in the model.

Frequency	10-minute	5-minute	1-minute	30-second	20-second
Apple Inc	0.31%	2.97%	9.46%	3.88%	2.33%
Ping An Bank	2.23%	9.05%	12.53%	5.49%	1.63%

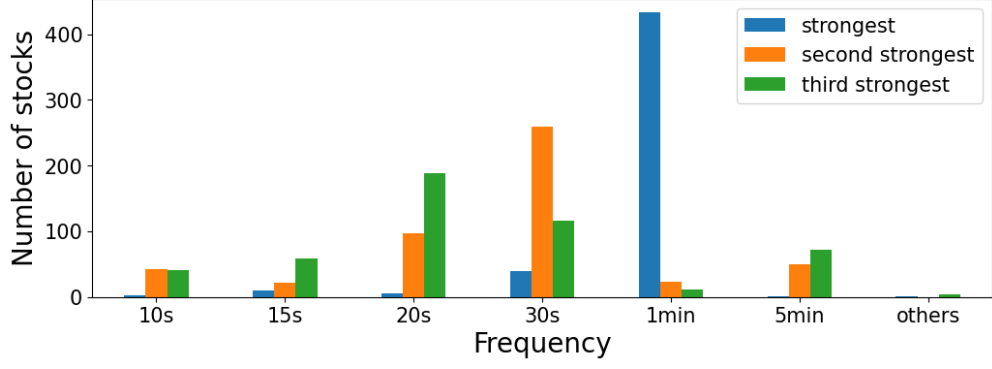
4.2 Market-wide Analysis

Universal Periodicities. After establishing a methodology for uncovering periodicities and measuring their relative importance for two representative stocks, we conduct the same analysis for all 2,573 stocks in our sample.

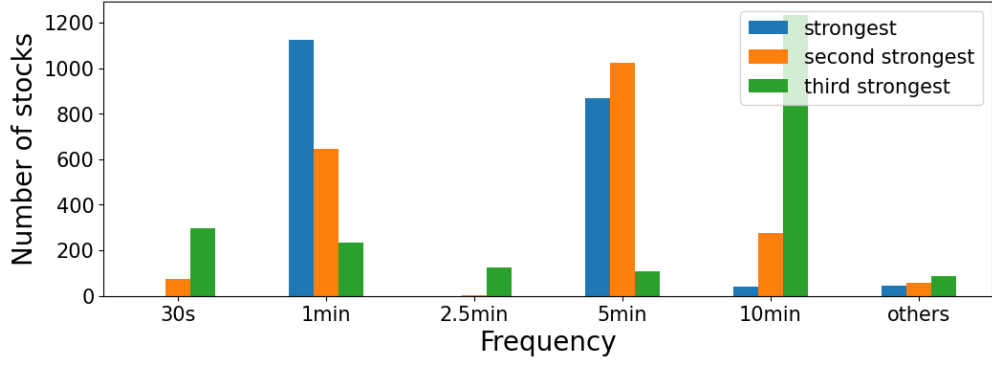
We first analyze which frequencies are the most important across all stocks in both the US and the Chinese markets. For each stock, we record the top three frequencies with the highest intensity coefficients, and summarize the number of occurrences for each frequency in Figure 3.

We observe strikingly consistent market-wide patterns. In the US market, almost all 492 stocks in our sample have their strongest frequencies in a small set: 10-second, 15-second, 20-second, 30-second, 1-minute, and 5-minute. Similarly, in the Chinese market, the three strongest frequencies of almost all 2,081 stocks belongs to a small set: 30-second, 1-minute, 2.5-minute, 5-minute, and 10-minute. These frequencies happen to be multiples of five seconds or one minute. The vast majority of individual stocks in the US have 1-minute as the strongest frequency and 30-second as the second strongest frequency. In the Chinese market, around half of all individual stocks have 1-minute as their strongest frequency, and another half have 5-minute as the strongest frequency.

We also summarize the fVRs across all stocks to quantify the relative contribution to total variance from each frequency. Figure 4 shows the box plot of fVRs across stocks for selected frequencies, which is the market-wide version of Table 1. For the vast majority of stocks, the fVRs at the selected frequencies are significantly higher than the benchmark level of 0.2 percent. In the Chinese market, the median fVRs at the 5-minute frequency and the 1-minute frequency are both around 10 percent, and the median fVR at the 10-minute frequency is around 5 percent. The fVRs in the US market are lower compared to the Chinese market, implying more dispersed trading at different frequencies. Nonetheless, the median fVRs at the 1-minute and 30-second frequencies are around 5 and 3 percent, respectively, both at least ten-fold higher than the benchmark level.



(a) The US market.

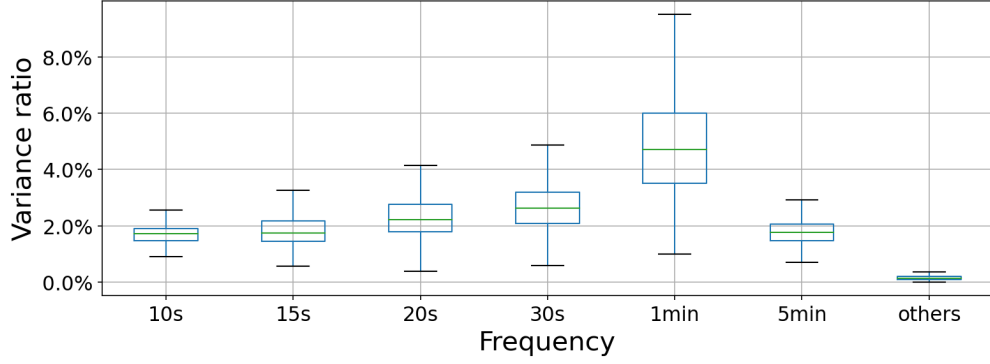


(b) The Chinese market.

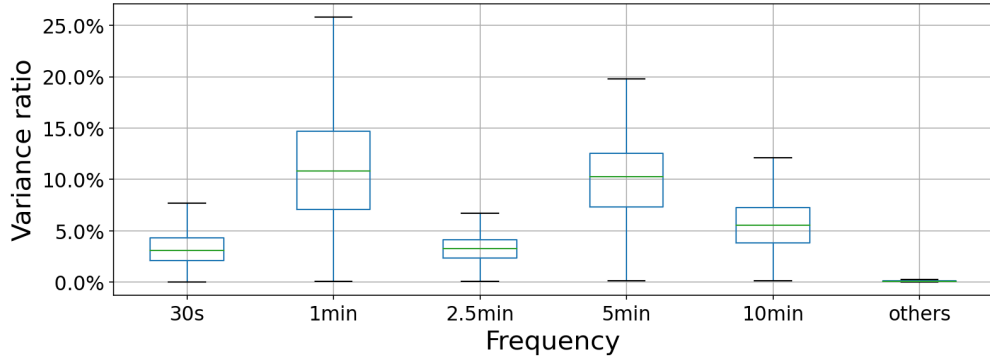
Figure 3: The number of occurrences of each frequency as the top three strongest frequencies in terms of estimated intensity coefficients across all stocks in 2019–2021. Selected frequencies are included and everything else is grouped into the “others” category.

The results in Figures 3–4 confirm that the periodicity we observed in Section 4.1 is not a coincidence for the individual stocks we consider. Instead, it is a universal pattern of trading activities in both the US and the Chinese markets. In addition, although both markets show strong periodicities, we also confirm that the dominant frequencies are different. The US market shows stronger intensity coefficients at higher frequencies, which is consistent with stylized facts that the US market contains more algorithmic and high-frequency traders as opposed to the Chinese market driven by retail investors (Li and Wang, 2010; Jones et al., 2020), and that the Chinese market discourages intraday trading by imposing the “T+1 rule” that requires a minimum of one-day holding period for any position.

We conduct two robustness checks. The first expands our main analysis (based on data in 2019–2021) to a 31-year period (1993–2023) in the US market and a 10-year period (2014–2023) in the Chinese market. The results are provided in Section 5.1.3. The second applies the wavelet



(a) The US market.



(b) The Chinese market.

Figure 4: Box plot of frequency variance ratios (fVR)—the ratio of the variance explained by selected frequencies over the total variance of the de-trended time series—for all stocks in 2019–2021. All other frequencies are grouped into the “others” category. The benchmark fVR is $\frac{1}{500} = 0.2\%$ if no periodicity exists because we included $n = 500$ periodic components in the model.

variance estimator (see also Chinco and Ye (2017)) to offer additional evidence of the existence of the universal high-frequency periodicity. The results are provided in Appendix E.

Other Financial Time Series. It is worth highlighting that although we focus on the trading volume of stocks in this article, our framework in Section 3 can be broadly applied to other intraday time series that potentially contain periodicities. As an example, in Appendix F we apply our framework to the intraday time series of bid-ask spread, volatility, and order imbalance, which are also studied by Heston, Korajczyk, and Sadka (2010). In general, we find periodicities in bid-ask spread, volatility, but not in the order imbalance. We also check the robustness of our results by winsorizing the intraday trading volume data at the 95th percentile for each stock and each trading day to reduce the impact of extreme data on fVR. We find robust periodicities and, in certain cases, even stronger fVRs.

5 Understanding Periodic Trading Behavior

In this section, we conduct additional analyses to understand the periodic trading behavior we document. We acknowledge that in order to definitively pinpoint the source of periodic trading behavior, it requires investors’ identity labels, which we cannot access. Therefore, we conduct seven different analyses *without* relying on identity information which, taken together, suggest that the periodicity in trading volume likely reflects the behaviors of trading algorithms with repeated and regular trading instructions.

5.1 Variation in Measures, Cross-section, and Time-series of Trading Volume

5.1.1 Different Measures of Trading Volume

In our main analysis, we choose the number of trades as our measure of trading volume, instead of the number of traded shares or the monetary volume. This section shows that periodic trading behavior is robust across the three different measures of trading volume. However, it is the strongest when measured in the number of trades. This is consistent with the interpretation that algorithmic traders drive this phenomenon because they tend to send orders in small quantities by splitting large orders (O’Hara, Yao, and Ye, 2014; Johnson, Van Ness, and Van Ness, 2017).

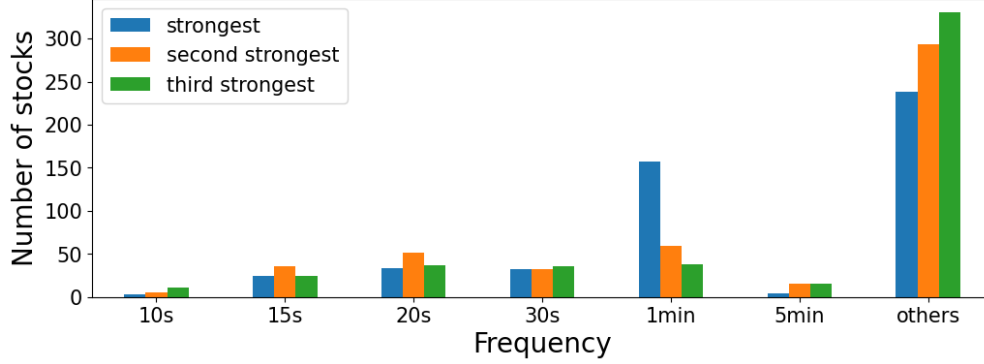
We run analysis in parallel with those in Section 4 for both the number of traded shares and the monetary volume, and find consistent periodicities in these alternative definitions of trading volume with those in Section 4. Figures A.8–A.11 in Appendix G summarize these results in detail. These findings are also consistent with the results of Broussard and Nikiforov (2014) who study volume in traded shares and the number of trades.

However, we also observe that the periodicities in both the number of traded shares and the monetary volume are slightly weaker than the periodicities in traded shares. To understand what drives the difference between these measures, we observe the following relationships:

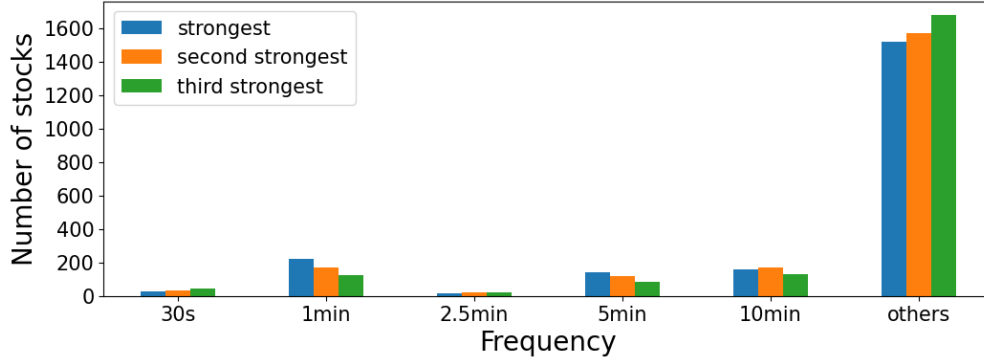
$$\begin{aligned} \text{number of trades} \times \text{average shares per trade} &\approx \text{number of traded shares}, \\ \text{number of trades} \times \text{average monetary value per trade} &\approx \text{monetary volume}. \end{aligned} \tag{8}$$

Therefore, we follow the same methodology as in Section 4 and estimate the intensity coefficients by

replacing the number of trades with the *average shares per trade* and the *average monetary value per trade*, respectively. Figure 5 shows the top three frequencies when using the *average shares per trade*. In sharp contrast to Figure 3, very few stocks show periodicities at the selected frequencies.



(a) The US market.



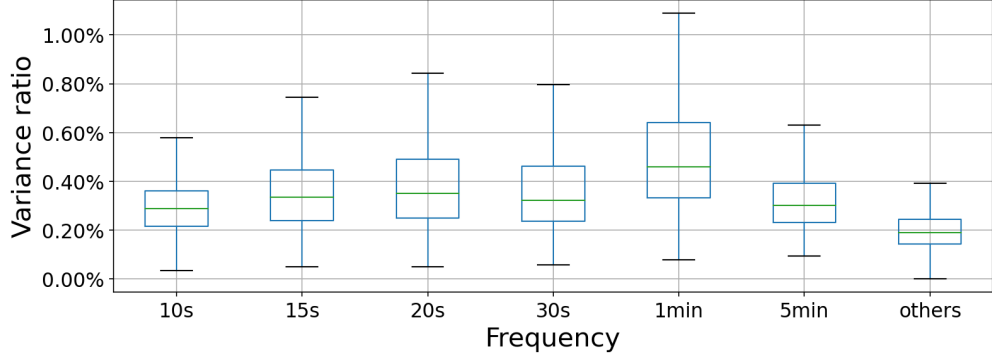
(b) The Chinese market.

Figure 5: The number of occurrences of each frequency as the top three strongest frequencies in terms of estimated intensity coefficients across all stocks in 2019–2021, when the average shares per trade is used.

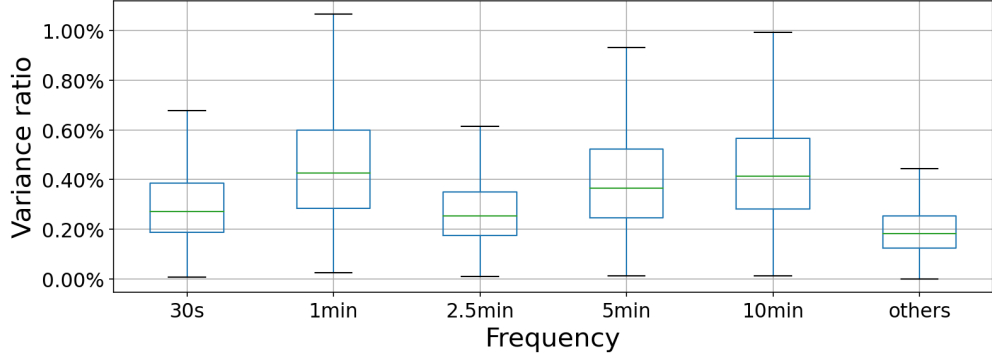
Furthermore, Figure 6 shows the fVRs for the *average shares per trade*. In contrast to Figure 4, the fVRs are much closer to the benchmark level of 0.2 percent at all frequencies for most stocks, which implies very weak evidence of periodicity in this measure.¹⁵

In Figures A.12–A.13 in Appendix G, we further confirm that these results are similar for the *average monetary value per trade*, in the sense that very few stocks show periodicities at the selected frequencies, and that the fVRs are much closer to the benchmark level of 0.2 percent at all frequencies for most stocks.

¹⁵We also find that signed volumes do not show periodicities, which suggests that this phenomenon is not driven by one side of trading alone.



(a) The US market.



(b) The Chinese market

Figure 6: Box plot of frequency variance ratios (fVR) when the average shares per trade is used. The benchmark fVR is $\frac{1}{500} = 0.2\%$ if no periodicity exists because we included $n = 500$ periodic components in the model.

Together, these results provide compelling evidence that the periodicities in trading activities come from the number of trades, rather than the average shares per trade or the average monetary value per trade. This is also consistent with the fact that the strongest periodicities are observed for the number of trades, compared to the number of traded shares or the monetary volume (see Figures A.8–A.11 in Appendix G). The reason that periodicities still exist in the latter two measures is that they can be represented as the product of a periodic term and another non-periodic term, as shown in Equation (8). But the non-periodic terms simply add noise.

A common practice in algorithmic and high-frequency trading is to split large orders into smaller ones (O’Hara, Yao, and Ye, 2014; Johnson, Van Ness, and Van Ness, 2017). The fact that periodicities are the strongest in the number of trades suggests that they may reflect the behaviors of investors who adopt algorithmic trading and algorithmic execution.

5.1.2 Correlation with Stock Characteristics

In this section, we show that stocks and days with higher levels of algorithmic trading tend to have stronger periodicities in trading volume.

To achieve this goal, we need a statistic to summarize the strength of periodicity for each stock on each day. We have documented in Section 4 that the frequencies with the highest fVRs are at 10-second, 15-second, 20-second, 30-second, 1-minute, and 5-minute in the US market, and 30-second, 1-minute, 2.5-minute, 5-minute, and 10-minute in the Chinese market (see Figures 3–4). Therefore, for each stock s on trading day d , we define the following proxy for the strength of periodicity:

$$peri_{s,d} := \begin{cases} fVR_{10s,s,d} + fVR_{15s,s,d} + fVR_{20s,s,d} + fVR_{30s,s,d} + fVR_{1min,s,d} + fVR_{5min,s,d}, \\ \quad \text{if } s \text{ is a stock in the US market,} \\ fVR_{30s,s,d} + fVR_{1min,s,d} + fVR_{2.5min,s,d} + fVR_{5min,s,d} + fVR_{10min,s,d}, \\ \quad \text{if } s \text{ is a stock in the Chinese market.} \end{cases} \quad (9)$$

In contrast to the methodology in Section 4, which estimates the fVR based on a spectral decomposition of the intraday trading volume time series averaged over all trading days in 2019–2021, here we need the fVR for each day because certain stock characteristics change on a daily basis. As a result, for stock s and trading day d , we estimate $VR_{j,s,d}$ based on a spectral decomposition of the intraday trading volume time series on that day, where the first index in these fVR’s represents the corresponding frequency. Therefore, for each stock s , Equation (9) defines a new time series, $\{peri_{s,d}\}$, for $d = 1, 2, \dots, D$.

On the other hand, we compute several characteristics to measure activity level, trading efficiency, and fraction of algorithmic trading at the stock-day level by following widely used proxies in the literature. For example, Hendershott, Jones, and Menkveld (2011) first suggest that higher levels of the number of messages per day and the negative trading volume (in monetary terms) divided by the number of messages can be taken as proxies for higher fractions of algorithmic trading. This idea has been further developed by Boehmer, Fong, and Wu (2021) and Hatch et al. (2021). O’Hara, Yao, and Ye (2014) suggest that a sizable fraction of odd lots are placed by algorithmic traders. Based on their idea, Weller (2018) and Erhard and Sloan (2020) use odd lots volume ratios

as a proxy of algorithmic trading. In addition to these metrics, we also add several volatility-related metrics to our list of explanatory variables, which are summarized below:

- number of messages per day (*mess*),
- negative of monetary trading volume divided by the number of messages (*nvdmess*),
- odd lot volume ratios (*oddrat*),
- volatility of 1-minute intraday return (*vol*),
- the distance between 2 and the ratio of the variance of 2-minute return and the variance of 1-minute return (*varrat*).

We offer several remarks for the list of explanatory variables we chose. The *mess* describes a stock’s level of trading activities in the market. It is also a proxy for algorithmic trading because algorithmic traders tend to send larger numbers of orders than other investors. The *nvdmess* is the negative average trading volume per message, which is positively related to the fraction of algorithmic trading because algorithmic traders tend to place small orders. Similarly, the *oddrat* is positively associated with the fraction of algorithmic trading, because it measures the fraction of volume from odd lots, which are commonly believed to be placed most often by high-frequency/algorithmic traders to reduce order impact. The variable *vol* characterizes volatility. Finally, *varrat* is based on the variance ratio test of Lo and MacKinlay (1988). The ratio of variances in 2-minute returns and 1-minute returns characterizes price deviations from a random walk,¹⁶ and the deviation of the ratio from 2 measures the price efficiency of the underlying stock. Because algorithmic trading tends to improve price efficiency (Hendershott, Jones, and Menkveld, 2011; Conrad, Wahal, and Xiang, 2015; Clark-Joseph, Ye, and Zi, 2017; Boehmer, Fong, and Wu, 2021), *varrat* tends to be negatively associated with the fraction of algorithmic trading.

With these explanatory variables and the response variable $peri_{s,d}$, we run the following regression with different fixed effects:

$$peri_{s,d} \sim \beta_1 mess_{s,d} + \beta_2 nvdmess_{s,d} + \beta_3 oddrat_{s,d} + \beta_4 vol_{s,d} + \beta_5 varrat_{s,d}. \quad (10)$$

Table 2 summarizes the regression results of (10) for stocks in both the US (Panel A) and Chinese (Panel B) markets.

¹⁶In efficient markets where prices follow random walk, this ratio should be close to 2.

Table 2: Regression results of the strength of periodicity, $peri_{s,d}$, on stock-level characteristics, with different fixed effects (F.E.). All variables are normalized by their mean and standard deviations. Estimated coefficients are significant at the 1 percent (***), 5 percent (**), or 10 percent (*) levels. Standard errors are computed based on pooling at the stock level for the stock F.E., at the daily level for the daily F.E., and at both the stock and daily level for the stock and daily F.E. Here we report the “within R^2 ” which measures the variance explained within each panel of fixed effects. The last column shows signs of coefficients implied from the interpretation that periodic trading behaviors are positively associated with algorithmic trading.

	No F.E.	Stock F.E.	Daily F.E.	Stock & daily F.E.	Signs implied from “algo trader” interpretation
Panel A: The US market					
<i>mess</i>	-9.51×10^{-4}	$5.93 \times 10^{-4***}$	$9.42 \times 10^{-4***}$	$8.74 \times 10^{-4***}$	+
<i>nvdmess</i>	$1.14 \times 10^{-3**}$	-2.59×10^{-5}	$2.10 \times 10^{-4***}$	$5.67 \times 10^{-5*}$	+
<i>oddrat</i>	$1.67 \times 10^{-3**}$	$2.67 \times 10^{-4***}$	$4.66 \times 10^{-4***}$	$5.62 \times 10^{-4***}$	+
<i>vol</i>	-2.22×10^{-4}	-2.33×10^{-5}	2.35×10^{-5}	3.45×10^{-5}	+
<i>varrat</i>	$-2.92 \times 10^{-3***}$	$-7.03 \times 10^{-5***}$	$-5.95 \times 10^{-5***}$	$2.76 \times 10^{-5**}$	−
R^2	3%	1%	2%	1%	
Panel B: The Chinese market					
<i>mess</i>	$3.01 \times 10^{-3***}$	$3.00 \times 10^{-3***}$	$2.88 \times 10^{-3***}$	$3.13 \times 10^{-3***}$	+
<i>nvdmess</i>	$1.19 \times 10^{-3***}$	$3.14 \times 10^{-4***}$	$4.88 \times 10^{-5*}$	$4.83 \times 10^{-4***}$	+
<i>oddrat</i>	$7.49 \times 10^{-4**}$	$6.95 \times 10^{-4***}$	$1.24 \times 10^{-3***}$	$6.62 \times 10^{-4***}$	+
<i>vol</i>	$7.59 \times 10^{-4***}$	$1.94 \times 10^{-3***}$	$2.16 \times 10^{-3***}$	$2.17 \times 10^{-3***}$	+
<i>varrat</i>	$-6.80 \times 10^{-4***}$	$-5.67 \times 10^{-5***}$	$-1.60 \times 10^{-4***}$	$-1.38 \times 10^{-4***}$	−
R^2	13%	16%	15%	16%	

Most variables in the US market and all variables in the Chinese market suggest that algorithmic trading is positively associated with periodic patterns in trading volumes. In particular, we observe stronger periodicities for stocks and days with higher fractions of algorithmic trading—measured by *mess*, *nvdmess*, and *oddrat*—and higher levels of price efficiency—measured by negative *varrat*.¹⁷ Specifically, the standardized *peri* value increases by approximately 5×10^{-5} – 3×10^{-3} on average as the standardized *mess*, *nvdmess*, *oddrat*, or negative *varrat* increases by one unit, with *mess* having the largest impact.

¹⁷Trade-to-order ratio, denote as *traderat*, is another widely used algorithmic trading proxy in the literature (Hagströmer and Nordén, 2013; Brogaard, Hendershott, and Riordan, 2014; Weller, 2018). As a robustness check, we conduct the following alternative regression in the US market with both stock and daily fixed effects, by replacing *mess* and *nvdmess* with *traderat* because they are highly correlated:

$$peri_{s,d} \sim \beta_1 traderat_{s,d} + \beta_2 oddrat_{s,d} + \beta_3 vol_{s,d} + \beta_4 varrat_{s,d}.$$

The estimated coefficients are $\hat{\beta}_1 = -2.66 \times 10^{-7}$, $\hat{\beta}_2 = 2.52 \times 10^{-5}$, $\hat{\beta}_3 = 3.96 \times 10^{-5}$, and $\hat{\beta}_4 = -6.28 \times 10^{-6}$, all of which are consistent with our “algo trader” interpretation. On the other hand, *traderat* may not serve as a good proxy in the Chinese market because of the $T + 1$ rule that prohibits sales of a stock bought on the same day.

Overall, the estimated regression coefficients for nearly all variables are statistically significant, and their signs suggest that the strength of periodicity is positively correlated with the activity level, trading efficiency, and the fraction of algorithmic trading of a stock. We emphasize that this analysis does not provide a causal statement for whether the explanatory variables cause periodicity because, for example, the level of algorithmic trading could also be endogenously associated with liquidity and price impact in the market. Nonetheless, this analysis is consistent with the interpretation that algorithmic trading is positively associated with periodic trading behaviors in both markets.

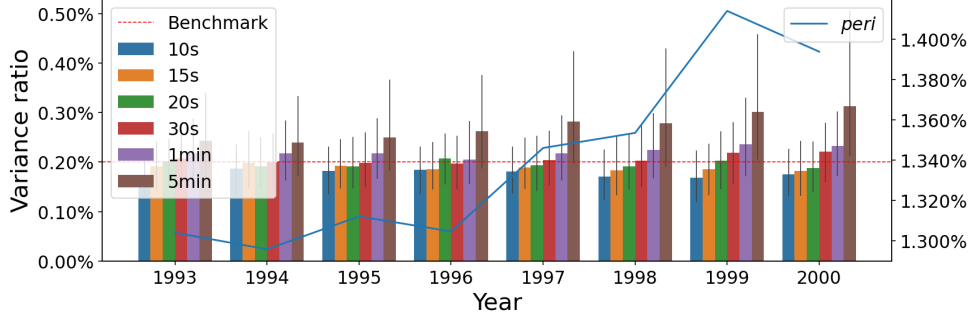
5.1.3 Robustness and Time-series Patterns of Periodicity

As a robustness check, we expand our main analysis to a 31-year period (1993–2023) in the US market and a 10-year period (2014–2023) in the Chinese market. We summarize the market-wide strength of periodicity (fVR) each year in Figure 7. In the US, we plot 1993–2000, 2001–2010, and 2011–2023 separately because the fVRs have very different scales in these periods, but we always use the red dashed line to represent the benchmark fVR if no periodicity exists.

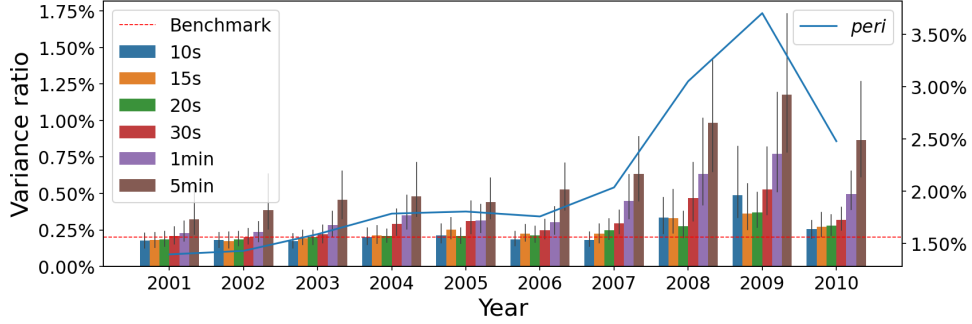
We observe robust periodicities in the US market since at least 2008, as indicated by the fVRs that are consistently and significantly higher than the benchmark level of 0.2%. In contrast, no significant periodicities are observed in the pre-decimalization period before 2001, when trading algorithms were not widely adopted. In the Chinese market, the periodicities have existed consistently since at least 2014. Although they seem weak before 2018 due to the scale of the plot because the fVRs start to increase sharply since 2019, the fVRs for some lower frequencies, such as 10 minutes, are still consistently higher than the benchmark level since 2014.

In addition, the strength of periodicity in both markets increases over time, driven by trading at higher frequencies. For example, in the US, the fVR for 5 minutes has been strong and stable ever since 2008, while the fVRs for 1 minute and 30 seconds both increased from below 1% in 2008 to around 3-4% for the median stock in 2023. In China, the strongest frequency was 10 minutes in the early years before switching to 5 minutes in 2019 and 1 minute in 2021.

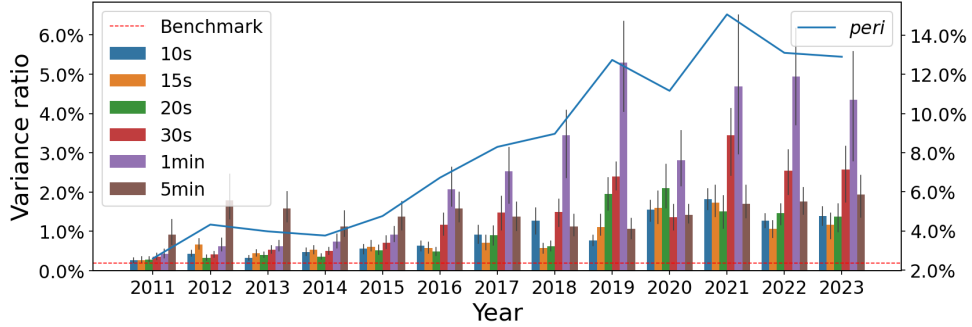
Furthermore, we document an increase in the strength of periodicity in the Chinese market since the first week of October 2024 following the release of multiple economic stimulus policies by the Politburo meeting of the Central Committee. The weekly *peri* in the market increased by more than half, from an average of approximately 0.3 to 0.45, when trading volume significantly



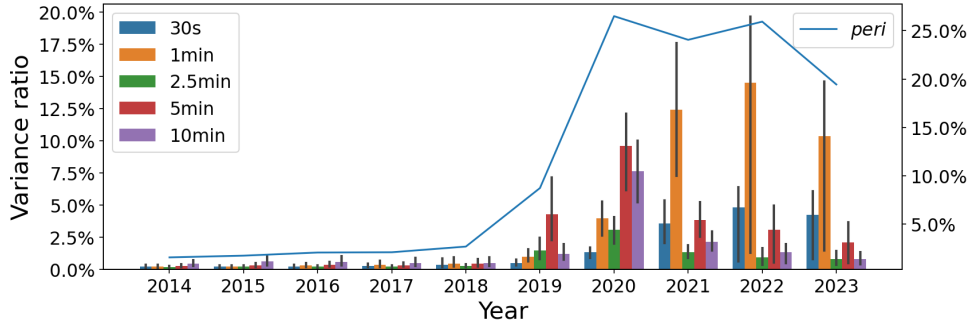
(a) 1993–2000, the US market.



(b) 2001–2010, the US market.



(c) 2011–2023, the US market.



(d) The Chinese market.

Figure 7: Barplot of frequency variance ratios (fVR) for both markets in different years. The error bar represents the first to the third quartile across stocks. The red dashed line marks the benchmark fVR, $\frac{1}{500} = 0.2\%$, if no periodicity exists because we included $n = 500$ periodic components in the model. The blue line shows an aggregate measure of the strength of periodicity across different frequencies (right axis), which is formally defined in (9) in Section 5.

increased and high-frequency trading became more active and profitable,¹⁸ as shown in Figure 8.

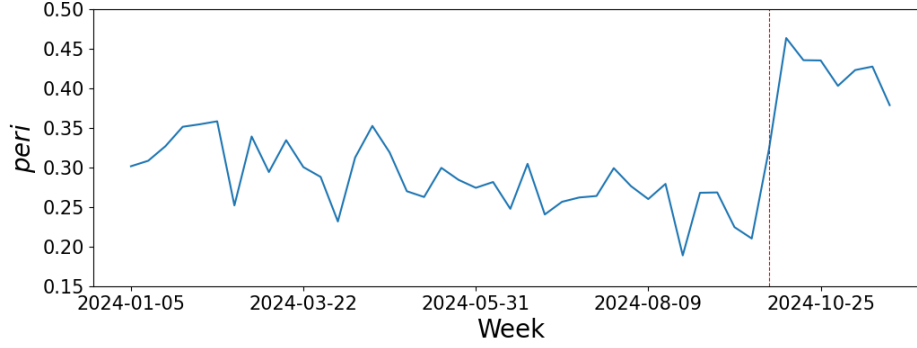


Figure 8: Weekly estimated *peri* values in the Chinese market in 2024, the red dashed line marks the first week of October.

These time-series patterns are also highly consistent with the evolution of the popularity of algorithmic trading. First, the strength of periodicity increases over time, which is consistent with the stylized fact that algorithms have been increasingly adopted in both markets, particularly those at higher frequencies. Second, in the US market, the 5-minute fVRs have been stable since 2008. The 1-minute fVRs become stronger since 2016 and the sub-minute fVRs become stronger since approximately 2019. This is also consistent with the stylized fact that trading generally happens faster and at higher frequencies over time. Third, in the Chinese market, the strength of periodicity becomes stronger particularly since 2019, with a similar pattern of slower frequencies showing up first, followed by faster frequencies. This also coincides perfectly with the period of rapid growth of the quantitative trading industry in China in terms of asset under management (AUM).¹⁹ The event analysis around the first week of October 2024 in China provides additional support that the periodicities are driven by algorithmic trading. Finally, the significant frequencies of periodicity in the US are faster than those in the Chinese market, consistent with the fact that the US market has a much higher fraction of high-frequency and algorithmic trading activities. Despite this fact, the fVRs in the Chinese market are higher than those in the US market after 2020.²⁰

¹⁸See <https://www.hibor.com.cn/data/c9adb25f551342744c72c1bb9d593ae0.html>, accessed January 28, 2025.

¹⁹According to research reports by CITIC Securities, one of the largest investment banks in China, the AUM in quantitative trading in China was around 300 billion RMB before 2018, and rose to 460 billion RMB in 2019, 860 billion RMB in 2020, and over 1 trillion RMB after 2021. See <https://research.citics.com/home>, accessed September 22, 2024.

²⁰Although the US market has more algorithmic trading, it is also much more sophisticated than China. In addition, the unique regulations in the Chinese market—such as the $T+1$ rule that prohibits sales of a stock that was bought on the same day, and the difficulty of short sales of individual stocks—make intraday arbitrage generally much more difficult than in the US, which may lead to more intraday predictable patterns in the Chinese stock market.

5.2 Profitability of Common Strategies and Price Impact from Trades

5.2.1 Profitability of Common Algorithmic and Execution Strategies

We directly link the strength of periodicity to the level of algorithmic trading in the market through two canonical trading strategies. We consider a VWAP execution strategy (Konishi, 2002) and an intraday contrarian strategy (Grant, Wolf, and Yu, 2005; Khandani and Lo, 2011)—the former representing a common volume-based execution strategy (Frei and Westray, 2015) that accounts for a large share of institutional trading (Białkowski, Darolles, and Le Fol, 2008), and the latter a popular return-based statistical arbitrage strategy frequently used by hedge funds and mutual funds (Wei, Wermers, and Yao, 2015), both widely studied in the literature and adopted in the industry. We show that the strength of periodicity in trading volume is positively associated with the level of trading activities of these two example strategies, proxied by their profitability.

We first describe how profitability is measured for these strategies. The VWAP execution strategy equally divides the intraday curve of the expected trading volume into several intervals and executes at the middle of each interval. For each stock s on trading day d , we quantify the performance of this strategy by the relative VWAP loss defined as $\text{Loss}_{s,d} = |\hat{\text{VWAP}}_{s,d} - \text{VWAP}_{s,d}| / \text{VWAP}_{s,d}$, where $\text{VWAP}_{s,d}$ is the true VWAP and $\hat{\text{VWAP}}_{s,d}$ is the realized VWAP of stock s on day d for the execution strategy; the latter uses intraday volume estimated by the moving average over the past 5 days. Higher profitability corresponds to lower $\text{Loss}_{s,d}$.

The intraday contrarian strategy constructs a long-short dollar-neutral portfolio that buys underperforming stocks and sells outperforming stocks relative to the market average every 5 minutes. The portfolio weight $\omega_{s,k}$ of stock s in the k -th 5-minute non-overlapping interval is given by

$$\omega_{s,k} = \begin{cases} -\frac{r_{s,k-1} - \bar{r}_{k-1}}{\sum_{r_{i,k-1} > \bar{r}_{k-1}} (r_{i,k-1} - \bar{r}_{k-1})}, & r_{s,k-1} > \bar{r}_{k-1}, \\ \frac{r_{s,k-1} - \bar{r}_{k-1}}{\sum_{r_{i,k-1} < \bar{r}_{k-1}} (r_{i,k-1} - \bar{r}_{k-1})}, & r_{s,k-1} < \bar{r}_{k-1}, \end{cases} \quad (11)$$

where $r_{s,k}$ is the return of stock s and $\bar{r}_k$ is the equal-weighted average return of all stocks. Higher profitability corresponds directly to the strategy's higher daily return Contra_d .

For the VWAP execution strategy, we have a matrix of $\{\{\text{Loss}_{s,d}\}_{d=1}^D\}_{s=1}^S$ for S stocks over D different days. We perform a similar analysis as in Section 5.1.2 by regressing $\text{peri}_{s,d}$ on the variable

$\text{Loss}_{s,d}$ controlling for other algorithmic trading proxies with different fixed effects:

$$\text{peri}_{s,d} \sim \beta_1 \text{Loss}_{s,d} + \beta_2 \text{mess}_{s,d} + \beta_3 \text{nvdmess}_{s,d} + \beta_4 \text{oddrat}_{s,d} + \beta_5 \text{vol}_{s,d} + \beta_6 \text{varrat}_{s,d}. \quad (12)$$

Table 3 summarizes results of (12) for stocks in both the US (Panel A) and Chinese (Panel B) markets, and the majority of the regression results show that the relative loss of the VWAP execution strategy is significantly negatively associated with periodicities in trading volumes. This suggests that periodic patterns are stronger when the VWAP execution strategy is more profitable and active. The only exception is the regression in the Chinese market with no fixed effect, which likely reflects omitted variable bias.²¹

Table 3: Regression results of the strength of periodicity, $\text{peri}_{s,d}$, on the loss of VWAP execution strategy and stock-level characteristics, with different fixed effects (F.E.). All variables are normalized by their mean and standard deviations. Estimated coefficients are significant at the 1 percent (***), 5 percent (**), or 10 percent (*) levels. Standard errors are computed based on pooling at the stock level for the stock F.E., at the daily level for the daily F.E., and at both the stock and daily level for the stock and daily F.E. Here we report the “within R^2 ” which measures the variance explained within each panel of fixed effects.

	No F.E.	Stock F.E.	Daily F.E.	Stock & daily F.E.
Panel A: The US market				
Loss	$-5.00 \times 10^{-4***}$	-7.35×10^{-4}	$-8.98 \times 10^{-5***}$	$-5.87 \times 10^{-5***}$
<i>mess</i>	$5.59 \times 10^{-4***}$	$3.14 \times 10^{-4***}$	$4.93 \times 10^{-4***}$	$4.23 \times 10^{-4***}$
<i>nvdmess</i>	$2.40 \times 10^{-4*}$	$1.30 \times 10^{-4***}$	$1.61 \times 10^{-4***}$	$1.14 \times 10^{-4***}$
<i>oddrat</i>	$4.88 \times 10^{-4**}$	$2.32 \times 10^{-4***}$	$3.10 \times 10^{-4***}$	$3.40 \times 10^{-4***}$
<i>vol</i>	$-2.30 \times 10^{-3***}$	3.53×10^{-6}	4.32×10^{-5}	4.34×10^{-5}
<i>varrat</i>	$1.57 \times 10^{-2***}$	-4.03×10^{-5}	$-8.60 \times 10^{-5**}$	3.88×10^{-5}
R^2	2%	1%	2%	1%
Panel B: The Chinese market				
Loss	$2.05 \times 10^{-4***}$	$-6.58 \times 10^{-5***}$	$-1.06 \times 10^{-4***}$	$-6.10 \times 10^{-5***}$
<i>mess</i>	$2.60 \times 10^{-3***}$	$1.31 \times 10^{-3***}$	$1.33 \times 10^{-3***}$	$1.15 \times 10^{-3***}$
<i>nvdmess</i>	$6.36 \times 10^{-4***}$	$1.13 \times 10^{-3***}$	$8.75 \times 10^{-4***}$	$1.03 \times 10^{-3***}$
<i>oddrat</i>	$1.01 \times 10^{-3***}$	$1.15 \times 10^{-3***}$	$1.05 \times 10^{-3***}$	$1.06 \times 10^{-3***}$
<i>vol</i>	$-7.15 \times 10^{-4***}$	$3.53 \times 10^{-4***}$	$2.27 \times 10^{-4***}$	$4.19 \times 10^{-4***}$
<i>varrat</i>	$1.47 \times 10^{-2***}$	$-3.87 \times 10^{-5***}$	$-1.11 \times 10^{-4***}$	$-8.66 \times 10^{-5***}$
R^2	13%	12%	10%	8%

²¹When large numbers of shares are traded, investors must balance liquidity concerns against price goals (Białkowski, Darolles, and Le Fol, 2008), which inevitably drives the performance of the VWAP execution strategy to relate to stock- and time-level characteristics such as liquidity. Therefore, the deviation in the Chinese market with no fixed effect underscores the importance of controlling for stock and daily level fixed effects.

For the intraday contrarian strategy, we only have the time series of $\{contra_d\}_{d=1}^D$ over D different days. We therefore run a time series regression for the strength of periodicity $peri_d$ in the market on the explanatory variable $Contra_d$ controlled for aggregated algorithmic trading proxies:

$$peri_d \sim \beta_1 Contra_d + \beta_2 mess_d + \beta_3 nvdmess_d + \beta_4 oddrat_d. \quad (13)$$

Here $mess$, $nvdmess$, and $oddrat$ are the number of messages per day, negative of monetary trading volume divided by the number of messages, and odd lot volume ratios of the market. We run regressions for all data in 2019–2021 and for each year separately as a robustness check.

Table 4 summarizes results of (13) for stocks in both the US (Panel A) and Chinese (Panel B) markets, which again show that the profitability of the intraday contrarian strategy is positively associated with the strength of periodicities. However, the coefficients of $Contra$ are less statistically significant compared to the results for VWAP execution strategies. This is partly because the contrarian strategy is a cross-sectional long-short strategy and, therefore, analysis can only be performed as a time-series regression rather than as a stock-day panel regression for the VWAP execution strategy, which reflects challenges in the statistical power of this analysis.

Table 4: Regression results of the strength of periodicity, $peri_d$, on the daily return of the intraday contrarian strategy and stock-level characteristics in different years. All variables are normalized by their mean and standard deviations. Estimated coefficients are significant at the 1 percent (***), 5 percent (**), or 10 percent (*) levels.

	2019–2021	2019	2020	2021
Panel A: The US market				
$Contra$	2.58×10^{-2}	3.14×10^{-2}	$9.09 \times 10^{-2*}$	2.90×10^{-2}
$mess$	-2.55×10^{-2}	$-3.24 \times 10^{-1***}$	$-3.32 \times 10^{-1***}$	$-1.90 \times 10^{-1**}$
$nvdmess$	$-1.55 \times 10^{-1***}$	$-2.41 \times 10^{-1***}$	-8.86×10^{-2}	$1.73 \times 10^{-1*}$
$oddrat$	$4.52 \times 10^{-1***}$	-4.79×10^{-2}	$-3.27 \times 10^{-1***}$	-8.01×10^{-3}
R^2	16%	30%	36%	2%
Panel B: The Chinese market				
$Contra$	$1.17 \times 10^{-1***}$	3.46×10^{-2}	7.71×10^{-2}	2.23×10^{-2}
$mess$	$2.05 \times 10^{-1***}$	$4.55 \times 10^{-1***}$	4.01×10^{-3}	$6.62 \times 10^{-1***}$
$nvdmess$	-8.87×10^{-2}	$-4.03 \times 10^{-1**}$	$2.35 \times 10^{-1*}$	-4.24×10^{-2}
$oddrat$	-4.37×10^{-2}	$5.61 \times 10^{-1***}$	$-3.08 \times 10^{-1**}$	$1.48 \times 10^{-1*}$
R^2	10%	20%	4%	37%

Overall, the estimated regression coefficients consistently suggest that the strength of periodicity

is positively associated with the profitability of both strategies. Periodicities in trading volume tend to be stronger when these strategies are more profitable and active. Similar to Section 5.1.2, we emphasize that this analysis does not provide a causal statement for whether those two strategies cause periodicity because they are just two examples of algorithmic strategies widely adopted in the market. Nonetheless, this analysis is consistent with the interpretation that algorithmic trading, such as the VWAP execution strategy and the intraday contrarian strategy, can drive periodic patterns in volume.

5.2.2 Price Impact During Spikes of Trading Volume

In this section, we show that trades during spikes of volume have a higher price impact than those at other times, which suggests that they contain more information.²² This is consistent with the interpretation that algorithmic traders drive these trade clusters because they tend to be informed.

We adopt the influential price impact regression first proposed by Glosten and Harris (1988) to estimate the price impact, which decomposes the quote spread into the adverse selection cost component and the order-processing cost component. In particular, we take the raw tick-level quote and trade data and model the price change between two consecutive trades by:

$$p_n - p_{n-1} = \lambda_0 d_n + \lambda_1 q_n + \gamma_0 (d_n - d_{n-1}) + \gamma_1 (q_n - q_{n-1}) + \varepsilon_n, \quad (14)$$

where p_n , d_n , and q_n are the price, direction, and signed quantity of the n -th trade, respectively. If a trade is buyer-initiated, $d_n = 1$ and $q_n > 0$; otherwise $d_n = -1$ and $q_n < 0$. Here the parameters γ_0 , γ_1 , λ_0 , and λ_1 capture different components that determine price changes, and we summarize their interpretations in Table 5.

Table 5: Interpretation of parameters in the price impact regression of Glosten and Harris (1988).

Parameter	Intepretation
λ_0	the impact of the trading direction
λ_1	the impact of the trading quantity
γ_0	the constant part of order-processing cost
γ_1	the order-processing cost that varies with the trading quantity

²²Proxies of information content are closely related to price impact. For example, Hasbrouck (1991a,b) models the information effect of a trade by the ultimate price impact of the trade innovation. Easley, De Prado, and O'Hara (2016) develop a Bayesian model of inferring information from trade executions and price change.

To compare the price impact between trades at round times and those at non-round times, we introduce a dummy variable, $\mathbf{1}_{5min,n}$, that indicates whether the n -th trade belongs to a 3-second interval at the 5-minute round times such as 9:35:00 am, 9:40:00 am, etc., when strong spikes in volume have been documented in Section 4. Our adjusted price impact regression is therefore:

$$\begin{aligned} p_n - p_{n-1} = & \lambda_0 d_n + \tilde{\lambda}_0 d_n \mathbf{1}_{5min,n} + \lambda_1 q_n + \tilde{\lambda}_1 q_n \mathbf{1}_{5min,n} \\ & + \gamma_0 (d_n - d_{n-1}) + \tilde{\gamma}_0 (d_n - d_{n-1}) \mathbf{1}_{5min,n} + \gamma_1 (q_n - q_{n-1}) + \tilde{\gamma}_1 (q_n - q_{n-1}) \mathbf{1}_{5min,n} + \varepsilon_n. \end{aligned} \quad (15)$$

For each individual stock in our dataset of both the US and the Chinese stock markets, we estimate a price impact regression (15).²³ Figure 9 summarizes the regression results for the US and the Chinese markets. Because the estimated parameters of (15) depend on the magnitude of stock prices, we normalize all parameter estimates by a stock's price in these figures.

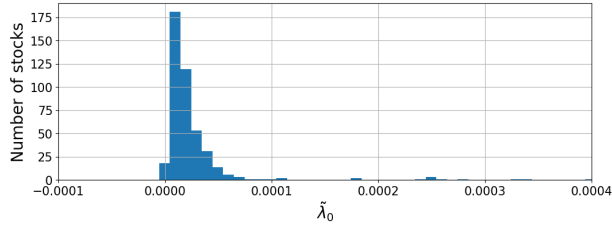
The vast majority of stocks yield positive parameter estimates in the price impact regression. In particular, the fractions of stocks with positive estimated values of $\tilde{\lambda}_0$, $\tilde{\lambda}_1$, $\tilde{\gamma}_0$, and $\tilde{\gamma}_1$ are 99.2%, 74.2%, 99.6%, and 81.5% in the US market. The results are similar in the Chinese market.

Positive values of $\tilde{\lambda}_0$ and $\tilde{\lambda}_1$ imply that the price impact of a unit trade at round times is higher than that at non-round times on average. Similarly, positive values of $\tilde{\gamma}_0$ and $\tilde{\gamma}_1$ imply that the spread and the trading cost at round times are higher than those at non-round times. Both suggest that these volume spikes are associated with higher information content.²⁴

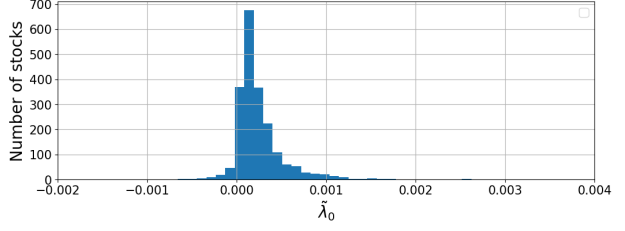
These trades are likely triggered by trading algorithms that exploit high-frequency information based on input data and signals that are updated at round times, or persistent trading from broker's trading algorithms on behalf of institutional investors, which can also be amplified by the order anticipation strategies documented by Sağlam (2020).

²³Regression (15) requires information about the direction of each trade in the data, which we have in the Chinese market. We adopt Lee and Ready's (1991) tick test to identify the direction of each trade in the US market.

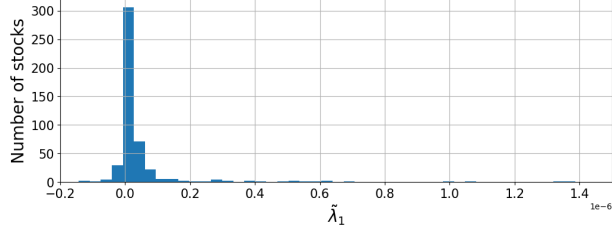
²⁴We emphasize that our findings do *not* necessarily imply that traders at round times would realize lower trading costs *if* they switch to non-round times, which needs counterfactual experiments to assess.



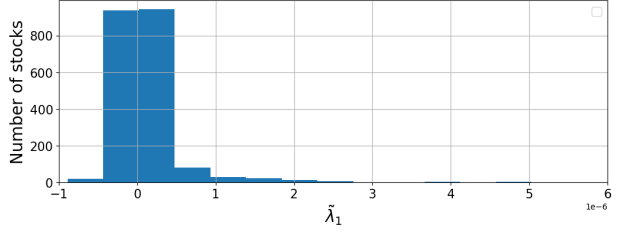
(a) Histogram of $\tilde{\lambda}_0$ across stocks. US market.



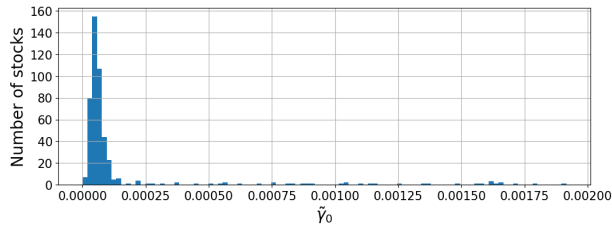
(b) Histogram of $\tilde{\lambda}_0$ across stocks. Chinese market.



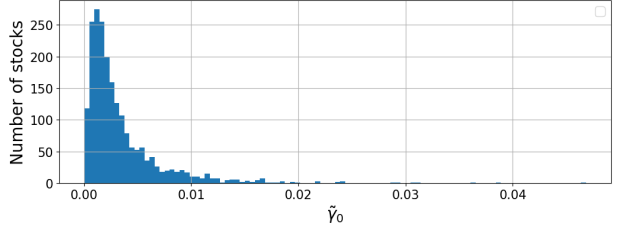
(c) Histogram of $\tilde{\lambda}_1$ across stocks. US market.



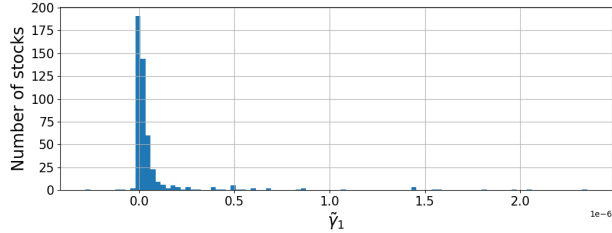
(d) Histogram of $\tilde{\lambda}_1$ across stocks. Chinese market.



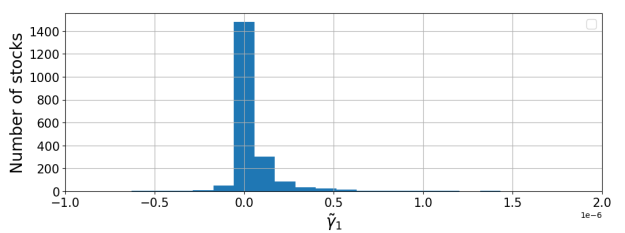
(e) Histogram of $\tilde{\gamma}_0$ across stocks. US market.



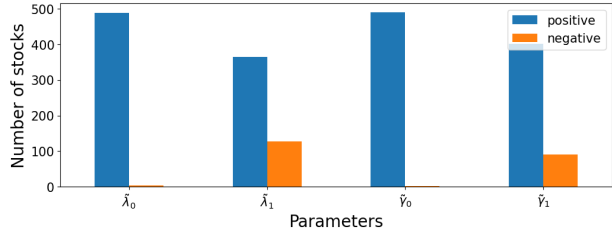
(f) Histogram of $\tilde{\gamma}_0$ across stocks. Chinese market.



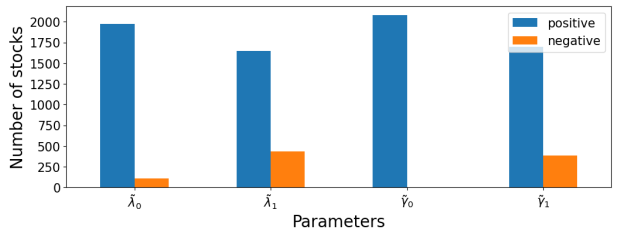
(g) Histogram of $\tilde{\gamma}_1$ across stocks. US market.



(h) Histogram of $\tilde{\gamma}_1$ across stocks. Chinese market.



(i) Histogram of parameter signs. US market.



(j) Histogram of parameter signs. Chinese market.

Figure 9: Summary of price impact regressions for stocks in the US market (five plots on the left-hand side) and the Chinese market (five plots on the right-hand side). In (a)–(h), parameter values are normalized by stock prices.

5.3 Alternative Channels

5.3.1 Retail Trading Control

In this section, we show that the periodic trading behaviors we document are robust after controlling for several retail trading proxies, including the removal of data of retail trades, the removal of odd-lot trades, comparing Robinhood outage days with normal days, and the time-series pattern during the Covid outbreak in 2020.

First, the strength of periodicity in volume remains robust after excluding retail trades in the US market identified by the BJZZ algorithm of Boehmer et al. (2021). Figure 10a shows the boxplots of fVRs across different stocks in the US market when retail trades are excluded. The results are almost identical to those in Figure 4, demonstrating that the periodicity in trading volume cannot be driven by retail trades at least to the extent of those identified by the BJZZ algorithm.

Second, the strength of periodicity in volume remains robust after excluding odd-lot trades in the US market with strictly less than 100 shares.²⁵ As noted by O’Hara, Yao, and Ye (2014), a decent fraction of odd-lot trades may also come from retail investors. Figure 10b shows the boxplots of fVRs in the US market when odd-lot trades are excluded. The results are again almost identical to those in Figure 4.

Third, the strength of periodicity in volume remains robust after using Jiang et al.’s (2022) proxy to exclude small trades in the Chinese market with strictly less than 10000 RMB.²⁶ Figure 10c shows the boxplots of fVRs in the Chinese market when small trades are excluded. The results are again almost identical to those in Figure 4.

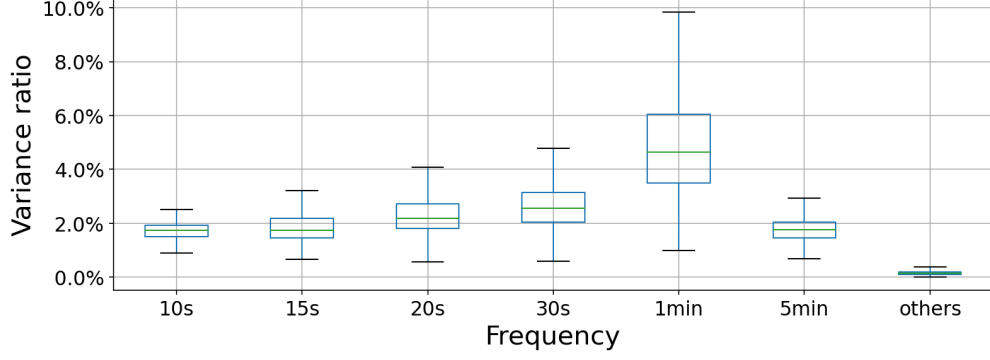
Fourth, the strength of periodicity in volume remains robust on, before, and after Robinhood outage days. In particular, we identified 43 platform outages at Robinhood during 2019–2021 from its official website,²⁷ and calculate the strength of periodicity for the outage days as well as for the one-week period immediately preceding and following these days, summarized in Figure 11. The strength of periodicity on outage days, when fewer retail investors are able to trade, is similar to, and even slightly higher than, the other days around them.

Finally, we document a drop in the strength of periodicity in the US market in April–June 2020

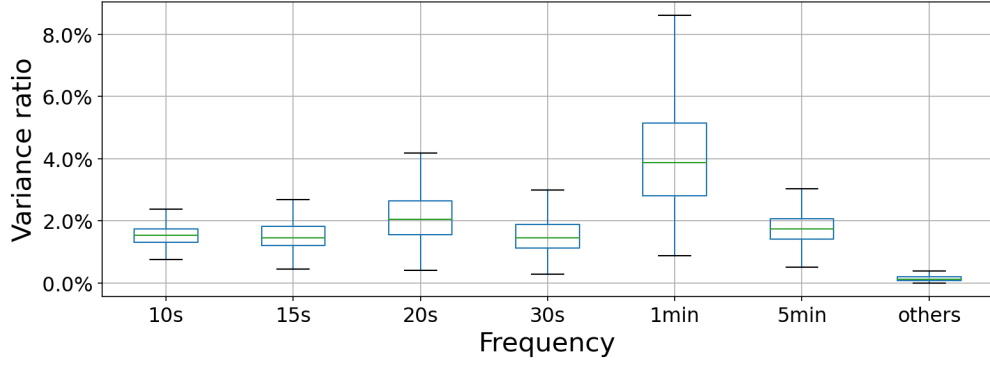
²⁵We identify these trades by “I” of the order condition variable in the TAQ dataset.

²⁶Jiang et al. (2022) adopt this proxy because retail investors account for a significant portion of small trades.

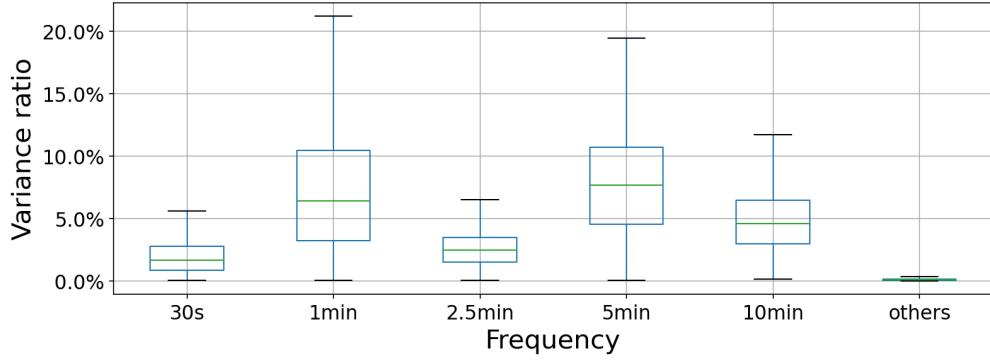
²⁷See <https://status.robinhood.com/history>, accessed May 11, 2024.



(a) Excluding retail trades identified by the BJZZ algorithm, the US market.



(b) Excluding odd-lot trades, the US market.



(c) Excluding small trades, the Chinese market.

Figure 10: Box plot of frequency variance ratios (fVR) for different measures when certain types of trades are excluded. The benchmark fVR is $\frac{1}{500} = 0.2\%$ if no periodicity exists because we included $n = 500$ periodic components in the model.

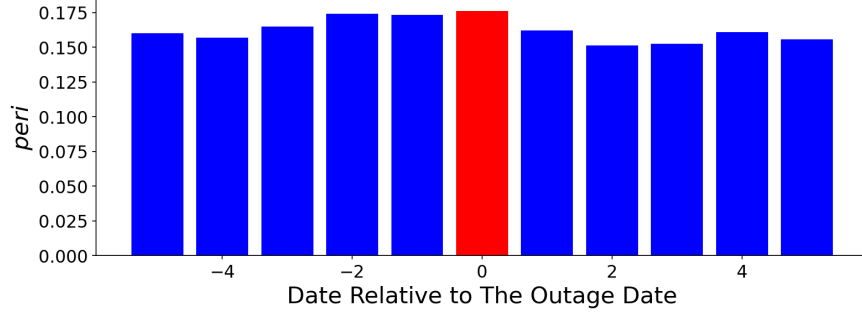


Figure 11: Bar plot of *peri* values for the dates on which outages occurred, as well as for the one-week period preceding and following these dates. For instance, the bar 0 (red bar) is the *peri* value for the averaged intraday trading volume across those outage dates, the bar 1 is the *peri* value for the averaged intraday trading volume across the dates following all outage dates.

during the initial months of the Covid outbreak, the market’s monthly *peri* decreased by more than half, from an average of approximately 0.12-0.16 to around 0.06, when a large number of retail investors poured into the market (Eaton et al., 2022), as shown in Figure 12. This offers another supporting evidence that retail trading is likely not the driving force behind the periodicities.

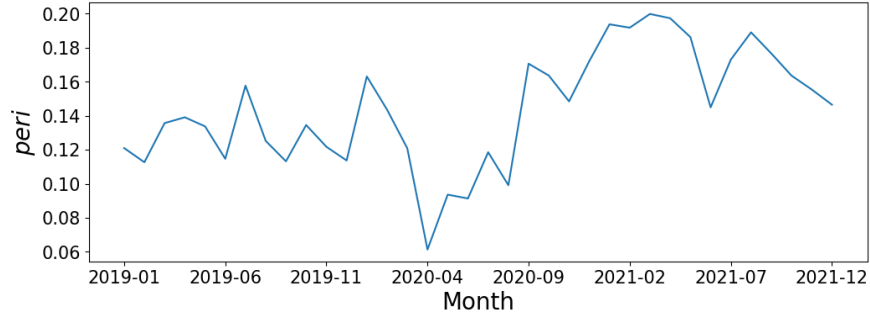


Figure 12: Monthly estimated *peri* values in the US market in 2019–2021.

5.3.2 Are News Articles Published Periodically?

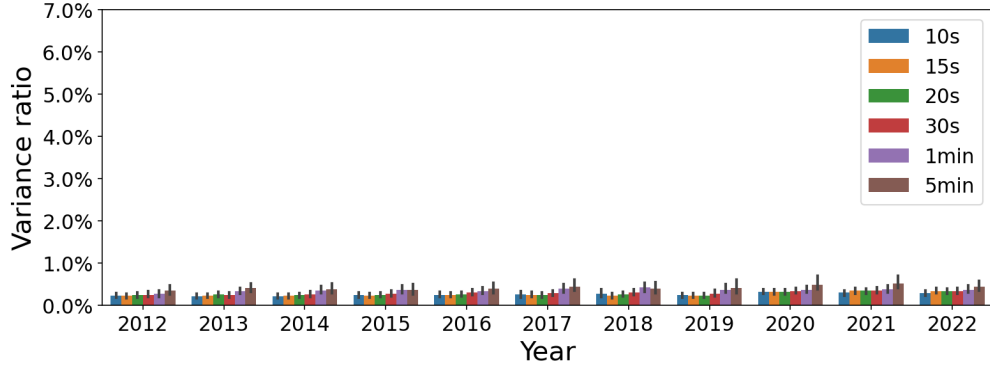
So far we have shown that the strength of periodicity is positively associated with proxies of algorithmic trading, and that volume spikes generate higher price impact and are unlikely to be driven by retail trading alone. In this section, we explore whether periodic trading behavior can be driven by the consumption of periodic news articles.

We obtain data on published news articles from Ravenpack, a popular real-time news database adopted by 70% of the largest hedge funds and asset managers.²⁸ It contains news articles from

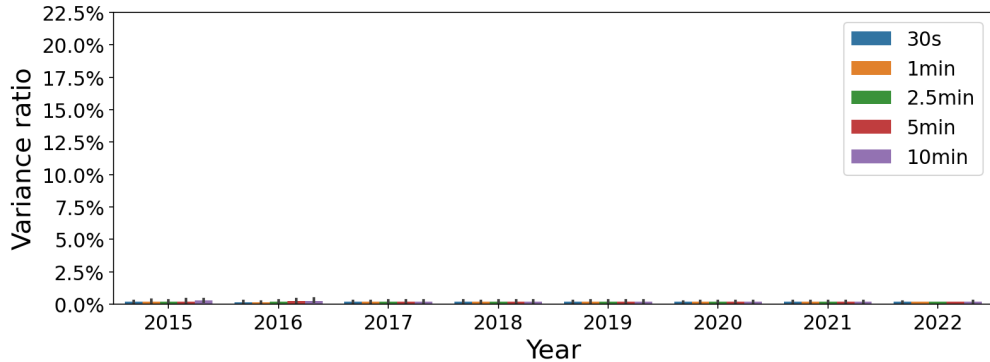
²⁸See <https://www.ravenpack.com/products/edge/data/news-analytics>, accessed Sep 20, 2024.

more than 40,000 global sources, including premium news providers, regulatory and press wires. It provides a millisecond-level timestamp of each article and determines which companies are mentioned in the article. This dataset has been widely used in the literature to study the influence of published news on the financial market (von Beschwitz, Keim, and Massa, 2020; Ben-Rephael et al., 2022; Chen and Li, 2023).

We extract news articles from Ravenpack that mention the companies in our dataset and analyze their periodicities. Details of the matching process are provided in Appendix H. For each stock, we form a time series of the number of published news articles and perform spectral decomposition using the methodology in Section 3. Figure 7 summarizes the distribution of fVRs for published news articles in each year, in which we intentionally align their y-axes with the y-axes in Figure 7 for ease of comparison. In sharp contrast to the consistently increasing periodicity in the trading volume, the published news articles show no periodicities throughout our sample period.



(a) The US Market.



(b) The Chinese market.

Figure 13: Barplot of frequency variance ratios (fVR) for the published news articles for both markets in different years. The benchmark fVR is $\frac{1}{500} = 0.2\%$ if no periodicity exists because we included $n = 500$ periodic components in the model. The y-axes are aligned with those in Figure 7.

This result demonstrates that the strong and persistent periodicities that we document in trading volume are unlikely to be driven by information in published news articles *directly*, at least to the extent captured by the Ravenpack dataset. However, we also caution against generalizing the implications of our findings too broadly.

Our analysis shows that published news articles do not show periodicity, but this does not rule out the possibility of post-processing information in news articles and consuming it in periodic ways. For example, an investor may have computer programs that read news articles in batches of regular time intervals and act on the output of these batches. To study this possibility conclusively would require private information and data within the operations of asset managers, which we do not have access to and is left to future work.

5.4 Summary

Overall, the results in Sections 5.1.1–5.3.2 are consistent with the interpretation that periodic trading activities reflect the behaviors of trading algorithms with repeated and regular trading instructions. Given the popularity of various forms of algorithms in financial markets today, a wide range of investors can contribute to this phenomenon, including both sophisticated hedge funds that employ algorithmic trading strategies such as the contrarian strategy and traditional asset managers that use brokers’ execution algorithms such as VWAP to buy and sell.

We provide three remarks to rationalize these periodic trading behaviors. First, although we do not find periodicity in published news articles as an example of an information source, investors may still receive and process information periodically in practice, including, for example, fixed-window price-volume bars, regular data feeds such as social media posts and credit card transactions, any and other signals that arrive or are generated at regular time intervals.²⁹ In addition, algorithmic traders typically develop return forecasts (or alphas) with a fixed horizon such as 10 seconds or 5 minutes, and execute trades based on forecasts with these fixed horizons. These heuristics in practice imply that algorithms tend to send repeated trading instructions in loops with round start times and fixed time increments. After all, it is hard to imagine anyone using a 3.1415926-minute price-volume bar as input to algorithms or forecasting the return 3.1415926-minute ahead.

²⁹For example, Sağlam (2020) finds evidence of order anticipation strategies by examining predictable patterns in large-order trades.

Second, our findings are consistent with the implications from equilibrium models of regular or clustered trading volumes. For example, investors may trade at regular times as a form of sunshine trading to reduce trading cost (Admati and Pfleiderer, 1991), and may cluster trades in one exchange to get access to better liquidity (Pagano, 1989). Instead of trade concentration in one out of several trading venues, our findings imply trade concentration along the time dimension. In fact, for informed algorithmic traders with large market impact, trading at regular volume clusters when liquidity spikes may help reduce their trading costs.

Finally, Bogousslavsky (2016) shows that infrequent rebalancing can lead to autocorrelations in returns and volumes. There is also substantial evidence that fund flows from certain institutional investors exhibit strong autocorrelation and periodic patterns at the daily, monthly, and quarterly frequencies (see, for example, Guercio and Tkac (2002), Frazzini and Lamont (2008), Campbell, Ramadorai, and Schwartz (2009), Lou (2012), and Kamstra et al. (2017)). Our findings can be regarded as the high-frequency version of these behaviors.

6 Applications

6.1 Volume Prediction

We have documented universal periodic trading activities. In this section, we demonstrate that incorporating the periodic terms significantly improves intraday volume predictions and leads to meaningful economic gains. Specifically, the improvement in prediction R^2 for the median stock in most months ranges from 1-5% in the US and 1-2% in the Chinese market. The dollar cost saving in VWAP execution is positive for most stocks, and the magnitude is 2-5% in the US and 3-7% in the Chinese market compared to the benchmark for the median stock.³⁰

6.1.1 Model Formulation

Because VWAP execution strategies are determined by relative trading volume throughout the day, we focus on predicting the normalized intraday volume the next day using the normalized intraday volume of the last twenty days, denoted by $v_{t,s,d} = V_{t,s,d} / \left(\sum_{i=1}^T V_{i,s,d} \right)$ following notations from

³⁰In a recent study, Goyenko et al. (2024) model the economic benefits of predicting volume through a portfolio framework and find that the benefits of predicting volume can be substantial.

(5). We build three different models to predict future intraday volume: the baseline model, the Periodic Moving Average model, and the Periodic Neural Network model.

The baseline model. The most straightforward model for intraday volume prediction is to simply use the U-shaped (or double U-shaped) trend. We therefore use the moving average of normalized intraday volume over the past twenty days as our baseline. Specifically, we calculate:

$$v_{t,s,d}^{\text{MA}} := \frac{1}{20} \sum_{i=1}^{20} \frac{1}{2q+1} \sum_{j=-q}^q v_{t+j,s,d+1-i}, \quad (16)$$

where $q = 100$. We then fit a linear model to scale the moving average properly for next day: $\hat{v}_{t,s,d+1}^{\text{MA}} = \beta \cdot v_{t,s,d}^{\text{MA}}$, where $\beta = \arg \min_{\beta} \sum_{t,s,d} \left(\beta \cdot v_{t,s,d}^{\text{MA}} - v_{t,s,d+1} \right)^2$ is the coefficient estimated from the data of all stocks over the training data and $\hat{v}_{t,s,d+1}^{\text{MA}}$ is the prediction of the baseline model.

The Periodic Moving Average model. As shown in Section 4.2, the main periodic frequencies are 10-second, 15-second, 20-second, 30-second, 1-minute, and 5-minute in the US market, and 30-second, 1-minute, 2.5-minute, 5-minute, and 10-minute in the Chinese market. Therefore, we add these periodic terms to the baseline to build the Periodic Moving Average model (Periodic MA):

$$\hat{v}_{t,s,d+1}^{\text{Periodic MA}} = \beta_{\text{MA}} \cdot v_{t,s,d}^{\text{MA}} + \sum_{i=1}^I \beta_i \cdot \mathbf{cos}_t^{(i)} \cdot v_{t,s,d}^{\text{MA}}, \quad (17)$$

where $\mathbf{cos}_t^{(i)} \in \mathbf{cos}_{\text{USA}}$ for stocks in the US market and $\mathbf{cos}_t^{(i)} \in \mathbf{cos}_{\text{CHN}}$ for stocks in the Chinese market. Here

$$\begin{aligned} \mathbf{cos}_{\text{USA}} &:= \left\{ \cos \frac{3\pi t}{5}, \cos \frac{2\pi t}{5}, \cos \frac{3\pi t}{10}, \cos \frac{\pi t}{5}, \cos \frac{\pi t}{10}, \cos \frac{\pi t}{50} \right\} \\ \mathbf{cos}_{\text{CHN}} &:= \left\{ \cos \frac{\pi t}{5}, \cos \frac{\pi t}{10}, \cos \frac{\pi t}{25}, \cos \frac{\pi t}{50}, \cos \frac{\pi t}{100} \right\} \end{aligned}$$

are two sets of periodic basis for stocks in the US and the Chinese markets, respectively, and $\beta = (\beta_{\text{MA}}, \beta_1, \dots, \beta_I)^\top$ are coefficients estimated from training data:

$$\beta = \arg \min_{\beta} \sum_{t,s,d} \left(\beta_{\text{MA}} \cdot v_{t,s,d+1}^{\text{MA}} + \sum_{i=1}^I \beta_i \cdot \mathbf{cos}_t^{(i)} \cdot v_{t,s,d+1}^{\text{MA}} - v_{t,s,d+1} \right)^2. \quad (18)$$

The Periodic Neural Network The Periodic Neural Network (Periodic NN) is similar to the Periodic MA model, except that the linear model is replaced by a non-linear neural network:

$$\hat{v}_{t,s,d+1}^{\text{Periodic NN}} = f(v_{t,s,d}^{\text{MA}}; \mathbf{cos}_t^{(1)}, \dots, \mathbf{cos}_t^{(I)}; v_{t,s,d-19}, v_{t,s,d-18}, \dots, v_{t,s,d}), \quad (19)$$

where f is a fully-connected neural network fitted on the training data, and $\hat{v}_{t,s,d+1}^{\text{Periodic NN}}$ is the prediction of the Periodic NN.

Overall, the baseline model is simple and popular for predicting intraday trading volume. The Periodic MA model explicitly incorporates the periodic information in a linear way, and the Periodic NN model exploits this periodic information using machine learning.

6.1.2 Model Performance

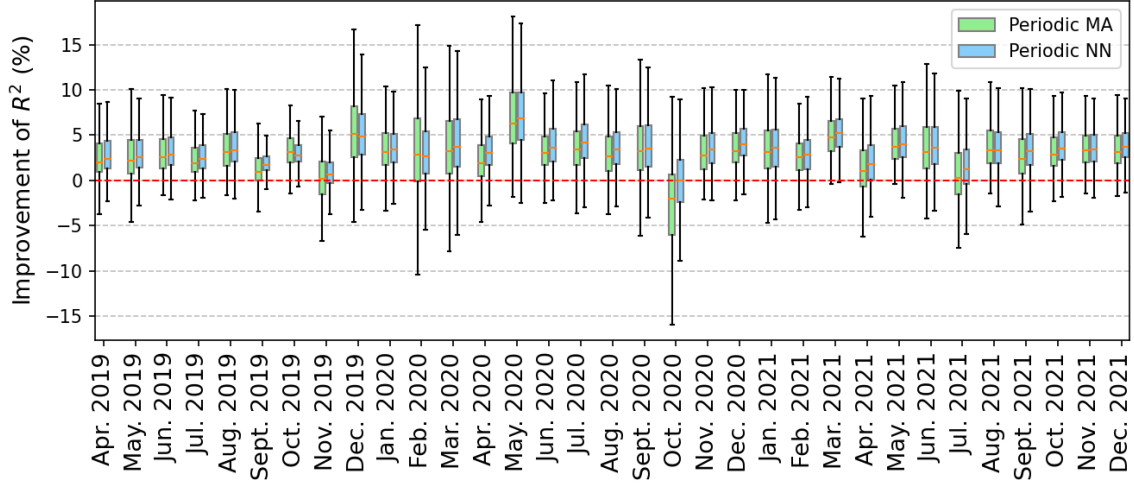
To compare the performance of the three models, we estimate parameters using a rolling window of three months and predict the daily intraday volume for the next month, for 492×33 stock months in the US and 2081×33 stock-months in China.

Figure 14 shows the improvement in out-of-sample R^2 from the Periodic MA and Periodic NN compared with the baseline model: $R_{\text{Periodic MA}}^2 - R_{\text{baseline}}^2$ and $R_{\text{Periodic NN}}^2 - R_{\text{baseline}}^2$. For each month, we show the distribution of improvements in R^2 across all stocks. Both models outperform the baseline in both markets, with average monthly improvements ranging from 1–5% for the median stock. The main improvement comes from incorporating the periodic terms into the model, and using the neural network model offers additional gains that are modest but consistent.

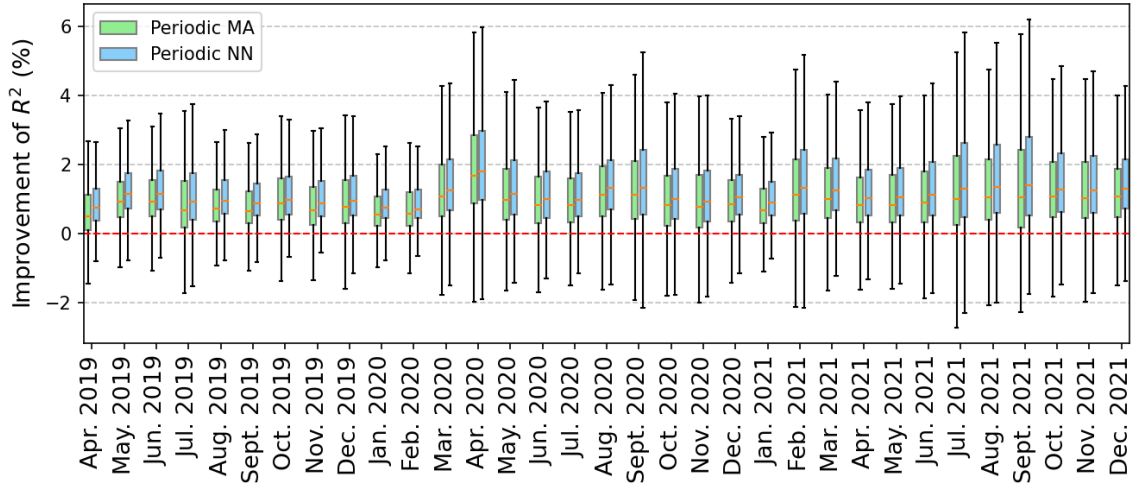
Furthermore, we quantify the economic value of these improvements in the context of VWAP execution. We define the relative VWAP loss of a specific model by $\text{Loss}_{\text{model}} = \frac{|\text{VWAP}_{\text{model}} - \text{VWAP}|}{\text{VWAP}}$ where VWAP is the true volume-weighted average price of a stock on one day, and $\text{VWAP}_{\text{model}}$ is the volume-weighted average price computed using the model’s predicted volume. Therefore, the relative improvement of a model with periodic information with respect to the baseline is:

$$\frac{\text{Loss}_{\text{baseline}} - \text{Loss}_{\text{Periodic MA}}}{\overline{\text{Loss}_{\text{baseline}}}} \quad \text{and} \quad \frac{\text{Loss}_{\text{baseline}} - \text{Loss}_{\text{Periodic NN}}}{\overline{\text{Loss}_{\text{baseline}}}},$$

where $\overline{\text{Loss}_{\text{baseline}}}$ is the average loss of the baseline model of all samples within each month.



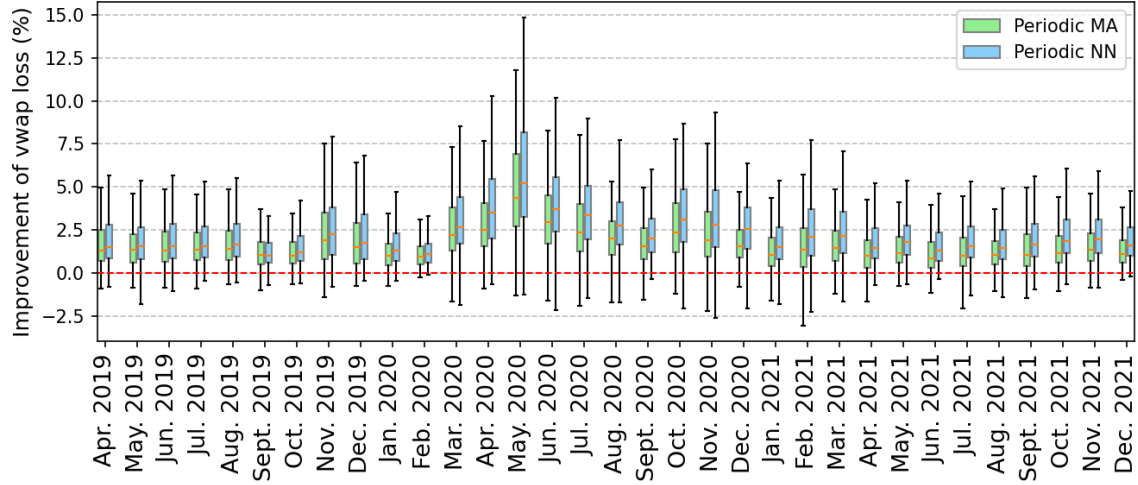
(a) The US market



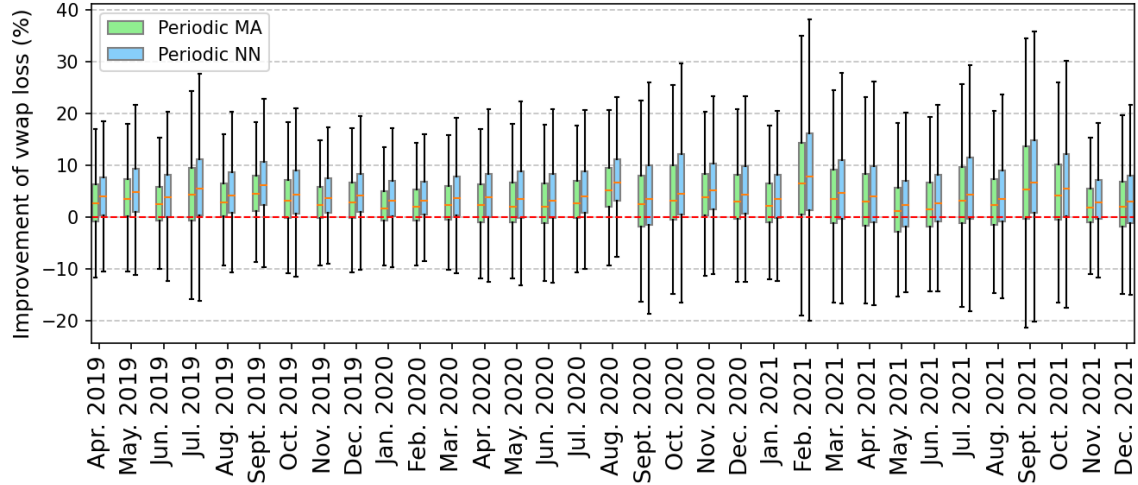
(b) The Chinese market

Figure 14: Boxplot of improvements in out-of-sample R^2 from the Periodic MA and Periodic NN compared to the baseline, $R^2_{\text{Periodic MA}} - R^2_{\text{baseline}}$ and $R^2_{\text{Periodic NN}} - R^2_{\text{baseline}}$, in each month. Each box represents the distribution of average improvements in R^2 across all stocks in a particular month. The median and mean of improvements in the two advanced models compared with the baseline in most months across the two markets are higher than 0.

Figure 15 shows the improvements in out-of-sample relative VWAP loss. In both markets, the two models with periodic terms decrease the relative VWAP loss by 2-7% compared to the baseline model. This can be directly translated into dollar cost savings for investors using VWAP execution algorithms, yielding a significant economic gain in improving execution quality.



(a) The US market



(b) The Chinese market

Figure 15: Boxplot of relative improvements in out-of-sample VWAP loss from the Periodic MA and Periodic NN compared to the baseline in each month. Each box represents the distribution of average relative improvements in VWAP loss across all stocks in a particular month. The median and mean of improvements in the two models compared with the baseline in most months are higher than 0.

In conclusion, our results provide compelling evidence for the potential of exploiting periodic patterns to improve intraday volume prediction and execution quality. Furthermore, applying

nonlinear machine learning models to incorporate these periodic patterns can provide additional improvements that are consistent over time.

6.2 Price Informativeness

Motivated by the recent literature that documents the tension between acquiring information (price informativeness) and incorporating it into asset prices (price efficiency), here we study the implications of periodic trading behaviors on the price informativeness of the underlying stock.

In particular, Weller (2018) shows that algorithmic trading is associated with reduced price informativeness despite improved price efficiency. Gider, Schmickler, and Westheide (2019) find consistent results by focusing on a longer-term measure of price informativeness based on Bai, Philippon, and Savov (2016) using international data from 18 stock markets. Dou, Goldstein, and Ji (2024) find a similar effect in a model of algorithmic collusion. Our analysis in Section 5.1.2 has documented that stocks with stronger periodicities in trading volume are associated with prices that more closely resemble a random walk, i.e., higher price efficiency.³¹ In this section, we show that stocks with stronger periodicities in trading volume are associated with lower price informativeness, thereby confirming Weller’s (2018) and Gider, Schmickler, and Westheide’s (2019) results from a different angle—intraday periodicities as a proxy of algorithmic trading.

6.2.1 Intraperiod Timeliness

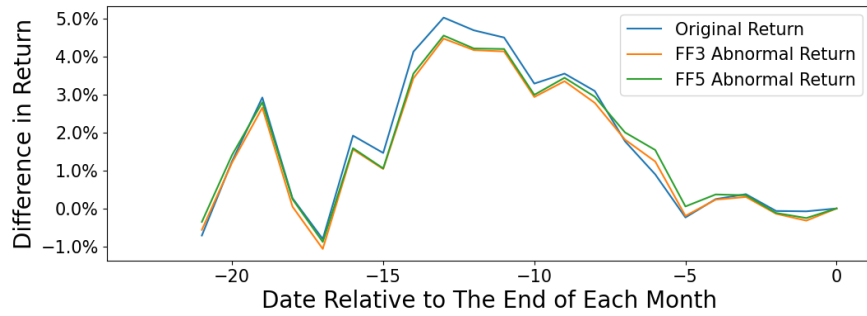
The intraperiod timeliness (IPT) measure compares the performance of several zero-cost “perfect-foresight” portfolios with respect to certain events for stocks with different characteristics. Higher perfect-foresight returns earlier in the pre-event period indicate more pre-event information acquisition, i.e., higher price informativeness in these stocks (Weller, 2018).

In particular, we first separate stocks into four quartiles each month based on their strength of periodicity in volume, *peri*, in that month. Within each quartile, we further separate stocks that increase/decrease in price in that month, and construct “perfect foresight” portfolios by investing $+1/N_{up}$ dollars in all N_{up} stocks that increase in price at the end of each month and $-1/N_{down}$ dollars in all N_{down} stocks that decrease in price at the end of each month. We look at the

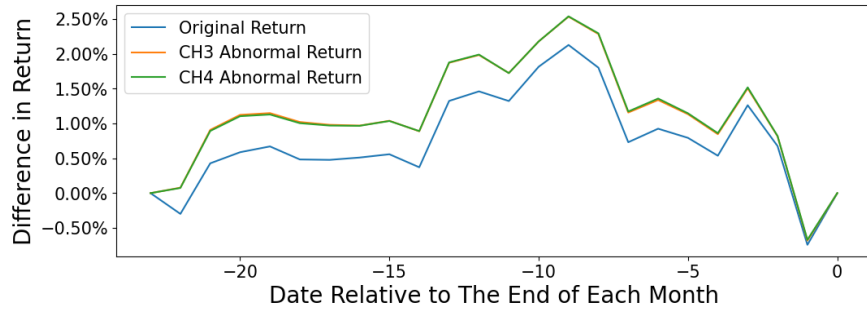
³¹The literature has documented that stronger algorithmic trading is associated with higher price efficiency. See, for example, Hendershott, Jones, and Menkveld (2011), Conrad, Wahal, and Xiang (2015), Clark-Joseph, Ye, and Zi (2017), and Boehmer, Fong, and Wu (2021).

cumulative returns of these “perfect foresight” portfolios as a function of time relative to the total returns at the end of the month, which measures how quickly the information at the end of the month is incorporated into prices.

We find consistent evidence that stocks with weaker periodicities in volume, a proxy for less algorithmic trading, are associated with higher price informativeness. Figure 16 shows the difference between the lowest *peri* and the highest *peri* quartiles in terms of their cumulative returns of the “perfect foresight” portfolios relative to total returns for each month, measured with respect to different asset pricing models. The lines are consistently above 0, which represents that stocks in the lowest *peri* quartile incorporate information into prices earlier in the month compared to stocks in the highest *peri* quartile. These results are consistent with the findings in Weller (2018) and Gider, Schmickler, and Westheide (2019).



(a) The US Market.



(b) The Chinese market.

Figure 16: The difference between the relative cumulative returns as of date k preceding the end of each month, on the lowest *peri* portfolio net of the highest *peri* portfolio. The returns are measured in original returns and the abnormal returns of different factor models.

6.2.2 Price Nonsynchronicity

Roll (1988) first proposed an R^2 -based measure of firm-specific information in prices, which is

further developed by Morck, Yeung, and Yu (2000), Durnev et al. (2003), and Weller (2018) into a price nonsynchronicity measure. In particular, they estimate price nonsynchronicity as $1 - R^2$ from the following regression:

$$r_{it} = \alpha + \beta_{im}r_{mt} + \gamma r_{ind,t} + \varepsilon_{it}, \quad (20)$$

where r_{it} is the return of the i -th stock on date t , r_{mt} is the return of market on date t , and $r_{ind,t}$ is the return of the industry to which the i -th stock belongs.³² Intuitively, lower price nonsynchronicity represents lower price informativeness because the stock's return is better approximated by aggregate information and less idiosyncratic information is produced (Weller, 2018).

Following Weller (2018), we first estimate the price nonsynchronicity measure $nonsynch_{im}^{event}$ as the $1 - R^2$ of regression (20) over a month, for each stock-month pair, (i, m) . We also normalize this measure by subtracting the average price nonsynchronicity over the last year, $nonsynch_{im}^{ave}$, estimated by the $1 - R^2$ of the same regression (20) over the $[T - 252, T - 1]$ period, where T represents the last day of month m .

We then run the following instrument variable (IV) panel regressions across all stocks and all months in our data:

$$\begin{aligned} peri_{im} &= \zeta + \eta lprice_{im} + \theta \times controls_{im} + \delta_{im}, \\ nonsynch_{im}^{event} - nonsynch_{im}^{ave} &= \alpha + \beta \widehat{peri}_{im} + \gamma \times controls_{im} + \varepsilon_{im}, \end{aligned} \quad (21)$$

for $i = 1, \dots, N$ and $m = \text{Jan 2019}, \dots, \text{Dec 2021}$. Here $lprice$ is the lagged log-price, and $controls$ include logged log-market capitalization, volatility, and lagged spread. Table 6 reports the correlations between $peri$ and the lagged log stock price instrument. The correlations are strengthened by orthogonalizing the lagged log price with respect to the control variables.

Table 7 provides the regression results for (21) under different control variables and different fixed effects in both the US and Chinese markets. The estimated coefficients of $peri$ in the Chinese market are consistently and significantly negative, while the results in the US market are less significant but are still consistent in general. Overall, these results suggest that stocks with weaker

³²The industry return is calculated by the value-weighted average return for stocks sharing the same three-digit SIC code in the US market or the same CSRC Industry Classification in the Chinese market.

Table 6: Correlations between the strength of periodicity *peri* and the lagged log-price *lprice*. The latter is orthogonalized with respect to market cap and other control variables. Correlations are significant at the 1 percent (***), 5 percent (**), or 10 percent (*) levels.

	The US Market	The Chinese Market
Full sample	−0.086***	0.339***
Net of market cap.	−0.243***	0.374***
Net of all controls	−0.208***	0.353***

periodicity, a proxy for less algorithmic trading, are associated with more firm-specific news in prices and higher price informativeness. They are again consistent with the findings in Weller (2018), Gider, Schmickler, and Westheide (2019), and Dou, Goldstein, and Ji (2024) that algorithmic trading reduces price informativeness despite its importance for translating available information into prices.

Table 7: Regression results for the price nonsynchronicity, *nonsynch*, on stock-level characteristics, with different model formulations and different fixed effects (FE). Estimated coefficients are significant at the 1 percent (***), 5 percent (**), or 10 percent (*) levels.

		(1)	(2)	(3)	(4)
The US market	<i>peri</i>	−0.2688	−1.4376*	0.0097	−0.7335
	Market cap.	0.0060	0.0142**	0.0043	0.0076
	Ret. vol.			−0.0024	−0.0296*
	Quoted spr.			0.0002	0.0002**
	Constant	−0.0899	−0.1958**	−0.0802	−0.2209**
	Stock FEs	No	Yes	No	Yes
The Chinese market	<i>peri</i>	−1.0533***	−0.9608***	−0.9270***	−1.0376***
	Market cap.	0.0186***	0.0182***	0.0181***	0.0185***
	Ret. vol.			−0.0052	0.0203*
	Quoted spr.			−0.0697	−0.1060*
	Constant	−0.4244***	−0.4180***	−0.4362***	−0.3455***
	Stock FEs	No	Yes	No	Yes

6.3 Excess Returns

We have shown in Section 5 that periodic trading activities are associated with higher price impact and information content from trades at round times compared to non-round times. Therefore, it is reasonable to hypothesize that stocks with stronger periodicities in trades also contain more information content in trades. This potentially creates more information asymmetry and adverse

selection when trading these stocks, and therefore investors must require a higher rate of return to compensate for the increased risk. In this section, we test whether there exist excess returns for stocks with strong periodicities in trades.

To confirm our hypothesis, we first evaluate the strength of periodicity for each stock s in each month m by estimating an aggregate measure $peri_{s,m}$ using Equation (9), which essentially summarizes the fraction of variance in de-trended trading volumes explained by the strongest periodic terms. To construct portfolios based on this measure, in each month, we sort stocks into five quintiles according to their $peri_{s,m}$ values. We then construct a long-short portfolio that buys the top quintile of stocks with the highest $peri_{s,m}$ values and sells the bottom quintile of stocks with the lowest $peri_{s,m}$ values. We refer to this portfolio as the *Periodic minus Smooth* (PmS) portfolio.

To assess the performance of these portfolios, we estimate the excess returns (alpha) with respect to several asset-pricing models using regressions based on *daily* portfolio and factor returns. In particular, we control for Fama-French three and five factors (Fama and French, 1993, 2015), momentum (Carhart, 1997), and liquidity risk factor.³³ We test both equal-weighted and value-weighted portfolios.

$$R_{\text{PmS},t} - R_{f,t} = \alpha + \beta_1(R_{M,t} - R_{f,t}) + \beta_2 SmB_t + \beta_3 HmL_t + \beta_4 RmW_t + \beta_5 CmA_t \\ + \beta_6 MoM_t + \beta_7 ILLQ_t + \epsilon_t,$$

where $R_{\text{PmS},t}$ is the return of the equal-weighted or value-weighted PmS portfolio on day t in month m , $R_{f,t}$ the risk-free rate on day t , and $R_{M,t}$, SmB_t , HmL_t , RmW_t , CmA_t , MoM_t , and $ILLQ_t$ the return of pricing factors on day t .

Table 8 summarizes *monthly* alphas of the PmS portfolios. In the US market, the alphas are around 0.4% to 0.5% equal-weighted and 0.8% to 0.9% value-weighted, which are quite significant

³³We use the liquidity risk factor ImL of Amihud et al. (2015) in the US and PmO of Liu, Stambaugh, and Yuan (2019) in China. In the Chinese market, we use the CH3 factors of Liu, Stambaugh, and Yuan (2019) which are designed specifically for the Chinese stock market to replace the Fama-French three factors. Liu, Stambaugh, and Yuan (2019) report no momentum effect in China so it is not included in the Chinese market. In terms of data sources, the Fama-French factor and momentum factor data is obtained from https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html for the US stock market and <http://sf.cufe.edu.cn/kydt/kyjg/zgzcglyjzx/xzzq.htm> for the Chinese stock market. The Chinese factor data of Liu, Stambaugh, and Yuan (2019) is obtained from <https://finance.wharton.upenn.edu/~stambaug/>. The Amihud's ImL liquidity risk factor data is calculated from the CRSP dataset.

economically. In the Chinese market, the PmS portfolios consistently generate statistically and economically significant alphas of 4% to 5% per month. Given that shorting individual stocks is difficult in China, the alphas of the top quintile may be more appropriate, which are still significantly positive and around 2% to 3% per month. This is consistent for both equal-weighted and value-weighted portfolios, and with respect to all factor models.

The difference between the two markets may reflect the fact that our dataset contains S&P 500 stocks in the US market but all 2,081 stocks in the SZSE of the Chinese stock market. Although the magnitudes are different, they are both economically significant and robust across different configurations of the asset-pricing model.

Table 8: Monthly alphas of portfolios sorted by the strength of periodicity, *peri*, controlling for different asset-pricing factors based on regressions of daily returns. The last column presents the Periodic minus Smooth (PmS) portfolio that buys/sells stocks in the top/bottom quintile. In the US market (Panel A and B), we control for the Fama-French three (FF3) and five (FF5) factors, Carhart’s (1997) momentum factor, and the liquidity risk factor (ImL) of Amihud et al. (2015). In the Chinese market (Panel C and D), we control for market, size, and value factors (CH3) of Liu, Stambaugh, and Yuan (2019), the liquidity risk factor (PmO) of Liu, Stambaugh, and Yuan (2019), and the Fama-French profitability (RmW) and investment (CmA) factors. Estimated alphas are significant at the 1 percent (***), 5 percent (**), or 10 percent (*) levels.

Asset-pricing model	Top	Second	Middle	Fourth	Bottom	PmS
Panel A: The US Market; Equal Weighted Portfolio						
FF3	0.60%***	0.16%	0.08%	0.15%	0.21%	0.39%
FF5	0.61%***	0.13%	0.01%	0.05%	0.13%	0.48%
FF5 + momentum	0.58%***	0.10%	-0.06%	-0.03%	0.05%	0.53%
FF5 + momentum + liquidity	0.58%***	0.11%	-0.11%	-0.02%	0.05%	0.53%
Panel B: The US Market; Value Weighted Portfolio						
FF3	0.74%**	-0.24%	-0.39%	-0.02%	-0.15%	0.90%*
FF5	0.64%***	-0.27%	-0.47%*	-0.12%	-0.20%	0.84%*
FF5 + momentum	0.64%***	-0.25%	-0.49%**	-0.14%	-0.20%	0.84%*
FF5 + momentum + liquidity	0.66%***	-0.30%	-0.50%**	-0.16%	-0.16%	0.83%*
Panel C: The Chinese Market; Equal Weighted Portfolio						
CH3	1.98%***	0.28%	-0.40%*	-1.14%***	-2.17%***	4.23%***
CH3 + liquidity (CH4)	2.05%***	0.27%	-0.43%**	-1.20%***	-2.22%***	4.36%***
CH4 + profitability + investment	2.26%***	0.43%**	-0.32%*	-1.13%***	-2.21%***	4.56%***
Panel D: The Chinese Market; Value Weighted Portfolio						
CH3	2.76%***	0.91%***	0.82%**	-0.53%	-2.02%***	4.87%***
CH3 + liquidity (CH4)	2.82%***	0.90%***	0.79%**	-0.56%	-2.09%***	5.01%***
CH4 + profitability + investment	2.70%***	0.76%***	0.65%*	-0.60%*	-2.06%***	4.86%***

In addition, Table 9 reports the estimated beta loadings of the regressions that lead to the alphas of the PmS portfolios in Table 8. The PmS portfolio has a positive loading on the market

and a negative loading on size, which implies that stocks with stronger periodicities in volume tend to be those with larger capitalizations in both markets. In the US market, the beta of investment is positive while the betas of several other factors remain statistically insignificant. In the Chinese market, the PmS portfolio has negative loadings on value and liquidity and positive loadings on profitability and investment. This implies, for example, stocks with stronger periodicities in volume tend to be growth stocks and liquid stocks.

Table 9: Beta loadings of the PmS portfolio with respect to known asset-pricing factors in the regressions that lead to the alphas in Table 8. Estimated betas are significant at the 1 percent (***), 5 percent (**), or 10 percent (*) levels.

(a) The US market

Factor	Equal Weighted Portfolio				Value Weighted Portfolio			
	FF3	FF5	FF5 + momentum	FF5 + momentum + liquidity	FF3	FF5	FF5 + momentum	FF5 + momentum + liquidity
Market	0.06***	0.07***	0.06***	0.07***	0.10***	0.11***	0.11***	0.11***
SmB	-0.04*	-0.08***	-0.07***	-0.07**	-0.09***	-0.07**	-0.07**	-0.08**
HmL	0.00	0.02	0.08***	0.08***	0.06***	0.01	0.00	0.00
RmW		-0.15***	-0.11**	-0.11**		0.02	0.02	0.02
CmA		0.15**	0.10*	0.10*		0.20***	0.20***	0.19***
UmD			0.07***	0.07***			-0.00	-0.00
ImL				0.00				0.03

(b) The Chinese market

Factor	Equal Weighted Portfolio			Value Weighted Portfolio		
	CH3	CH3 + liquidity (CH4)	CH4 + profitability + investment	CH3	CH3 + liquidity (CH4)	CH4 + profitability + investment
Market	0.30***	0.25***	0.28***	0.18***	0.13***	0.13***
SmB	-0.01	0.00	0.01	-0.61***	-0.59***	-0.37***
VmG	-0.43***	-0.36***	-0.49***	-0.74***	-0.65***	-0.64***
PmO		-0.32***	-0.32***		-0.35***	-0.39***
RmW			0.32***			0.24***
CmA			0.35***			-0.00

In Appendix I, we check the robustness of PmS excess returns by constructing the PmS portfolios on a daily or weekly basis and constructing double-sorted portfolios based on several stock-level characteristics. We find that the PmS returns are robust with different rebalancing frequencies and after controlling for known pricing factors and stock characteristics that are correlated with the strength of periodicities. We also provide statistics on the stability of the relative strength of periodicity for different stocks.

7 Conclusion

In this article, we develop a framework to systematically model, estimate, explain, and exploit the periodicity in intraday trading activities using spectral models and Fourier analysis. We represent the intraday time series using a trend term, multiple periodic terms, and an idiosyncratic noise term. We provide a methodology to consistently estimate the intensity coefficients of the periodic terms, by applying the inverse discrete Fourier transform (DFT) to the autocovariance function of the de-trended volume time series.

Our large-scale empirical analysis shows that, in both the US and Chinese markets, important and universal periodicities exist at the second and minute levels across a large sample of stocks. In particular, we find that the variance of the de-trended intraday trading volume is primarily explained by several periodic components at 10-second, 15-second, 20-second, 30-second, 1-minute, and 5-minute frequencies in the US market and 30-second, 1-minute, 2.5-minute, 5-minute, and 10-minute frequencies in the Chinese market.

To understand periodic trading behaviors, we provide evidence that suggests the phenomenon likely reflects the behavior of algorithmic trading with repeated and regular trading instructions. In practice, these regular instructions may reflect the increasingly periodic patterns in which data is delivered and processed through computer programs. This is consistent with Bogousslavsky’s (2016) model of infrequent rebalancing and can also be regarded as the high-frequency version of the well-known fund flow patterns at low frequency. In particular, we show that the strength of periodicity is positively associated with the level of algorithmic trading by exploring variations in the measures, cross-section, and time-series of trading volume. We directly link the strength of periodicity to the level of common algorithmic trading strategies in terms of profitability and price impact from trades. We also disprove two alternative channels—retail trades and published news articles—as drivers of observed periodic trading volume.³⁴

To demonstrate the practical value of incorporating these high-frequency periodicities in real-world applications, we add the periodic terms revealed by our spectral model to a benchmark model that uses the U-shaped intraday trend for volume predictions. Our new model outperforms

³⁴The NASDAQ high-frequency data set (see, for example, O’Hara, Yao, and Ye (2014)) provides categorical information for trades in terms of whether they come from high-frequency/non-high-frequency traders. We do not have access to this dataset, which future researchers can potentially use to verify our findings.

the benchmark in terms of out-of-sample R^2 over most months in our sample, leading to sizable economic gains in volume-weighted average price (VWAP) execution strategies.

Next, we show that stocks with stronger periodicities are associated with lower price informativeness, using the IPT measure and the price nonsynchronicity measure. These results add to the recent literature that documents the tension between acquiring information (price informativeness) and incorporating it into asset prices (price efficiency), and confirm the results of Weller (2018) and Gider, Schmickler, and Westheide (2019) from a different angle of periodic trading behaviors.

Finally, we show that excess returns exist for stocks with strong periodicities in trades after controlling for known asset-pricing factors. The fact that most of the PmS alphas are realized after the market open and that stocks with strong periodicities change frequently from month to month both suggest that such excess returns may reflect a risk factor related to information asymmetry and short-term trading, which is different from traditional risk factors based on firm characteristics.

Overall, our work provides a systematic framework to reveal and understand periodicity in financial time series, which can be applied not only to the trading volume but also more broadly to other quantities that potentially contain noisy, but important, periodic patterns.

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Online Appendix

A Proofs

Proof of Theorem 1. We first show that the time-series average $\bar{X}_\cdot$ converges as $T \rightarrow \infty$. Note that $0 < \lambda_j < \pi$ for all $j = 1, \dots, n$. Therefore,

$$\begin{aligned}\bar{X}_\cdot &= \frac{1}{T} \sum_{t=1}^T \sum_{j=1}^n a_j \cos(\lambda_j t) + \varepsilon_t \\ &= \frac{1}{T} \sum_{t=1}^T \varepsilon_t + \sum_{j=1}^n \frac{a_j}{T} \sum_{t=1}^T \cos(\lambda_j t) \\ &= \frac{1}{T} \sum_{t=1}^T \varepsilon_t + \sum_{j=1}^n \frac{a_j}{T} \sum_{t=1}^T \frac{\sin(\lambda_j(T+1)) + \sin(\lambda_j T) - \sin(\lambda_j)}{2 \sin(\lambda_j)} \\ &\xrightarrow{\text{a.s.}} 0.\end{aligned}$$

We then decompose $\gamma_T^X(h)$ into four terms:

$$\begin{aligned}\gamma_T^X(h) &= \frac{1}{T-h} \sum_{t=1}^{T-h} \left(\sum_{j=1}^n a_j \cos(\lambda_j t) + \varepsilon_t - \bar{X}_\cdot \right) \left(\sum_{k=1}^n a_k \cos(\lambda_k(t+h)) + \varepsilon_{t+h} - \bar{X}_\cdot \right) \\ &= \underbrace{\frac{1}{T-h} \sum_{t=1}^{T-h} \sum_{j=1}^n a_j \cos(\lambda_j t) \sum_{k=1}^n a_k \cos(\lambda_k(t+h))}_{(i)} - \underbrace{\frac{1}{T-h} \sum_{t=1}^{T-h} \bar{X}_\cdot X_t + \bar{X}_\cdot X_{t+h} - \bar{X}_\cdot^2}_{(ii)} \\ &\quad + \underbrace{\frac{1}{T-h} \sum_{t=1}^{T-h} \varepsilon_t \sum_{k=1}^n a_k \cos(\lambda_k(t+h)) + \varepsilon_{t+h} \sum_{j=1}^n a_j \cos(\lambda_j t)}_{(iii)} + \underbrace{\frac{1}{T-h} \sum_{t=1}^{T-h} \varepsilon_t \varepsilon_{t+h}}_{(iv)}.\end{aligned}$$

Because $\bar{X}_\cdot \xrightarrow{\text{a.s.}} 0$ as $T \rightarrow \infty$, it follows that $(ii) \xrightarrow{\text{a.s.}} 0$ as $T \rightarrow \infty$. Because

$$\left| \sum_{k=1}^n a_k \cos(\lambda_k(t+h)) \right| \quad \text{and} \quad \left| \sum_{j=1}^n a_j \cos(\lambda_j t) \right|$$

are both uniformly bounded by $\max_{j=1, \dots, n} |na_j|$, the strong law of large numbers (SLLN) can be applied. We therefore have $(iii) \xrightarrow{\text{a.s.}} 0$ as $T \rightarrow \infty$. SLLN also implies that $(iv) \xrightarrow{\text{a.s.}} \mathbf{1}_{h=0} \sigma^2$ as

$T \rightarrow \infty$. As for (i), we have

$$\begin{aligned}
(i) &= \frac{1}{T-h} \sum_{t=1}^{T-h} \sum_{j=1}^n a_j \cos(\lambda_j t) \sum_{k=1}^n a_k \cos(\lambda_k(t+h)) \\
&= \frac{1}{T-h} \sum_{t=1}^{T-h} \sum_{j=1}^n a_j^2 \cos(\lambda_j t) \cos(\lambda_j(t+h)) + \sum_{j \neq k} a_j a_k \cos(\lambda_j t) \cos(\lambda_k(t+h)) \\
&= \frac{1}{T-h} \sum_{t=1}^{T-h} \left[\sum_{j=1}^n \frac{1}{2} a_j^2 (\cos(\lambda_j h) + \cos(\lambda_j(2t+h))) \right. \\
&\quad \left. + \sum_{j \neq k} \frac{1}{2} a_j a_k (\cos(t(\lambda_k - \lambda_j) + \lambda_k h) + \cos(t(\lambda_k + \lambda_j) + \lambda_k h)) \right] \\
&\rightarrow \sum_{j=1}^n \frac{1}{2} a_j^2 \cos(\lambda_j h),
\end{aligned}$$

as $T \rightarrow \infty$. Combining the convergence results for (i)–(iv) completes the proof.

Proof of Corollary 1. By definition,

$$\begin{aligned}
\gamma_T^V(0) - \gamma_T^m(0) &= \frac{1}{T} \sum_{t=1}^T (m_t + X_t - \bar{m} - \bar{X})^2 - (m_t - \bar{m})^2 \\
&= \gamma_T^X(0) + \frac{2}{T} \sum_{t=1}^T (m_t - \bar{m}) (X_t - \bar{X}) \\
&= \underbrace{\gamma_T^X(0)}_{(v)} + \underbrace{\sum_{j=1}^n \frac{2a_j}{T} \sum_{t=1}^T (m_t - \bar{m}) \cos(\lambda_j t)}_{(vi)} + \underbrace{\frac{2}{T} \sum_{t=1}^T (m_t - \bar{m}) \varepsilon_t}_{(vii)}.
\end{aligned}$$

By taking $h = 0$ in Theorem 1, we have $(v) \xrightarrow{\text{a.s.}} \sigma^2 + \frac{1}{2} \sum_{j=1}^n a_j^2$ as $T \rightarrow \infty$. Assumptions 1–3 imply that $(vi) \rightarrow 0$ and $(vii) \xrightarrow{\text{a.s.}} 0$ as $T \rightarrow \infty$, which completes the proof.

B Summary Statistics of Volumes

Table A.1 provides summary statistics for our three-second intraday trading volume time series.

Table A.1: Summary statistics for the number of trades in three-second windows. We provide both the market-wide average and statistics for selected stocks with the highest/lowest trading volumes.

	mean	standard deviation	minimum	25% quantile	median	75% quantile	maximum
<i>Average across all stocks in the market</i>							
The US market	4.38	4.00	0	2.38	3.36	5.04	244.95
The Chinese market	2.72	2.24	0.32	1.57	2.20	3.16	187.13
<i>Five stocks with the highest average number of trades in the US market</i>							
AAPL	60.44	109.55	0	17	36	72	23113
AMD	36.35	54.87	0	9	21	45	7899
MSFT	34.58	55.71	0	11	22	41	13488
NVDA	24.30	48.59	0	5	12	27	8881
BAC	21.99	35.80	0	4	9	26	3683
<i>Five stocks with the lowest average number of trades in the US market</i>							
HII	0.84	3.26	0	0	0	0	550
NWS	0.79	3.70	0	0	0	0	852
TFX	0.78	3.28	0	0	0	0	206
MTD	0.67	3.03	0	0	0	0	279
NVR	0.41	1.86	0	0	0	0	170
<i>Five stocks with the highest average number of trades in the Chinese market</i>							
300059.SZSE	34.43	56.51	0	9	20	40	11198
000725.SZSE	29.35	82.00	0	3	11	29	13036
000063.SZSE	22.77	44.78	0	6	12	25	9534
002475.SZSE	22.70	37.76	0	7	14	26	9019
000100.SZSE	22.55	55.38	0	4	9	23	11131
<i>Five stocks with the lowest average number of trades in the Chinese market</i>							
002336.SZSE	0.20	1.23	0	0	0	0	166
000972.SZSE	0.19	1.20	0	0	0	0	233
002200.SZSE	0.17	0.93	0	0	0	0	271
002188.SZSE	0.16	0.92	0	0	0	0	171
002499.SZSE	0.15	0.92	0	0	0	0	387

C Validation on Synthetic Data

We first use synthetic data to validate our methodology and demonstrate how it reveals the periodicity of a time series. We consider three different time series generated by the following methods.

1. Non-periodic time series:

$$V_t^{\text{nonperiod}} = g(t) + 20\varepsilon_t, \quad t \leq 7800,$$

where $g(t) = \frac{1}{152100}(t - 3900)^2 + 100$ is a U-shaped trend term, and ε_t is a standard white noise.

2. Periodic time series with trigonometric basis:

$$\begin{aligned} V_t^{\cos} = & g(t) + 20\varepsilon_t + 3 \cos\left(\frac{t\pi}{10}\right) + 2 \cos\left(\frac{2t\pi}{10}\right) + 3 \cos\left(\frac{3t\pi}{10}\right) \\ & + 10 \cos\left(\frac{4t\pi}{10}\right) + 3 \cos\left(\frac{5t\pi}{10}\right) + 2 \cos\left(\frac{6t\pi}{10}\right) + 3 \cos\left(\frac{7t\pi}{10}\right) \\ & + 3 \cos\left(\frac{8t\pi}{10}\right) + 2 \cos\left(\frac{9t\pi}{10}\right) + 3 \cos\left(\frac{10t\pi}{10}\right), \quad t \leq 7800. \end{aligned}$$

The trigonometric basis aligns with our model specification in (1).

3. Periodic time series with non-trigonometric basis:

$$\begin{aligned} V_t^{\text{spike}} = & g(t) + 20\varepsilon_t + 3f\left(\frac{t\pi}{10}\right) + 2f\left(\frac{2t\pi}{10}\right) + 3f\left(\frac{3t\pi}{10}\right) \\ & + 10f\left(\frac{4t\pi}{10}\right) + 3f\left(\frac{5t\pi}{10}\right) + 2f\left(\frac{6t\pi}{10}\right) + 3f\left(\frac{7t\pi}{10}\right) \\ & + 3f\left(\frac{8t\pi}{10}\right) + 2f\left(\frac{9t\pi}{10}\right) + 3f\left(\frac{10t\pi}{10}\right), \quad t \leq 7800. \end{aligned}$$

The non-trigonometric basis is intentionally chosen to be different from our model specification in (1).

The first example only uses the U-shaped trend term and the white noise, therefore contains no periodic terms and serves as a control example. The second example uses the popular trigonometric basis to generate periodicity. The third example replaces the trigonometric basis by:

$$f(t) = \begin{cases} 3 \left(\frac{e^{-0.1} - e^{-2}}{0.1} \frac{t}{\pi} + e^{-2} \right) - 1, & t \leq 0.1\pi, \\ 3e^{-\frac{t}{\pi}} - 1, & 0.1\pi < t \leq 2\pi, \\ f\left(\frac{t}{\pi} - 2 \lfloor \frac{t}{2\pi} \rfloor\right), & t > 2\pi, \end{cases}$$

which is a function with period 2π that generates asymmetric periodic patterns that we empirically observe in Figure 1. Here $\lfloor \cdot \rfloor$ denotes the floor function.

In these three examples, we set the number of sample to be 7800 to match our intraday data in the US market. We set the coefficient of the noise term to be 20 to simulate the low signal-to-noise ratio in intraday volume time series. In fact, similar to our empirical observations in Figure 1, it is virtually impossible to detect any periodic patterns by visually inspecting the time series (see Figure A.1). Our methodology can recover these periodicities with low signal-to-noise ratios.

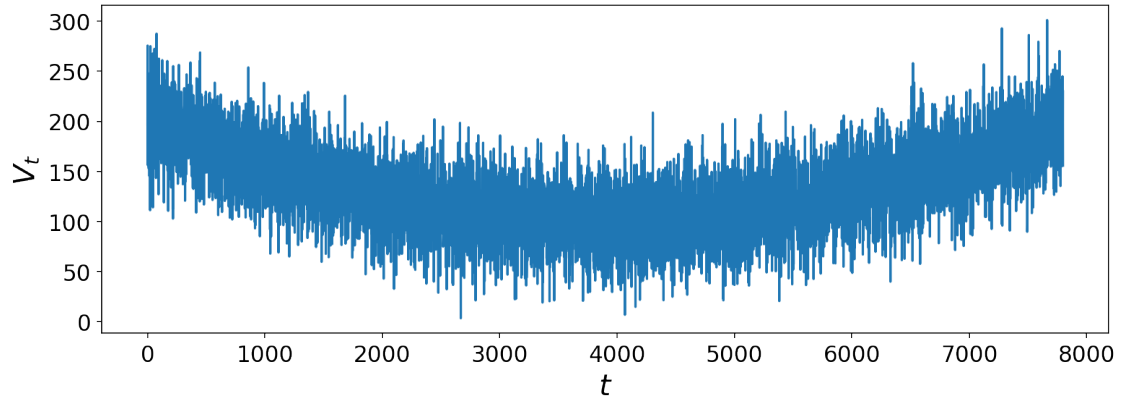
The synthetic time series, $V_t^{\cos}$ and V_t^{spike} , both have ten different periodic components. However, we do not know the true model in real datasets, so we set $n = 500$ in our estimation, the same choice as in Section 3.2.

Figure A.2 shows the estimated intensity coefficients for the three synthetic time series following the methodology in Section 3.2. The red dots represent the true values of squared intensity coefficients, a_j^2 , which we know from the construction of the synthetic time series. Figure A.2a shows the estimated $\hat{a}_j^2$'s for $V_t^{\text{nonperiod}}$, which are all close to 0. In other word, our method does not reveal any false periodicities in this control example.

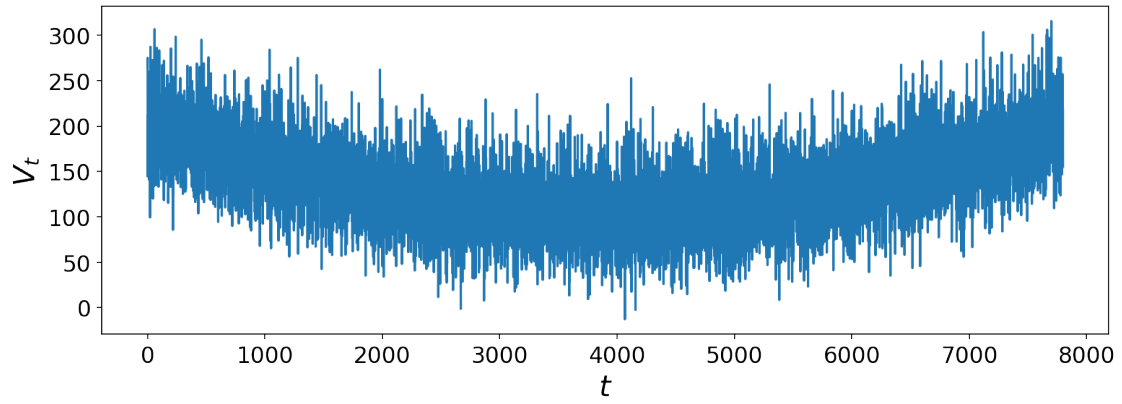
In Figure A.2b, our estimation for $V_t^{\cos}$ perfectly matches the ground truth at all periodic frequencies, and the estimated $\hat{a}_j^2$'s for other frequencies are close to 0. This example demonstrates that if the underlying time series is generated by the same trigonometric basis as the specification in (1), our estimation can recover all periodicities with an impressive precision.

Finally, Figure A.2c demonstrates the estimation results for a periodic time series whose periodicities are not in the same form as the model specification in (1). In this case, although the estimated $\hat{a}_j^2$'s do not perfectly match the ground truth, they are still able to recover these periodic terms reasonably well, compared with frequencies with no periodicity. This example demonstrates that our method can reveal the periodicity and extract the specific periodic terms even if the underlying model is not following the exact trigonometric formulation, which further validates the robustness of our methodology.

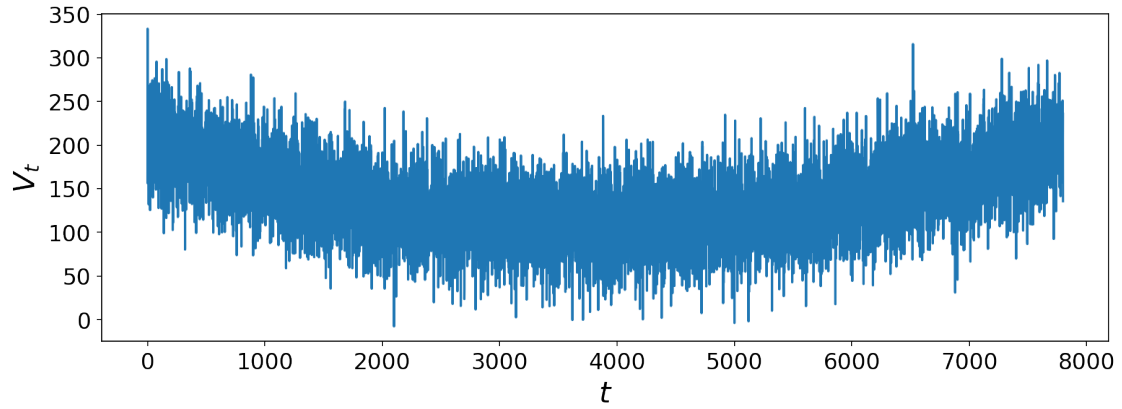
It is worth highlighting that our framework is developed without any restriction on the specific nature of the underlying time series. Therefore, it can be applied to not only the number of trades—our focus in the next section—but also other metrics of trading volumes and, more broadly, any financial time series that potentially contain important periodicities.



(a) Non-periodic time series.

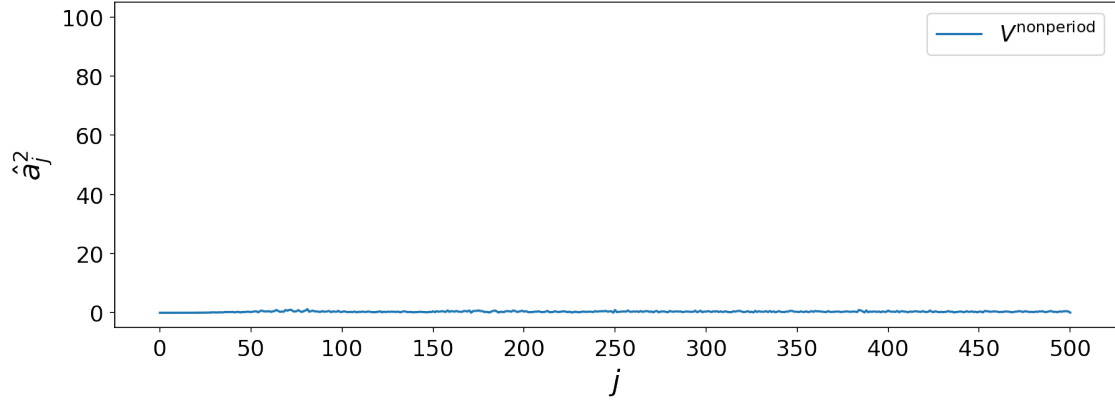


(b) Periodic time series with trigonometric basis.

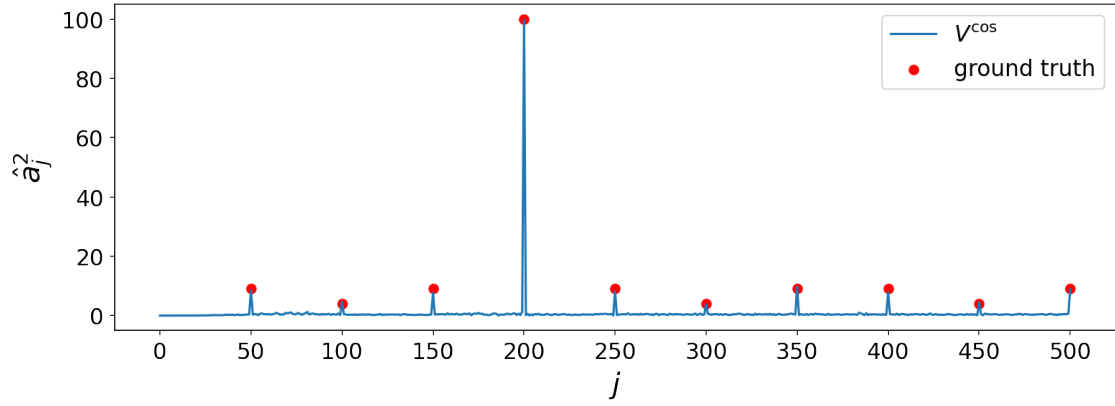


(c) Periodic time series with non-trigonometric basis.

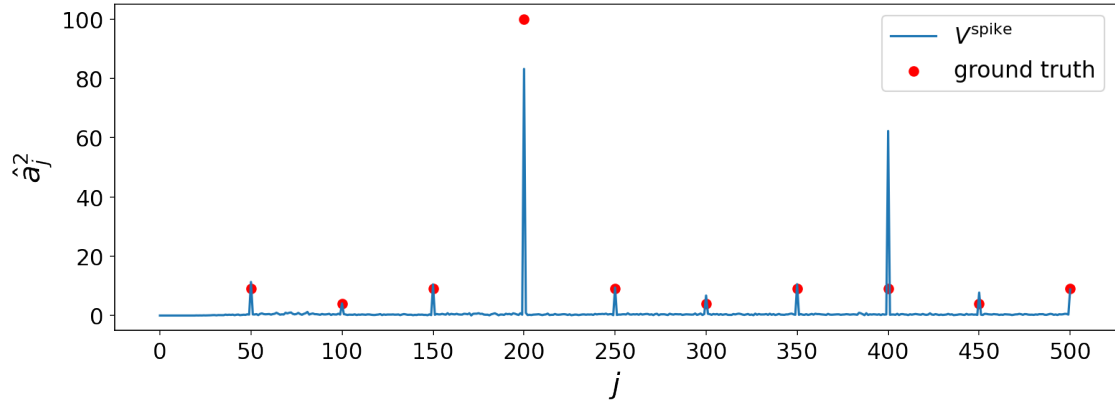
Figure A.1: Three synthetic time series generated based on different basis functions in Appendix C.



(a) Non-periodic time series.



(b) Periodic time series with trigonometric basis.



(c) Periodic time series with non-trigonometric basis.

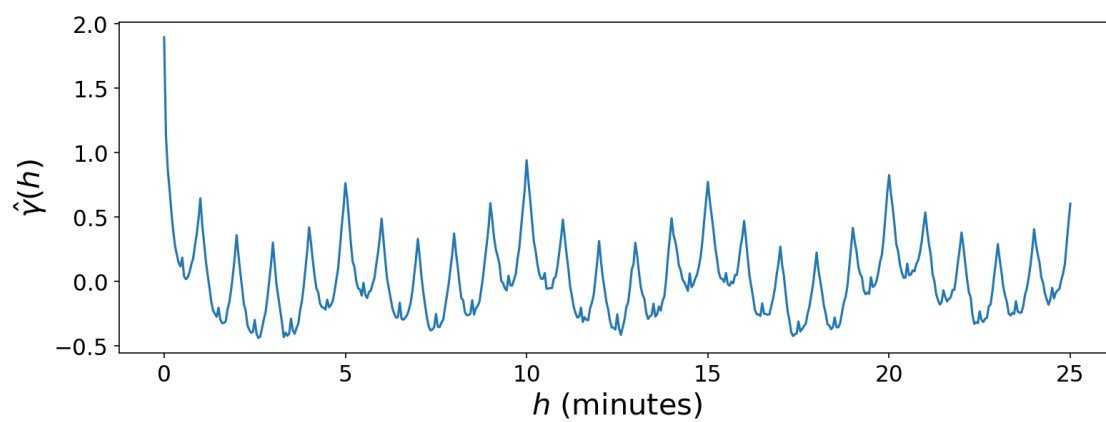
Figure A.2: Estimated intensity coefficients for the synthetic time series. The red dots represent the true squared intensity coefficients, a_j^2 , which we know from the construction of the synthetic time series.

D Estimated Autocovariance Functions for Single Stock

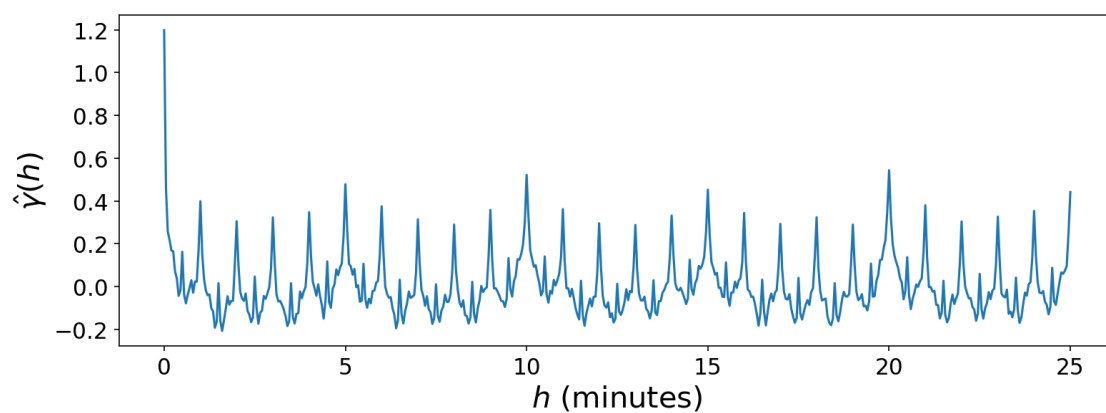
Figure A.3 shows the autocovariance functions estimated in Section 4.1, which show strong periodicities, though we observe no obvious patterns in the original time series of trading volume in Figure 1. In particular, the autocovariance function for Apple Inc demonstrates a strong period at the 1-minute frequency. Similarly, the autocovariance function for Ping An Bank shows more complicated cyclicity, with periods at both the 1-minute and 5-minute frequencies, especially in the morning session.

To further understand the spectral decomposition of our model in Section 3, we note that the estimated intensity coefficients are derived from the DFT of the autocovariance function, which implies that they can be considered as a second-order moment estimation. We plug the estimated intensity coefficients back into model (1), and compare the autocovariance function of the estimated time series with trigonometric basis with the autocovariance function of the original data.

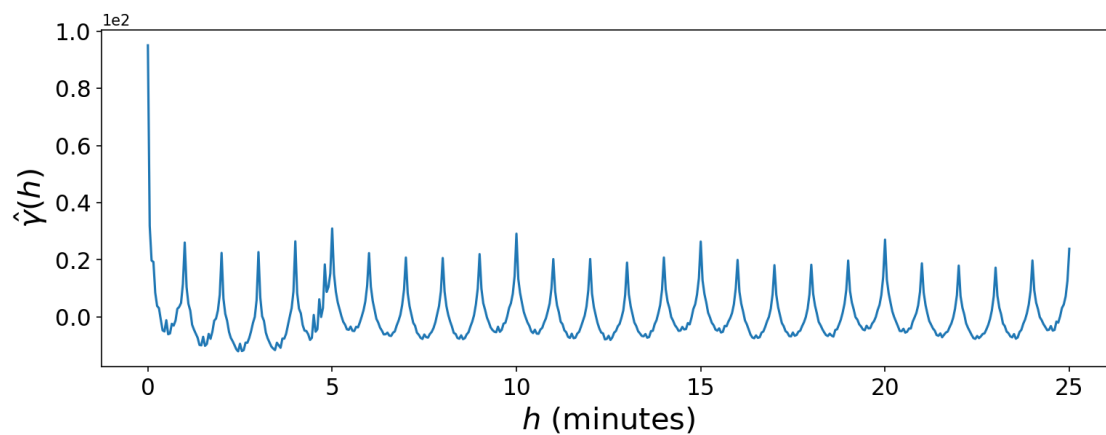
Figure A.4 shows that our estimated results perfectly recover the autocovariance functions of the trading volume time series, and therefore fully exploit the moment information of the original sample. This result further demonstrates that the simple cosine spectral basis functions, though may not exactly match the periodic patterns in real data, can still be very powerful in uncovering periodicities with low signal-to-noise ratio and capturing their autocovariance structures.



(a) Ping An Bank (morning session).

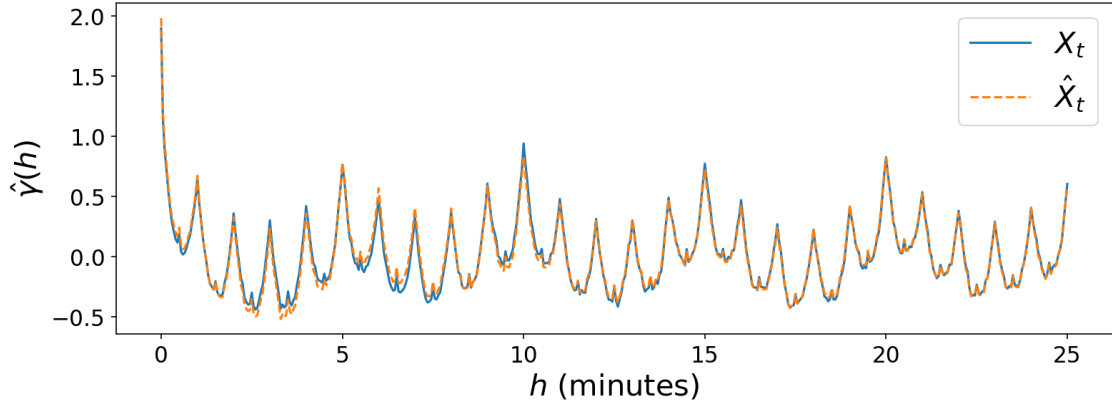


(b) Ping An Bank (afternoon session).

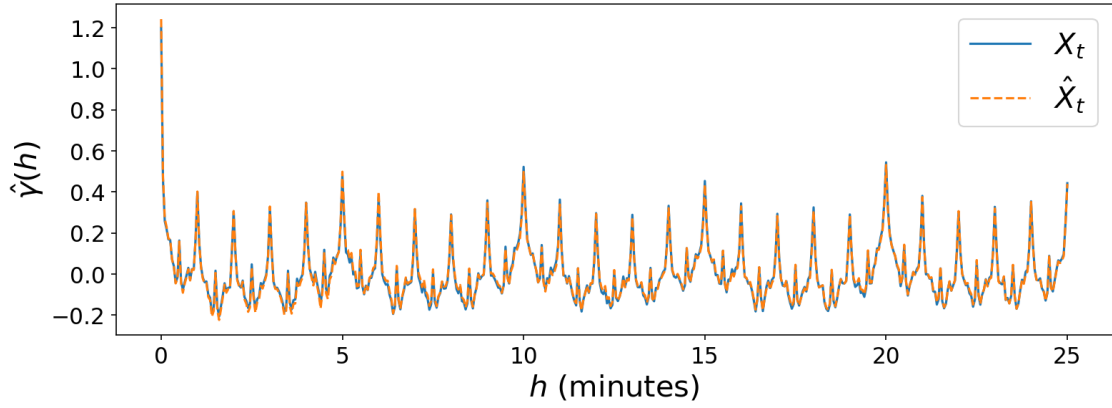


(c) Apple Inc.

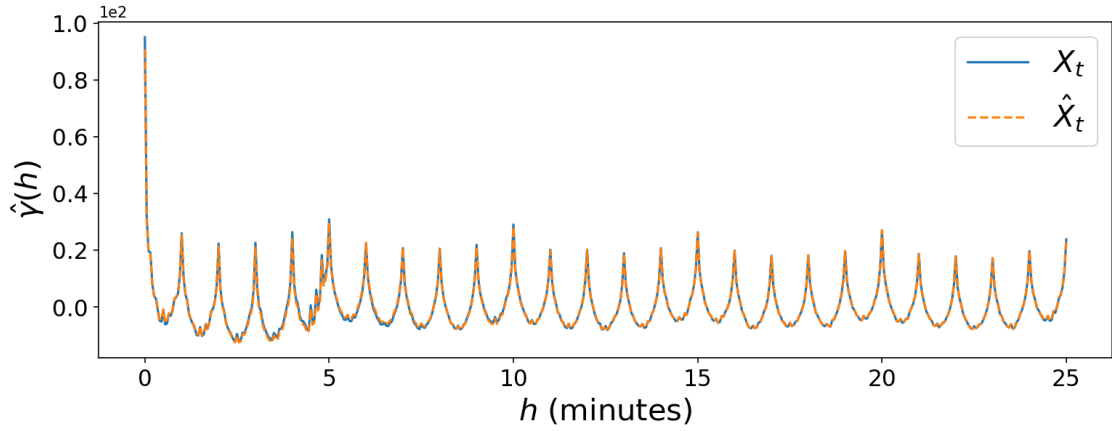
Figure A.3: Estimated autocovariance functions, $\hat{\gamma}(h)$, for two individual stocks in 2019–2021.



(a) Ping An Bank (morning session).



(b) Ping An Bank (afternoon session).



(c) Apple Inc.

Figure A.4: A comparison between sample autocovariance functions (solid lines) and the theoretical autocovariance functions for the estimated time series with trigonometric basis (dashed lines) for individual stocks.

E Validation with Wavelet-variance Estimator

As a robustness check, we follow Chinco and Ye (2017) to apply the wavelet-variance estimator to offer additional evidence of the existence of the universal high-frequency periodicity in trading volume. This analysis supports that fVR is a credible proxy for the strength of periodicity.

Suppose the demeaned trading volume series has the following wavelet representation:

$$V_t - \bar{V} = \sum_{f=1}^F \sum_{\ell=1}^{2^{f-1}} \hat{\theta}_{f,\ell} w_{f,\ell}(t),$$

where $F = \lceil \log_2 T \rceil$ and $w_{f,\ell}(t)$ denotes the Haar's wavelet defined as

$$w_{f,\ell}(t) := \begin{cases} \sqrt{2^{f-1}/T}, & \ell - 1 < t/T \cdot 2^{f-1} \leq \ell - 1/2, \\ -\sqrt{2^{f-1}/T}, & \ell - 1/2 < t/T \cdot 2^{f-1} \leq \ell, \\ 0, & \text{otherwise.} \end{cases}$$

Hence, the variance of the demeaned trading volume can be decomposed into F different frequency-specific components by looking at the average of squared wavelet coefficients at each frequency:

$$\text{Var}(V) = \text{Var} \left(\sum_{f=1}^F \sum_{\ell=1}^{2^{f-1}} \hat{\theta}_{f,\ell} w_{f,\ell}(t) \right) = \sum_{f=1}^F \sum_{\ell=1}^{2^{f-1}} \hat{\theta}_{f,\ell}^2 \text{Var}(w_{f,\ell}) = \sum_{f=1}^F \left(\frac{1}{T} \sum_{\ell=1}^{2^{f-1}} \hat{\theta}_{f,\ell}^2 \right).$$

The wavelet-variance estimator at frequency f is defined as

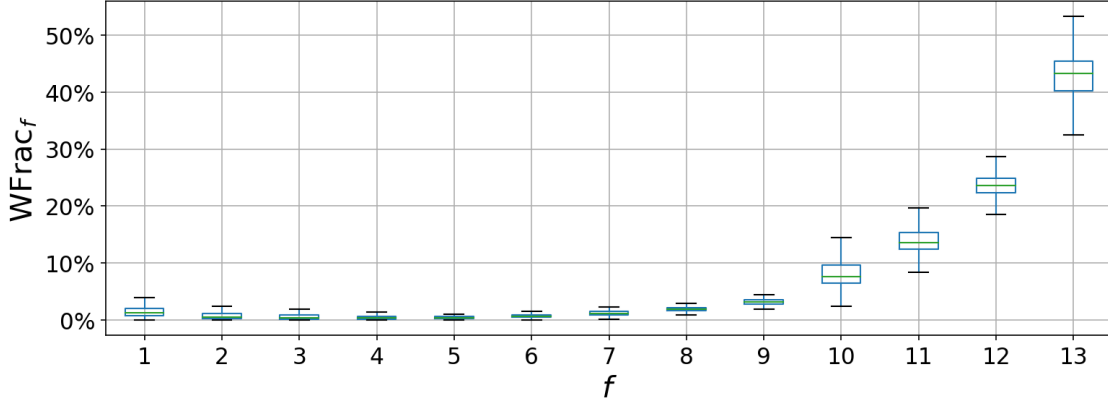
$$\text{WVar}_f(V) := \sum_{\ell=1}^{2^{f-1}} \hat{\theta}_{f,\ell}^2,$$

which captures fluctuations in trading volume from the $T/2^f$ horizon. The corresponding wavelet-variance fraction at frequency f is

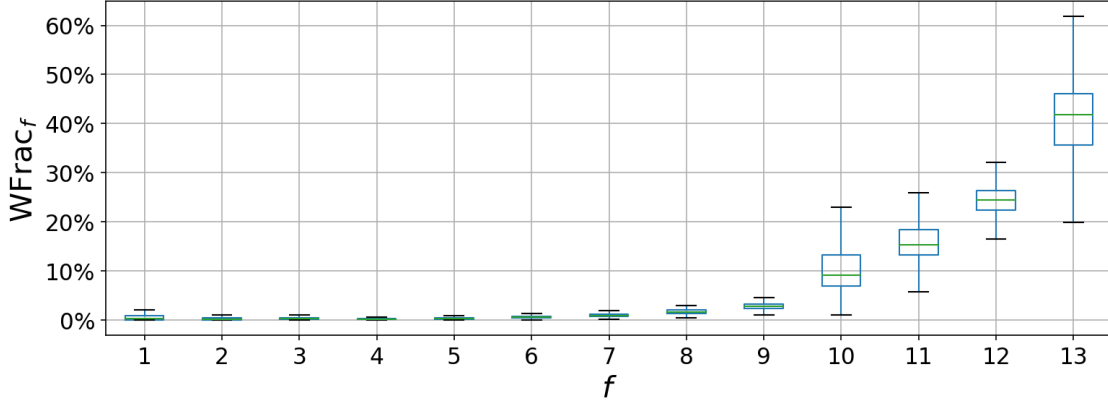
$$\text{WFrac}_f(V) := \frac{\text{WVar}_f(V)}{\sum_{f'=1}^F \text{WVar}_{f'}(V)}.$$

Following Chinco and Ye (2017), we apply the wavelet-variance estimator to each stock's one-minute trading volume data in each month, which leads to around $\frac{21 \text{ days}}{1 \text{ month}} \times \frac{390 \text{ mins}}{1 \text{ day}} = 8190$ observations in each time series (in the US market). Thus, fluctuations in volume at the 1-minute frequency roughly correspond to fluctuations at $f = 13$ since $2^{13} = 8192$. Figure A.5 shows the box plot of WFrac's across stocks and months for different frequencies f . For the vast majority of stocks and months, the WFrac's at the one-minute frequency are around 40% in both markets, which are significantly higher than the other frequencies. This provides another evidence for the existence of the universal high-frequency periodicities in trading volume and supports that the fVR serves as a

credible proxy for the strength of periodicity.



(a) The US market.



(b) The Chinese market

Figure A.5: Box plot of wavelet-variance fraction (WFract) for all stocks in 2019–2021. $WFract_f$ captures the fraction of fluctuations in trading volume from $T/2^f$ horizon. Particularly, the $WFract_{13}$ measures the fractions of the volume fluctuations explained by the one-minute frequency because each time series includes around $\frac{21 \text{ days}}{1 \text{ month}} \times \frac{390 \text{ mins}}{1 \text{ day}} = 8190 \approx 2^{13}$ observations of one-minute trading volume.

Although both fVR and WFract are designed to capture the contribution of different frequencies to the variance of trading volume, they are different in important aspects. The wavelet framework can only include frequencies with lengths that are powers of two (e.g., 1, 2, 4, etc.). As a result, it is not suitable for jointly analyzing the round-time frequencies such as 1-minute, 5-minute, or 10-minute within a single model. Instead, this framework is more appropriate for examining whether the variation in trading volume is primarily driven by high- or low-frequency components, which is the primary focus of Chinco and Ye (2017). In contrast, our spectral model can simultaneously incorporate a wide range of frequencies and identify the most dominant ones. Therefore, we mainly use the wavelet method here as a robustness check to provide additional evidence for the existence of universal high-frequency periodicity. However, the specific values of WFract and fVR are not directly comparable.

F Other Financial Time Series and Winsorized Volumes

Following Heston, Korajczyk, and Sadka (2010), we calculate the intraday time series of order imbalance, volatility, and spread on the same dataset we used in our paper. Figure A.6 are the boxplots of fVR for different measures in both the US market and the Chinese market. We find significant periodicities in volatility and spread, in both two markets, which is consistent with our results of price impact in Section 5.2.2 and the fact that the movement of volatility and spread are related to movements in volume (Heston, Korajczyk, and Sadka, 2010). However, we find no significant evidence on the periodicity of order imbalance. Our results for the three different measures are all consistent with Heston, Korajczyk, and Sadka (2010).

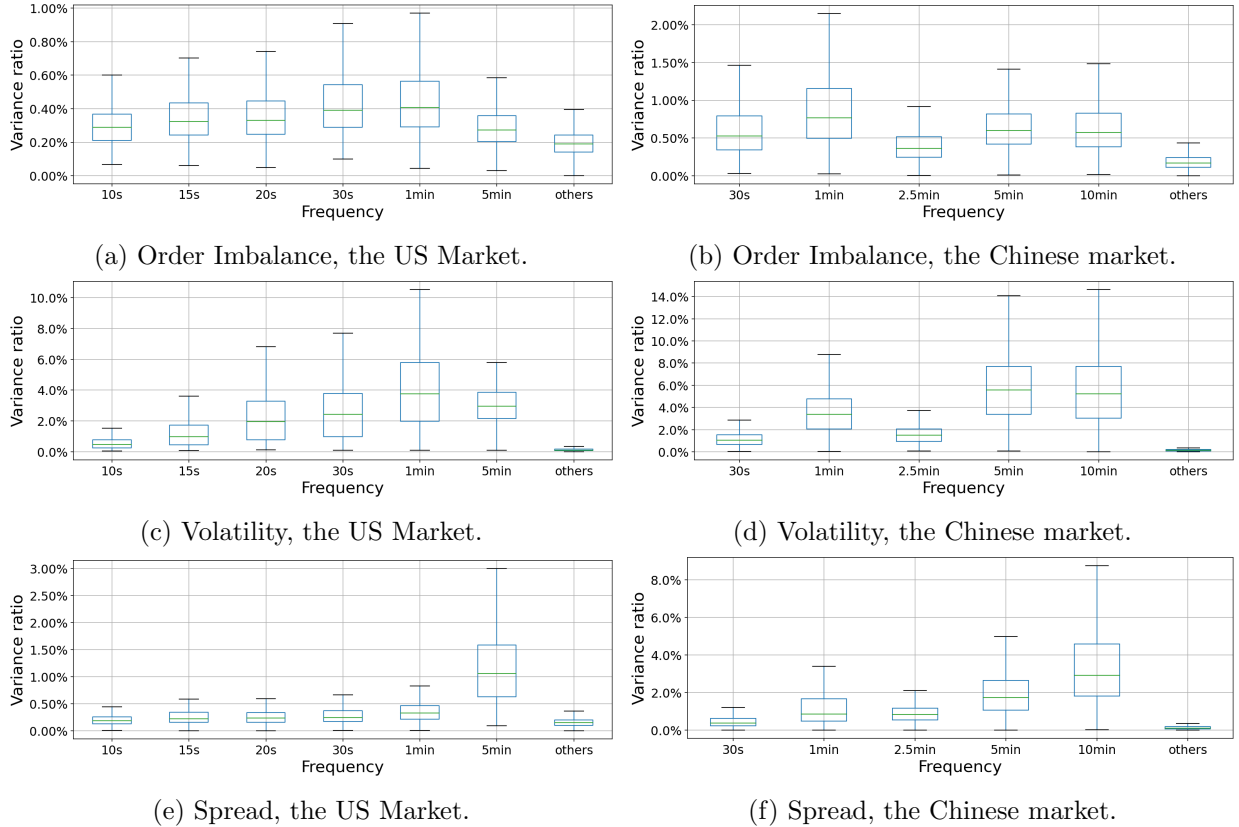
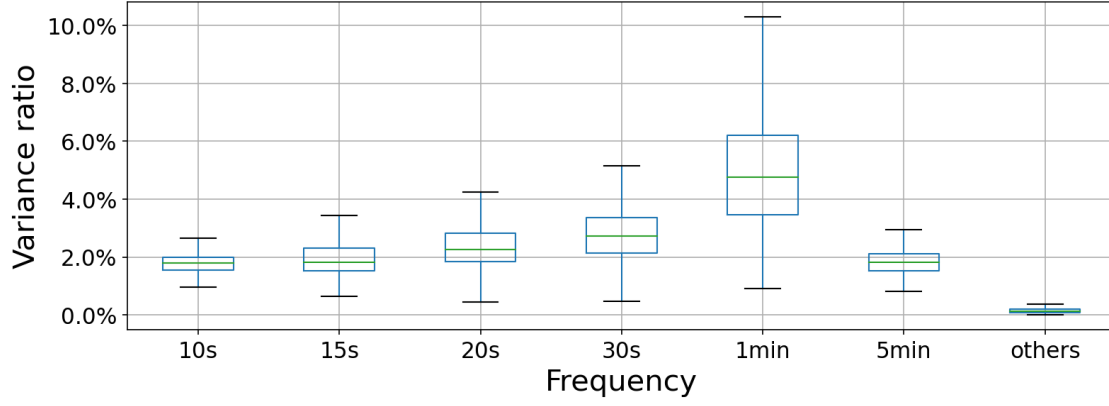
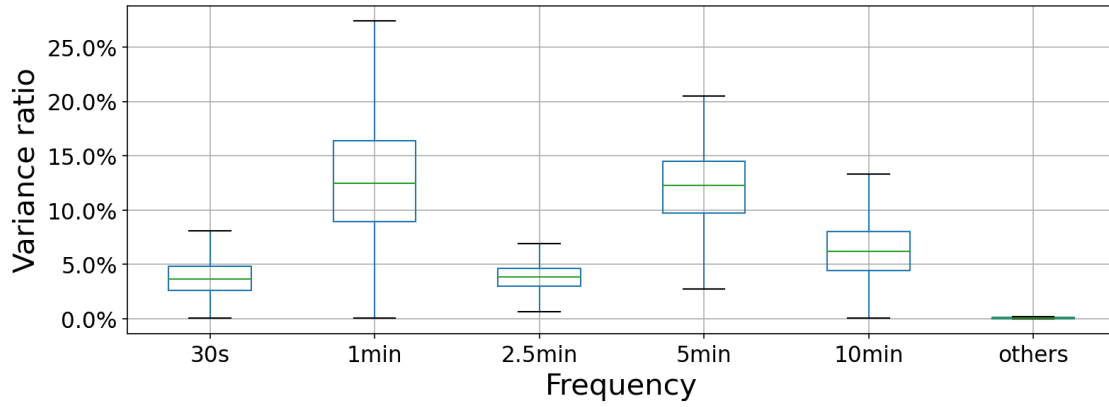


Figure A.6: Box plot of frequency variance ratios (fVR) for different measures. The benchmark fVR is $\frac{1}{500} = 0.2\%$ if no periodicity exists because we included $n = 500$ periodic components in the model.

As a robustness check, for each stock and each trading day, we winsorize the intraday trading volume data at the 95th percentile to reduce the impact of extreme data on fVR. Figure A.7 shows the box plot of fVRs after winsorization. The periodicities remain robust. The upper quartile bars are slightly higher than those before winsorization for 1-minute and 30-second frequencies in the US market and 10-minute, 5-minute, and 1-minute frequencies in the Chinese market. This result indicates that some stocks achieve higher fVRs due to the reduced noise from extreme data.



(a) The US market.



(b) The Chinese market.

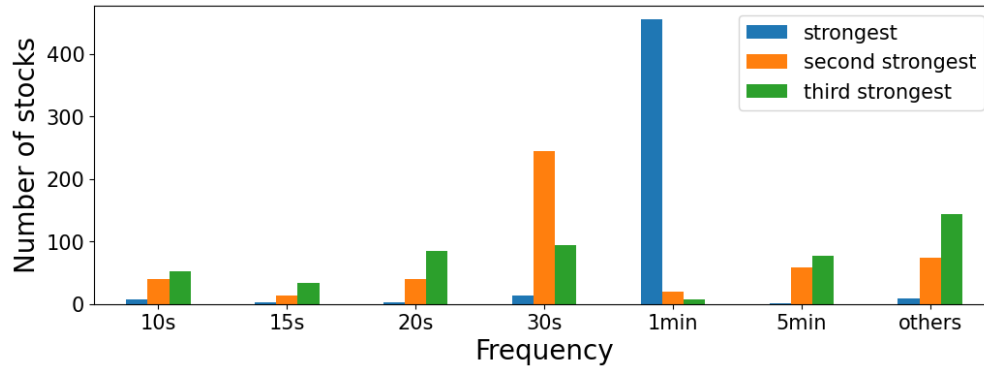
Figure A.7: Box plot of frequency variance ratios (fVR)—the ratio of the variance explained by selected frequencies over the total variance of the de-trended time series—for all stocks in 2019–2021. For each stock and each trading day, the intraday trading volume series is clipped at the 95th percentile. All other frequencies are grouped into the “others” category. The benchmark fVR is $\frac{1}{500} = 0.2\%$ if no periodicity exists because we included $n = 500$ periodic components in the model.

More generally, one may use our framework to study periodicities systematically in the ETF market, which is left for future work.

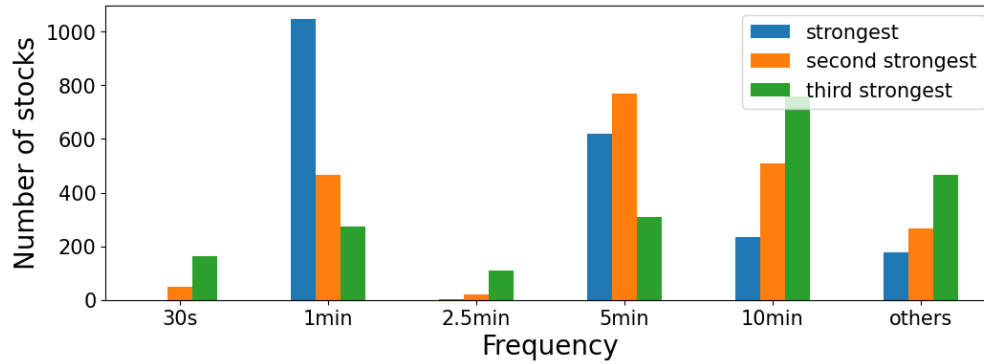
G Other Measures of Trading Volume

We follow the same methodology as in Section 4 and estimate the intensity coefficients by replacing the number of trades with the number of traded shares and the monetary volume, respectively.

Figures A.8–A.9 show the top three frequencies with the strongest intensity coefficients across all stocks. We can observe a significant decrease in the number of stocks showing periodicities at the selected frequencies compared to Figure 3.

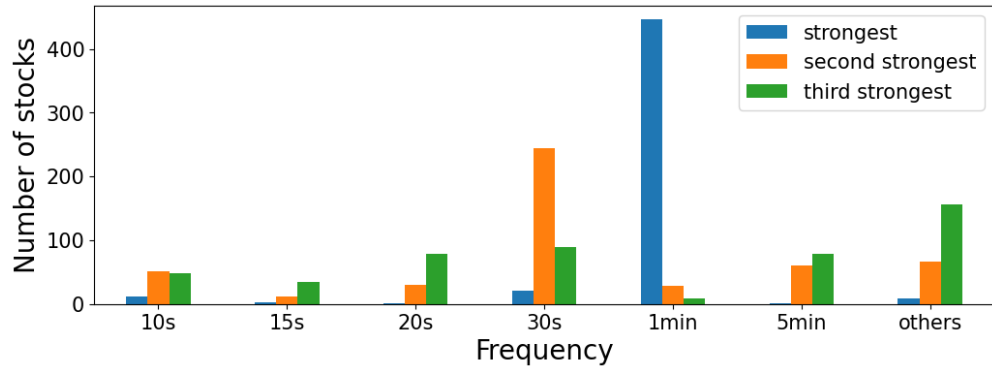


(a) The US market.

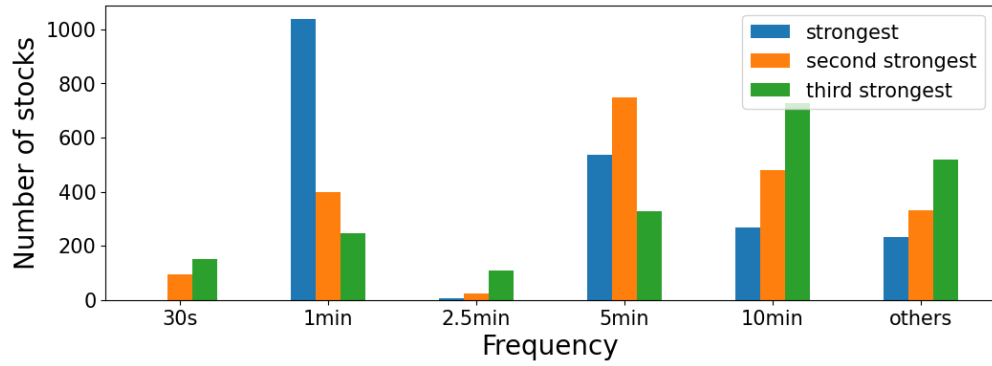


(b) The Chinese market.

Figure A.8: The number of occurrences of each frequency as the top three strongest frequencies in terms of estimated intensity coefficients across all stocks in 2019–2021, when the number of traded shares is used.



(a) The US market.

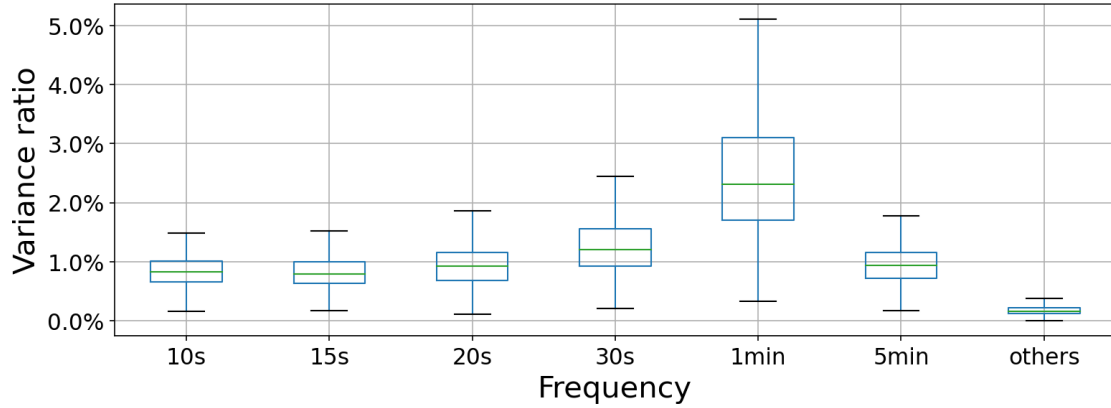


(b) The Chinese market.

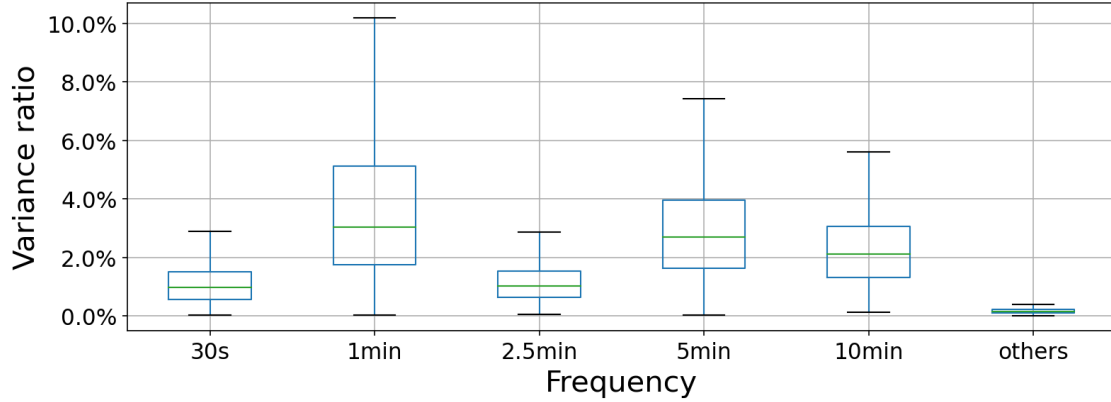
Figure A.9: The number of occurrences of each frequency as the top three strongest frequencies in terms of estimated intensity coefficients across all stocks in 2019–2021, when the monetary volume is used.

Furthermore, Figures A.10–A.11 show the fVRs for the traded shares and the monetary volume, respectively. In the US market, the fVRs for all selected frequencies are only around 1–2 percent, which are significantly less than those in Figure 4. Similarly, in the Chinese market, the average fVRs for 1-minute and 5-minute frequencies decreased from 10 percent to approximately 3 percent, and the fVRs for 30-second, 2.5-minute and 10-minute frequencies are around 1–2 percent only.

These results imply weaker periodicities in traded shares and monetary volumes than in the number of trades.

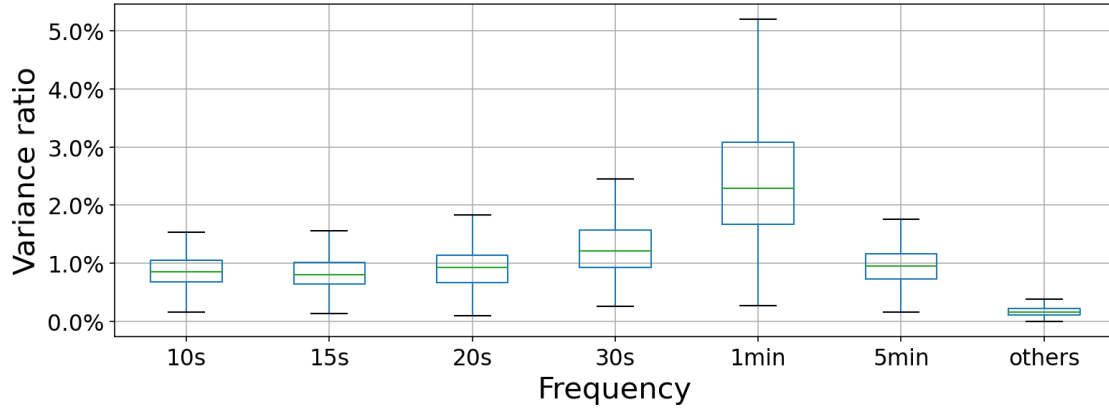


(a) The US market.

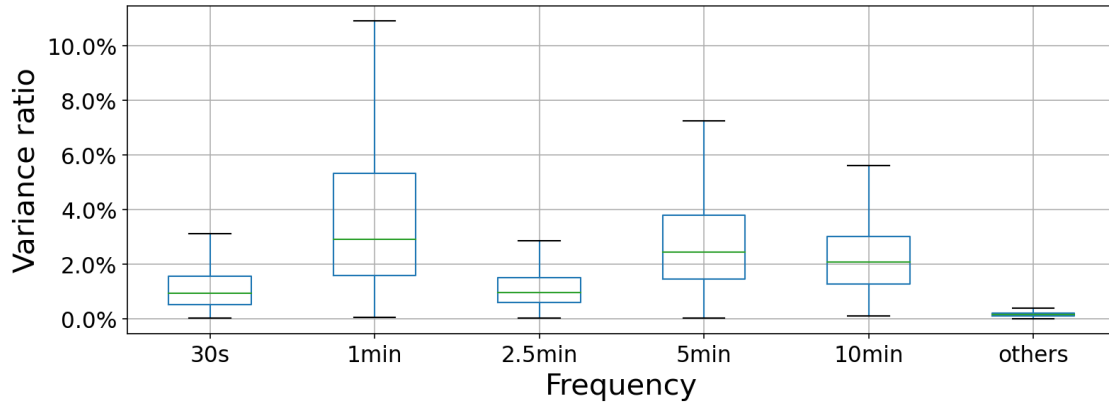


(b) The Chinese market

Figure A.10: Box plot of frequency variance ratios (fVR) when the number of traded shares is used. The benchmark fVR is $\frac{1}{500} = 0.2\%$ if no periodicity exists because we included $n = 500$ periodic components in the model.



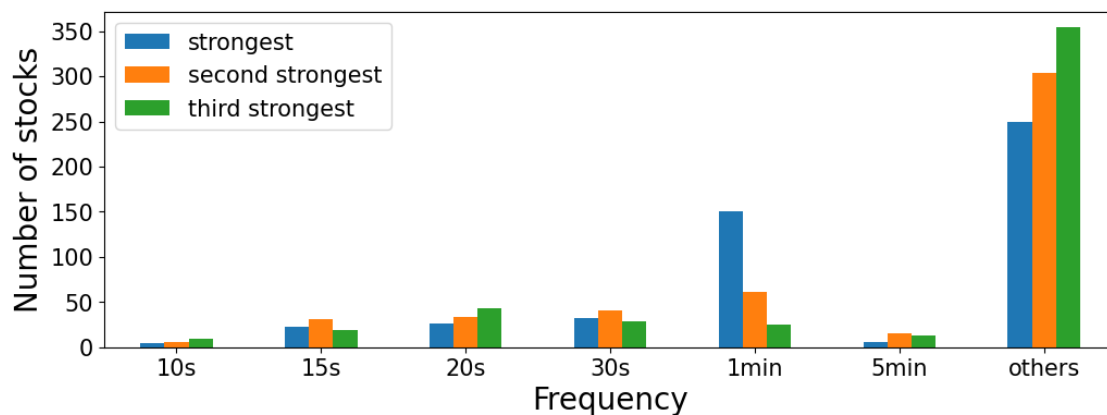
(a) The US market.



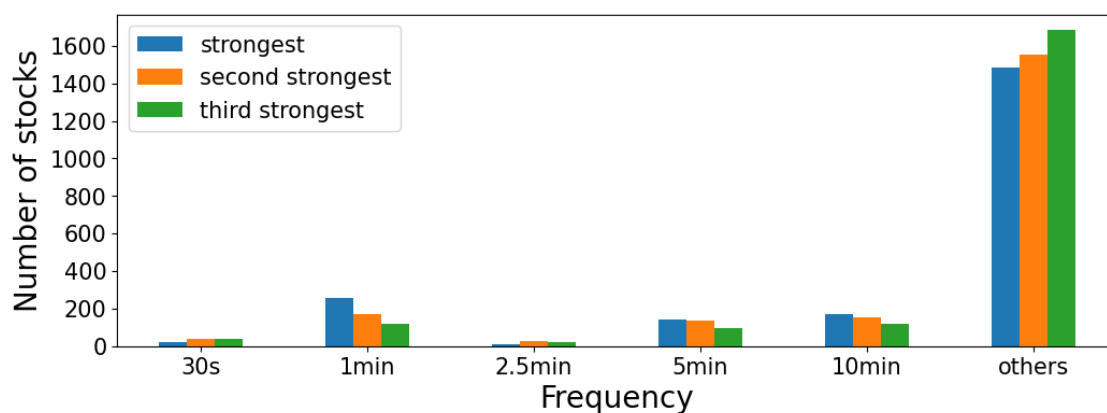
(b) The Chinese market.

Figure A.11: Box plot of frequency variance ratios (fVR) when the monetary volume is used. The benchmark fVR is $\frac{1}{500} = 0.2\%$ if no periodicity exists because we included $n = 500$ periodic components in the model.

Finally, similar to Figures 5 and 6, the results for the average monetary volume per trade in Figures A.12 and A.13 also show very weak evidence for periodicity in this measure.

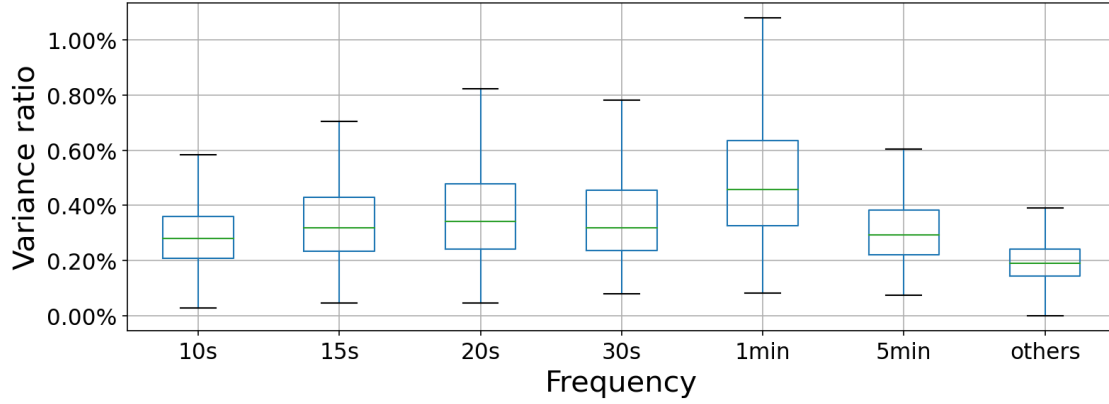


(a) The US market.

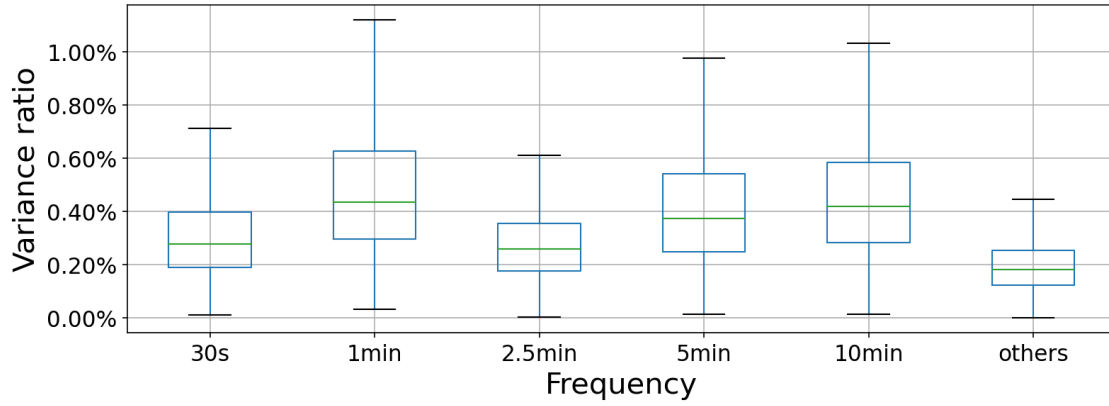


(b) The Chinese market.

Figure A.12: The number of occurrences of each frequency as the top three strongest frequencies in terms of estimated intensity coefficients across all stocks in 2019–2021, when the average monetary value per trade is used.



(a) The US market.



(b) The Chinese market

Figure A.13: Box plot of frequency variance ratios (fVR) when the average monetary value per trade is used. The benchmark fVR is $\frac{1}{500} = 0.2\%$ if no periodicity exists because we included $n = 500$ periodic components in the model.

H Data Processing Details for Ravenpack News

We only have access to the Ravenpack news data in 2012–2022 for the US market and in 2015–2022 for the Chinese market in the Ravenpack dataset. Each sample in the Ravenpack dataset contains several columns, including a millisecond-level timestamp and the companies it mentioned.

To find the number of published news articles related to each stock, we use the stock-company correspondence table provided by Wharton Research Data Services in the US and Wind database in China, respectively, to match news articles with stocks by their company names.³⁵ However, the exact formats of the company names in these three datasets are not the same, so we process them using the following procedure.

1. Select news articles published during the trading session, i.e. 9:30–16:00 (9:30–11:30 and 13:00–14:57) in trading days in the US (Chinese) market.
2. Remove all spaces, special characters, and numbers from the names of the company.
3. Capitalize all letters.
4. Calculate the similarity metric between every pair of company names using the Ratcliff-Obershelp algorithm (also known as the Gestalt pattern matching).
5. Match a news article with a stock if the similarity metric between their company names is greater than 0.8.

Based on this procedure, we form an intraday time series of the number of published news articles for each stock. Table A.2 provides the summary statistics of the total number of articles for each stock.

³⁵Wind is one of the largest provider of financial data in the Chinese market. See <https://www.wind.com.cn/>.

Table A.2: Summary statistics for the total number of articles for each stock. We provide both the market-wide average and statistics for selected stocks with the highest number of articles.

	mean	standard deviation	minimum	25% quantile	median	75% quantile	maximum
<i>Average across all stocks in the market</i>							
The US market	6657.52	17784.16	0	990	2452	5004	171863
The Chinese market	68.79	370.35	0	1	3	18	9309
<i>Five stocks with the highest number of articles in the US market</i>							
Ticker	Number of articles	Company name					
JPM	171863	JPMorgan Chase & Co.					
FB	161156	Facebook Inc.					
MSFT	132853	Microsoft Corp.					
CMCSK	124570	Comcast Corp.					
TWTR	121661	Twitter Inc.					
<i>Five stocks with the highest number of articles in the Chinese market</i>							
Ticker	Number of articles	Company name					
002415.SZSE	9309	Hangzhou Hikvision Digital Technology Co.,Ltd.					
000333.SZSE	5017	Midea Group Co., Ltd					
300750.SZSE	4489	Contemporary Amperex Technology Co., Ltd.					
000002.SZSE	4029	China Vanke Co.,Ltd.					
000725.SZSE	2826	BOE Technology Group Co., Ltd.					

I Additional Results for Excess Returns

Robustness Checks. We check the robustness of PmS excess returns by constructing PmS portfolios on a daily or weekly basis and constructing double-sorted portfolios based on several stock-level characteristics.

For daily PmS portfolios, we first evaluate the strength of periodicity for each stock s on each day d using (9), and then sort stocks into five quintiles according to their $peri_{s,d}$ values on each day. The PmS portfolio buys the top quintile of stocks with the highest $peri_{s,d}$ values and sells the bottom quintile of stocks with the lowest $peri_{s,d}$ values. The weekly PmS portfolio is constructed similarly. To align with the results in Table 8, we present the monthly alphas of the daily and weekly PmS portfolios in Table A.3. As expected, we obtain consistent and even higher excess returns if we rebalance the portfolios more frequently. However, we also emphasize that these results did not account for trading cost, which is particularly important when portfolios are rebalanced more frequently. The high alphas in the Chinese market for daily rebalancing are also unrealistic in practice due to restrictions on short-selling and the $T + 1$ rule.

In addition, as shown in Section 5.1.2, the $peri$ values of stocks are positively correlated with the level of algorithmic trading and volatility. Informed algorithmic traders contribute to information asymmetry in the market and affect stock returns.³⁶ Volatility is also known to affect the expected stock return. Therefore, returns of the PmS portfolio may be confounded with return effects from algorithmic trading and volatility³⁷.

We follow Fama and French (1993, 2015) and Amihud et al. (2015) to construct double-sorted portfolios by calculating the values of $mess$, $nvdmess$, $oddrat$, and vol as defined in Section 5.1.2 for each stock in each month. For each of these characteristics, we pre-sort stocks into three portfolios and, within each tercile, further sort stocks into quintiles based on their $peri$ values. We thus obtain $3 \times 5 = 15$ portfolios with an equal number of stocks in each portfolio. The top (bottom)- $peri$ quintile portfolio thus consists of the three top (bottom) portfolios across three levels of the selected characteristic. Table A.4 presents the monthly alphas of these double-sorted PmS portfolios, and we still obtain consistent excess returns after controlling for these pre-sorting characteristics.

Overall, our analysis shows that stocks with stronger periodicities in trading volume deliver positive excess returns on average, and these results are robust after controlling for known pricing factors and stock characteristics that are correlated with the strength of periodicities.

³⁶Bogousslavsky, Fos, and Muravyev (2023) develop a measure of informed trading based on machine learning.

³⁷In fact, as a sanity check, we also find positive excess returns by constructing long-short portfolios similar to those in Table 8 based on existing proxies of algorithmic trading rather than our measure $peri$.

Table A.3: Monthly alphas of portfolios sorted by the strength of periodicity, *peri*, under different rebalancing frequencies and controlling for different asset-pricing factors based on regressions of daily returns. The last column presents the Periodic minus Smooth (PmS) portfolio that buys/sells stocks in the top/bottom quintile. In the US market (Panel A, B, C, and D), we control for the Fama-French three (FF3) and five (FF5) factors, Carhart's (1997) momentum factor, and the liquidity risk factor (ImL) of Amihud et al. (2015). In the Chinese market (Panel E, F, G, and H), we control for market, size, and value factors (CH3) of Liu, Stambaugh, and Yuan (2019), the liquidity risk factor (PmO) of Liu, Stambaugh, and Yuan (2019), and the Fama-French profitability (RmW) and investment (CmA) factors. Estimated alphas are significant at the 1 percent (***), 5 percent (**), or 10 percent (*) levels.

Asset-pricing model	Top	Second	Middle	Fourth	Bottom	PmS
Panel A: The US Market; Equal Weighted Portfolio; Daily Rebalancing						
FF3	0.93%***	0.20%	-0.06%	0.13%	-0.05%	0.97%***
FF5	0.93%***	0.15%	-0.14%	0.09%	-0.14%	1.07%***
FF5 + momentum	0.88%***	0.11%	-0.19%	0.03%	-0.20%	1.08%***
FF5 + momentum + liquidity	0.85%***	0.07%	-0.19%	0.05%	-0.20%	1.05%***
Panel B: The US Market; Value Weighted Portfolio; Daily Rebalancing						
FF3	1.80%***	0.20%	0.29%	0.16%	-0.50%*	2.31%***
FF5	1.70%***	0.13%	0.15%	0.11%	-0.54%*	2.26%***
FF5 + momentum	1.70%***	0.14%	0.12%	0.09%	-0.58%**	2.29%***
FF5 + momentum + liquidity	1.74%***	0.08%	0.18%	0.14%	-0.56%**	2.31%***
Panel C: The US Market; Equal Weighted Portfolio; Weekly Rebalancing						
FF3	0.60%***	0.11%	0.36%	0.11%	-0.02%	0.62%*
FF5	0.59%***	0.08%	0.28%	0.04%	-0.08%	0.68%*
FF5 + momentum	0.57%***	0.05%	0.21%	-0.02%	-0.17%	0.74%**
FF5 + momentum + liquidity	0.55%***	0.03%	0.19%	-0.04%	-0.14%	0.70%**
Panel D: The US Market; Value Weighted Portfolio; Weekly Rebalancing						
FF3	1.35%***	0.43%*	0.25%	0.23%	-0.19%	1.54%***
FF5	1.24%***	0.37%	0.18%	0.13%	-0.21%	1.45%***
FF5 + momentum	1.23%***	0.33%	0.17%	0.11%	-0.20%	1.43%***
FF5 + momentum + liquidity	1.24%***	0.37%	0.14%	0.13%	-0.13%	1.38%***
Panel E: The Chinese Market; Equal Weighted Portfolio; Daily Rebalancing						
CH3	18.30%***	1.29%***	-3.89%***	-6.82%***	-8.24%***	28.80%***
CH3 + liquidity (CH4)	18.39%***	1.28%***	-3.93%***	-6.86%***	-8.29%***	28.96%***
CH4 + profitability + investment	18.65%***	1.41%***	-3.83%***	-6.79%***	-8.26%***	29.20%***
Panel F: The Chinese Market; Value Weighted Portfolio; Daily Rebalancing						
CH3	12.45%***	1.33%***	-2.96%***	-5.72%***	-7.46%***	21.43%***
CH3 + liquidity (CH4)	12.51%***	1.34%***	-2.97%***	-5.76%***	-7.52%***	21.56%***
CH4 + profitability + investment	12.45%***	1.20%***	-3.13%***	-5.84%***	-7.57%***	21.57%***
Panel G: The Chinese Market; Equal Weighted Portfolio; Weekly Rebalancing						
CH3	4.73%***	1.67%***	-0.55%***	-2.48%***	-4.59%***	9.74%***
CH3 + liquidity (CH4)	4.81%***	1.67%***	-0.58%***	-2.54%***	-4.65%***	9.89%***
CH4 + profitability + investment	5.03%***	1.83%***	-0.50%**	-2.47%***	-4.63%***	10.11%***
Panel H: The Chinese Market; Value Weighted Portfolio; Weekly Rebalancing						
CH3	4.44%***	2.12%***	-0.06%	-1.91%***	-4.09%***	8.88%***
CH3 + liquidity (CH4)	4.50%***	2.12%***	-0.08%	-1.96%***	-4.15%***	9.00%***
CH4 + profitability + investment	4.43%***	2.03%***	-0.26%	-2.06%***	-4.20%***	8.99%***

Table A.4: Monthly alphas of double-sorted PmS portfolios based on the strength of periodicity, *peri*, controlling for different pre-sorting characteristics and different asset-pricing factors, based on regressions of daily returns. In the US market (Panel A and B), we control for the Fama-French three (FF3) and five (FF5) factors, Carhart's (1997) momentum factor, and the liquidity risk factor (ImL) of Amihud et al. (2015). In the Chinese market (Panel C and D), we control for market, size, and value factors (CH3) of Liu, Stambaugh, and Yuan (2019), the liquidity risk factor (PmO) of Liu, Stambaugh, and Yuan (2019), and the Fama-French profitability (RmW) and investment (CmA) factors. Estimated alphas are significant at the 1 percent (***), 5 percent (**), or 10 percent (*) levels.

Asset-pricing model	Pre-sorting characteristic			
	<i>mess</i>	<i>nvdmess</i>	<i>oddrat</i>	<i>vol</i>
Panel A: The US Market; Equal Weighted Portfolio				
FF3	0.33%	0.47%	0.72%*	0.33%
FF5	0.31%	0.52%	0.78%*	0.44%
FF5 + momentum	0.40%	0.55%	0.84%**	0.45%
FF5 + momentum + liquidity	0.48%	0.56%	0.98%**	0.41%
Panel B: The US Market; Value Weighted Portfolio				
FF3	0.69%**	1.00%**	1.07%**	1.08%***
FF5	0.60%*	1.00%**	1.02%**	1.12%***
FF5 + momentum	0.63%**	1.04%**	1.04%**	1.09%***
FF5 + momentum + liquidity	0.66%**	1.08%***	1.11%**	1.07%***
Panel C: The Chinese Market; Equal Weighted Portfolio				
CH3	2.23%***	3.43%***	4.38%***	2.26%***
CH3 + liquidity (CH4)	2.27%***	3.54%***	4.52%***	2.32%***
CH4 + profitability + investment	2.33%***	3.72%***	4.87%***	2.44%***
Panel D: The Chinese Market; Value Weighted Portfolio				
CH3	2.62%***	3.17%***	4.65%***	2.42%***
CH3 + liquidity (CH4)	2.66%***	3.28%***	4.81%***	2.48%***
CH4 + profitability + investment	2.72%***	3.35%***	5.20%***	2.23%***

Intraday Patterns. To understand how PmS alphas are realized intraday, we run asset-pricing regressions by replacing the daily PmS portfolio returns in Table 8 with PmS portfolio returns within each of the 3-second intraday windows. Figure A.14 shows the cumulative alphas of the value-weighted PmS portfolio intraday. Most of the PmS alphas are realized in the morning, especially within the first 90 minutes in the US (60 minutes in China) after the market open while remaining mostly flat throughout the rest of the day. This is consistent with the interpretation that PmS excess returns reflect higher risks from information asymmetry and adverse selection when trading stocks with stronger periodicities in volume, because the level of informed trading and price discovery tends to be higher after market open relative to the rest of the day, as documented by Foster and Viswanathan (1993).

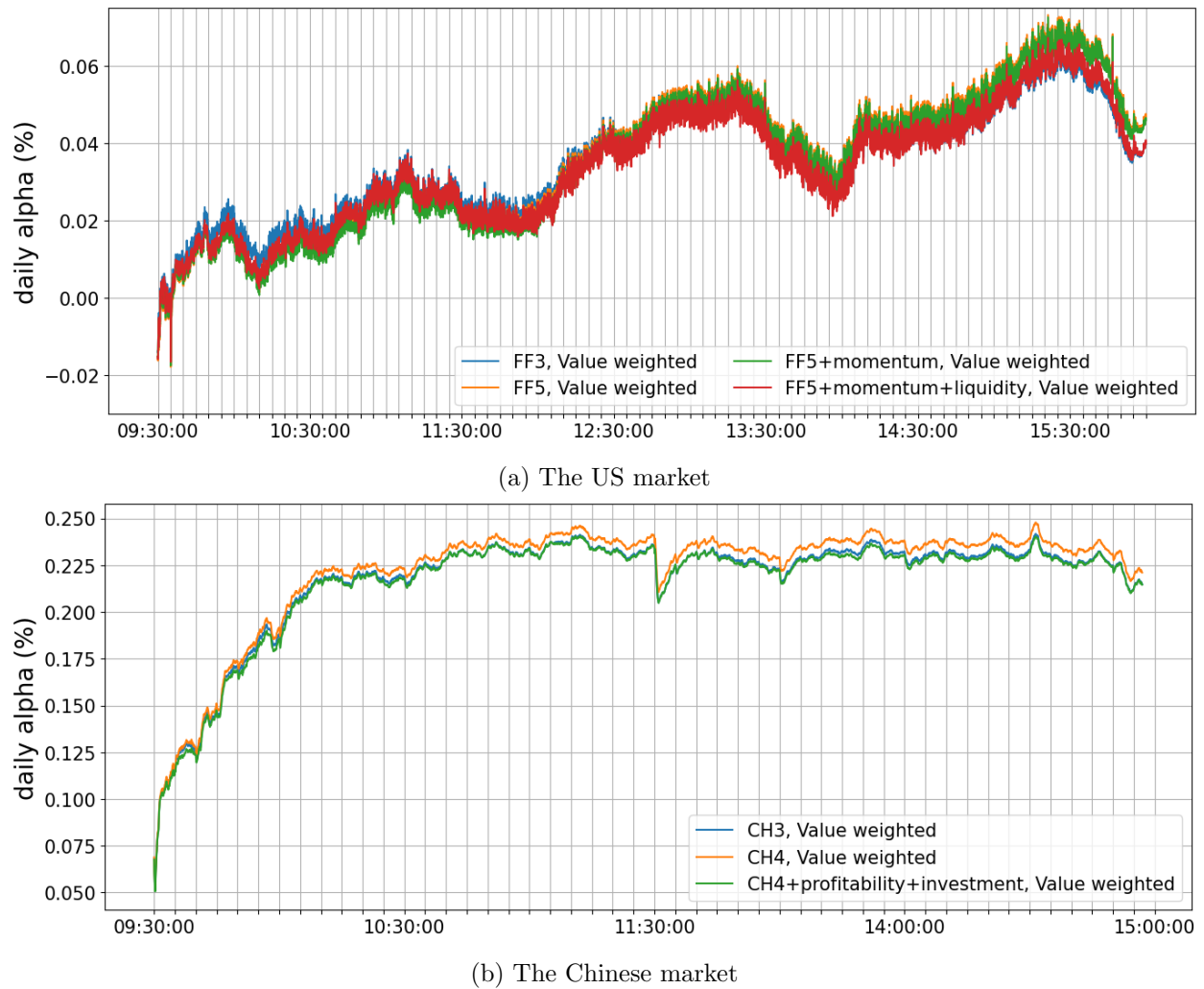


Figure A.14: The intraday cumulative alphas of the value-weighted PmS portfolio. The alphas at the market open (09:30:00 for both markets) are calculated by replacing the daily PmS portfolio returns with the overnight PmS portfolio returns in the regressions of Table 8. The intraday alphas are calculated by replacing the daily PmS portfolio returns with intraday 3-second returns in the regressions of Table 8.

Figure A.15 show the cumulative alphas of the equal-weighted PmS portfolio intraday. Similar to the patterns in Figure A.14, most of the PmS alphas are realized in the morning.

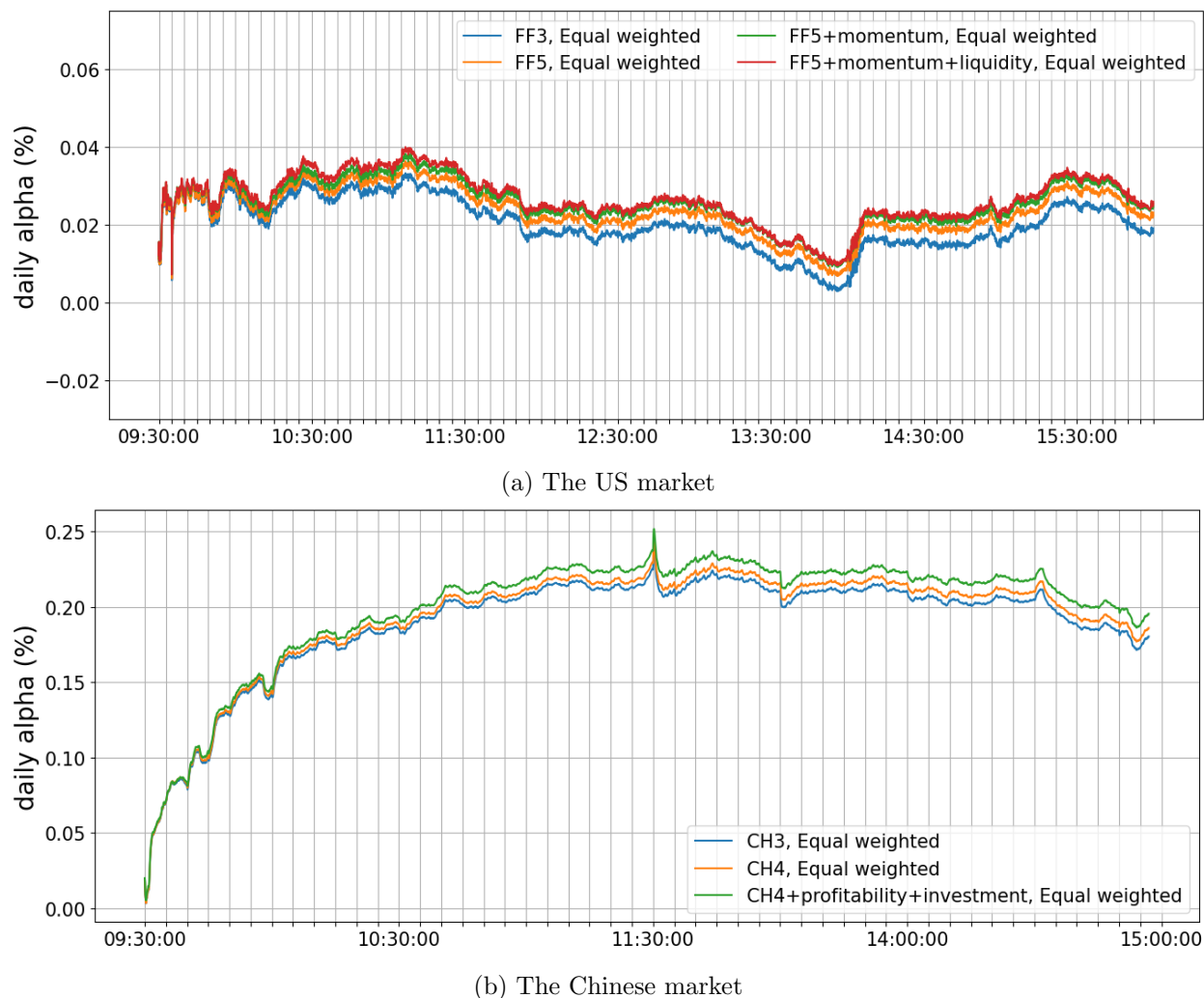


Figure A.15: The intraday cumulative alphas of the equal-weighted PmS portfolio. The alphas at the market open (09:30:00 for both markets) are calculated by replacing the daily PmS portfolio returns with the overnight PmS portfolio returns in the regressions of Table 8. The intraday alphas are calculated by replacing the daily PmS portfolio returns with intraday 3-second returns in the regressions of Table 8.

Stability in the Strength of Periodicity. Finally, Table A.5 shows the transition probabilities of stocks between the five quantiles. The stocks in the top/bottom quintiles with the highest/lowest *peri* have a chance of approximately 50% to stay in the same quintile in the next month. In other words, the relative rank of strengths in volume periodicity for individual stocks changes quickly over time. This finding is also consistent with Chinco and Ye (2017), who document that stocks dominated by short-run fluctuations in trading volume change frequently from month to month.

This result is consistent with our findings in Section 5 that the periodicity of trading volume

Table A.5: Transition probabilities of stocks between quintiles of the strength of periodicity. Each entry (i, j) in the matrices represents the probability that a stock in the i th quintile in month m moves to the j th quintile in the following month.

(a) The US market						
Month m	Quintile	Month $m + 1$				
		Top	Second	Middle	Fourth	Bottom
	Top	57.1%	24.4%	11.5%	5.5%	1.5%
	Second	25.0%	32.4%	24.0%	14.1%	4.6%
	Middle	11.3%	24.0%	29.6%	24.7%	10.3%
	Fourth	4.8%	14.4%	24.5%	33.2%	23.2%
	Bottom	1.9%	5.0%	10.9%	23.5%	58.7%
(b) The Chinese market						
Month m	Quintile	Month $m + 1$				
		Top	Second	Middle	Fourth	Bottom
	Top	49.1%	25.7%	13.6%	7.8%	3.8%
	Second	25.5%	29.0%	23.1%	14.5%	7.9%
	Middle	13.9%	22.5%	26.4%	23.2%	14.0%
	Fourth	7.7%	15.2%	22.9%	29.9%	24.2%
	Bottom	3.8%	7.7%	14.2%	24.8%	49.6%

reflects the behaviors of trading algorithms, and that stronger periodicities reflect more information content from trades at round times. Which stocks encompass more periodic trading and attract more volumes from algorithmic traders is likely dynamic over time. Therefore, the set of stocks with the strongest/weakest periodicities changes month over month. The fact that stocks with the strongest periodicity turn over frequently from month to month again suggests that our PmS factor likely reflects risk exposures different from traditional risk factors, which typically turn over at much slower rates. Nonetheless, our results in Table 8 and Figure A.14 show that they do lead to significant excess returns.

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